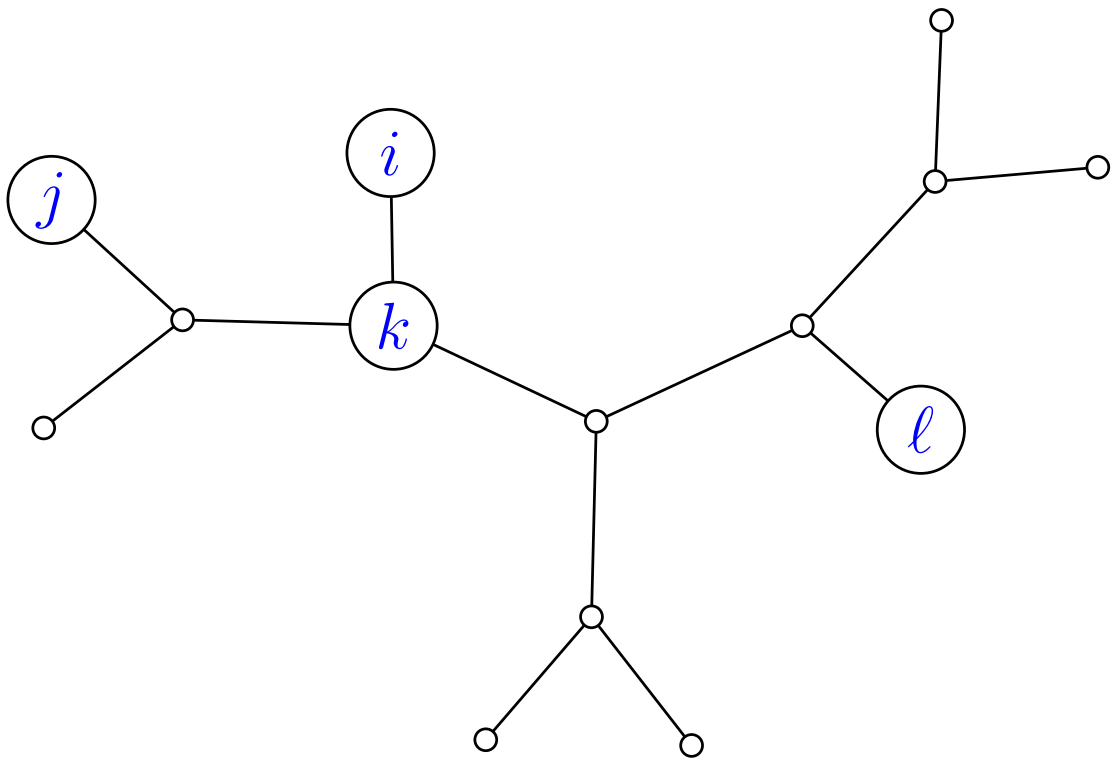
Computational Biology

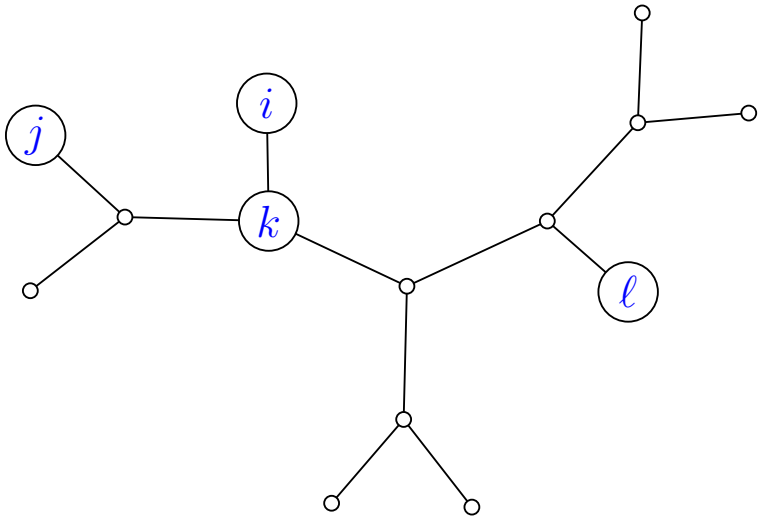
Lecture 14: *Evolutionary Trees*



Additive Trees and Parsimony

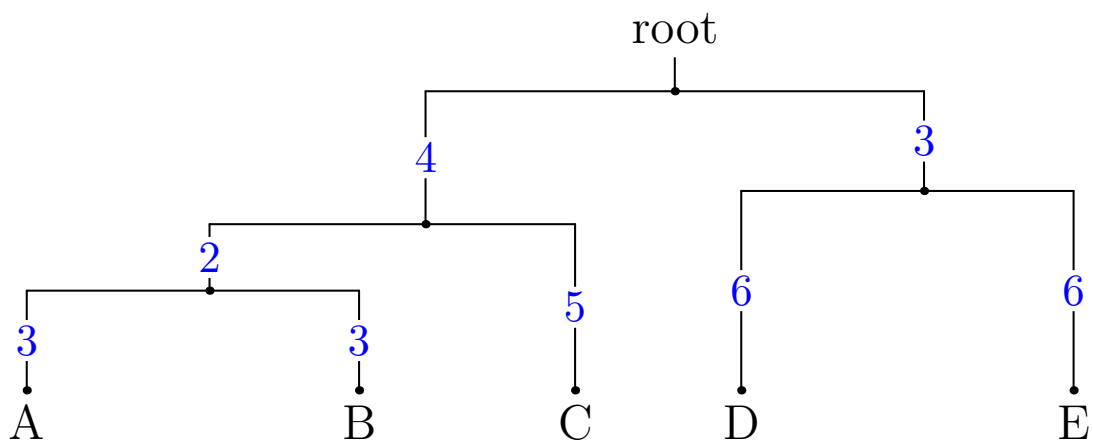
Outline

1. **Additive Trees**
2. Parsimony



Recap

- We looked at building ultrametric trees which would result if mutations accumulated at a regular rate



	A	B	C	D	E
A	0	3	5	9	9
B	3	0	5	9	9
C	5	5	0	9	9
D	9	9	9	0	6
E	9	9	9	6	0

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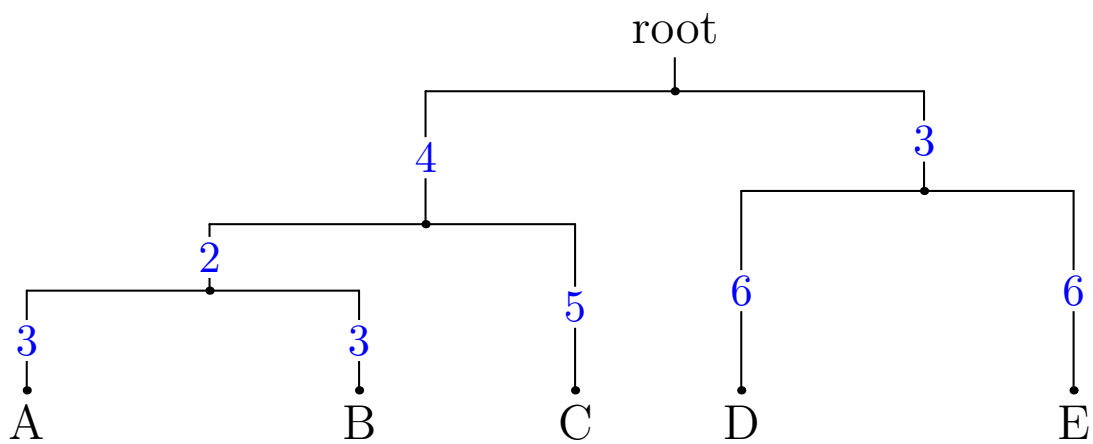

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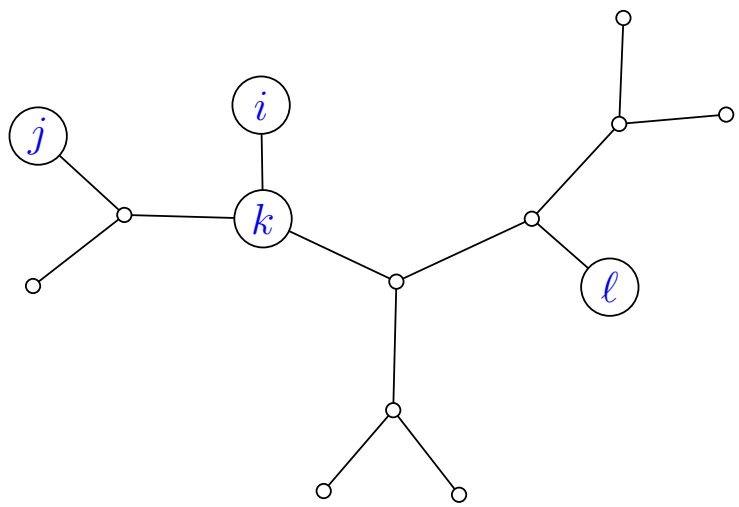
Additivity of Distances

- The condition of ultrametricity is very strong and only works with a perfect molecular clock

- A less strong condition is **additivity of distances**



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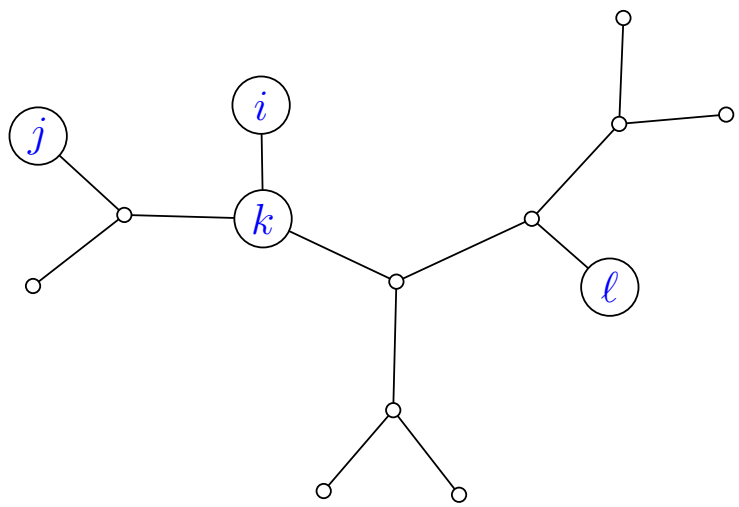

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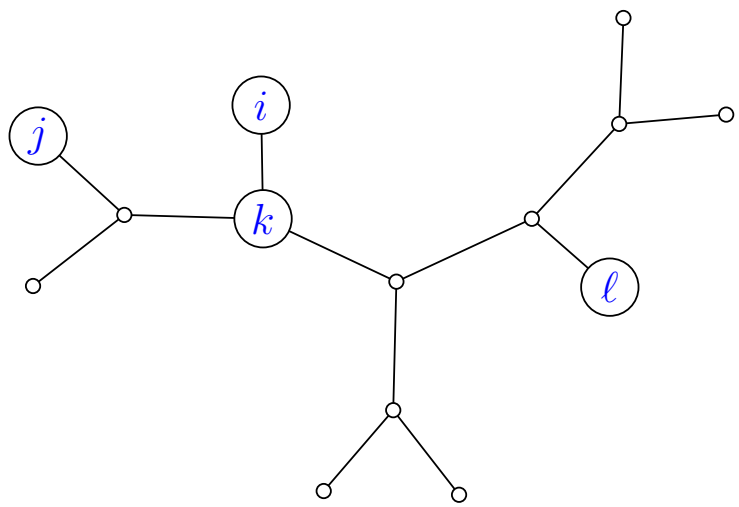

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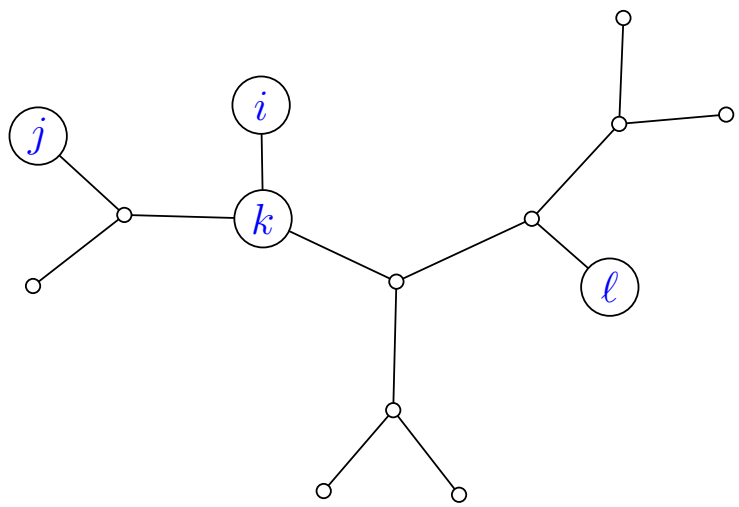

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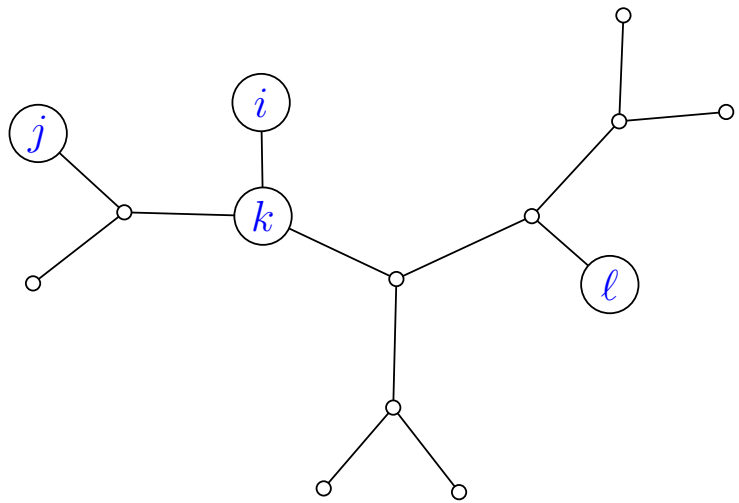
Additivity and Graphs

- Additivity does not assume a constant mutation rate, but only that mutations accumulate
- That is, the distances are consistent with an evolutionary graph



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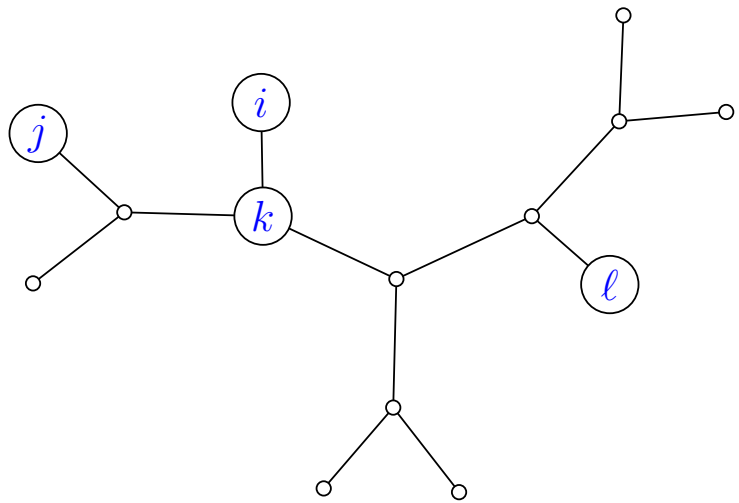

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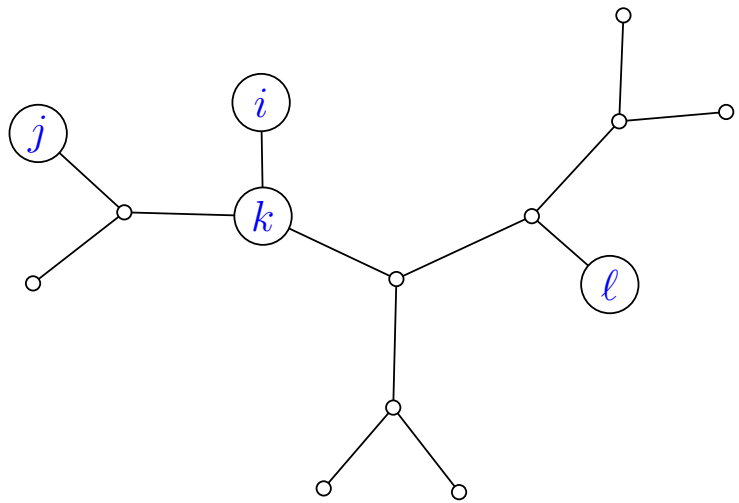

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Building Additive Trees

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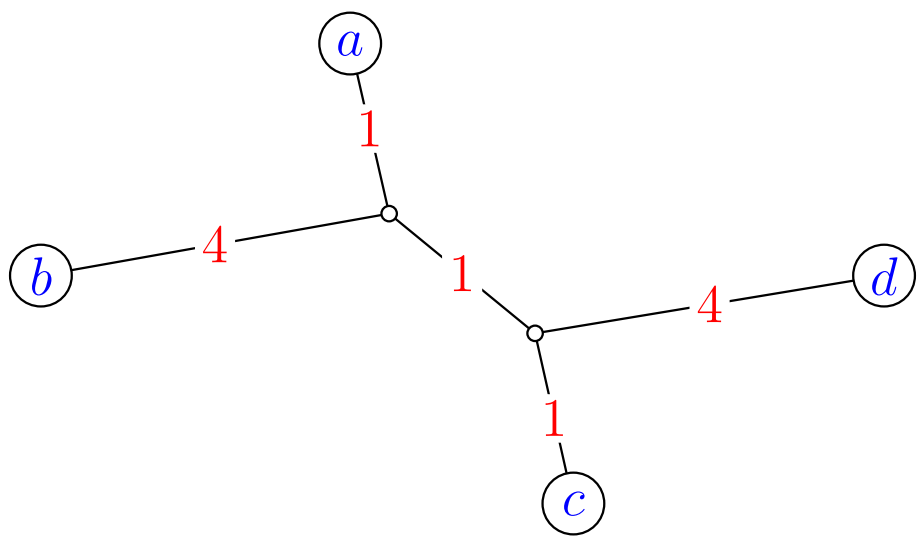

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$$D_{ij} = d_{ij} - r_i - r_j \qquad\qquad r_i = \frac{1}{|\mathcal{L}| - 2} \sum_{k \in \mathcal{L}} d_{ik}$$

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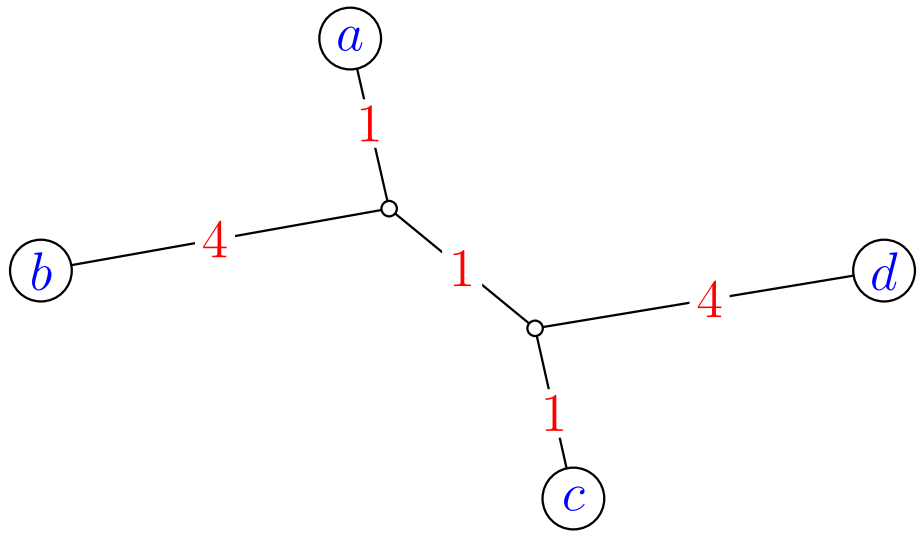

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Neighbour-Joining Algorithm

- From $d_{i,j}$ compute D_{ij}

d_{ij}	1	2	3	4	5	6
1	0	7	14	10	15	17
2	7	0	13	9	14	16
3	14	13	0	12	7	9
4	10	9	12	0	13	15
5	15	14	7	13	0	6
6	17	16	9	15	6	0

D_{ij}	1	2	3	4	5	6
1	0	-23.5	-15.5	-20.5	-14.5	-14.5
2	-23.5	0	-15.5	-20.5	-14.5	-14.5
3	-15.5	-15.5	0	-16.5	-20.5	-20.5
4	-20.5	-20.5	-16.5	0	-15.5	-15.5
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Joining Nodes

- Start with a forest of unconnected nodes
- We then create a new node k and join i and j to k
- We compute the distance $d_{km} = \frac{1}{2}(d_{im} + d_{jm} - d_{ij})$ between k and all other nodes m
- The distance from $d_{ki} = \frac{1}{2}(d_{ij} + r_i - r_j)$ and $d_{kj} = \frac{1}{2}(d_{ij} - r_i + r_j)$
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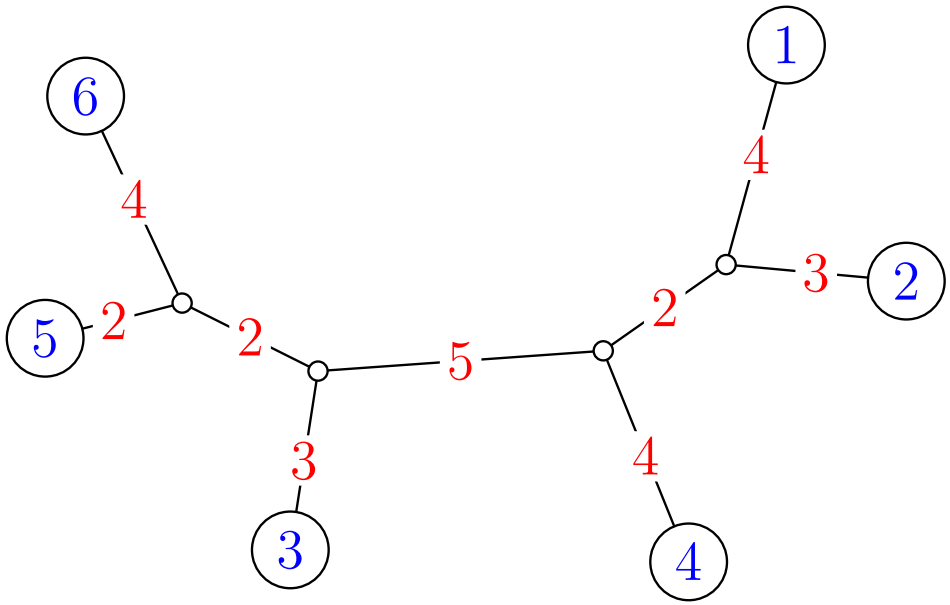

Broken Additivity

- If mutations just accumulated then additivity would follow
- However, backward mutations or two species finding the same mutation breaks additivity
- Thus, to use additivity we must look at large parts of the genome where we are not going to be thrown out by chance events like backward mutation

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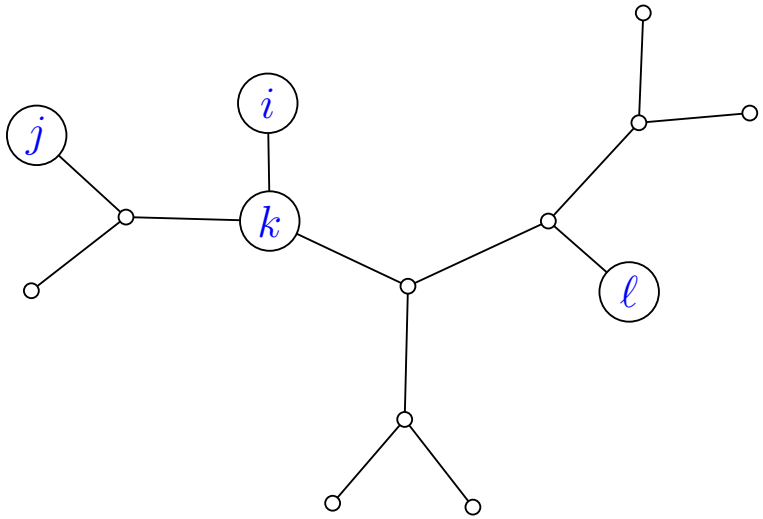
2. **Parsimony**



Character-Based Evolutionary Reconstruction

- A second major approach to tree construction is to find a tree that explains the evolutionary history of a set of sequences with as few mutations as possible
- These are known as **maximum parsimony trees**
- They are the most commonly used trees
- Here we are not given distances, but sequences

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Aligned Protein Sequences

```
           310       320       330       340       350       360       370       380
Human      KSHCIAEVENDEMPADLPSLAADFVESKDVCKNYAEAKDVFLGMFLYEYARRHPDYSVVLLLRLAKTYETTLEKCCAAAD
Bovine     KSHCIAEVEKDAIPENLPPLTADFAEDKDVCKNYQEAKDAFLGSFLYEYSRRHPEYAVSVLLRLAKEYEATLEECCAKDD
Horse      KSHCIAEVKEDDLPSDLPALAADFAEDKEICKHYKDAKDVFLGTFLYEYSRRHPDYSVSLLLRIAKTYEATLEKCCAEAD
Rabbit     KAHCIYGLHNDETPAGLPAVAEEFVEDKDVCKNYEEAKDLFLGKFLYEYSRRHPDYSVVLLLRLGKAYEATLKKCCATDD
Camel      KSHCIHEAERDEMPENLPAITEQFAEDKDVCKHYTEEKDVFLGMFLHEYARRHPEYAVSLLLRIAKEYEATLEDCCAKDD

           390       400       410       420       430       440       450       460
Human      PHECYAKVFDEFKPLVEEPQNLIKQNCELFEQLGEYKFQNALLVRYTKKVPQVSTPTLVEVSRNLGKVGSKCCKHPEAKR
Bovine     PHACYSTVFDKLKHLVDEPQNLIKQNCDQFEKLGEYGFQNALIVRYTRKVPQVSTPTLVEVSRSLGKVGTRCCTKPESER
Horse      PPACYRTVFDQFTPLVEEPKSLVKKNCDLFEEVGEYDFQNALIVRYTKKAPQVSTPTLVEIGRTLGKVGSRCCKLPESER
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Camel      PHACYATVFDKLQHLADEPQNLVKQNCELFEKLGEYGFQNDILVRYTKRLPQVSTPTLVEVARGLGRVGTKCCTLPESNR

           470       480       490       500       510       520       530       540
Human      MPCAEDYLSVVLNQLCVLHEKTPVSDRVTKCCTESLVNRRPCFSALEVDETYVPKEFNAETFTFHADICTLSEKERQIKK
Bovine     MPCTEDYLSLILNRLCVLHEKTPVSEKVTKCCTESLVNRRPCFSALTPDETYVPKAFDEKLFTFHADICTLPDTEKQIKK
Horse      LPCSENHLALALNRLCVLHEKTPVSEKITKCCTDSLAERRPCFSALELDEGYVPKEFKAETFTFHADICTLPEDEKQIKK
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Camel      MSCAEDYLSLILNRLCVLHEKTPVSPRVTKCCTESLVN-RPCFSSLTADETYEPKEFDEKTFTFHADLCSVSEF------
```

Algorithms for Parsimony

- There is no efficient algorithm for finding maximum parsimony trees
- We have to check all possible trees (although we can use branch and bound)
- This is feasible for a moderate number of sequences (< 50)
- Need to consider
  - ⋆ Assigning a cost to each tree
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Costing Trees

- We assume that we have a set of aligned sequences
- We treat each site independently!
- The total cost is the sum of the costs at each site
- We assume that we are given a binary tree whose leaves are the sequences
- Each node will either be a leaf or will have left and right children

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Costs of Sequence

- We assume that the element a at each site belongs to an alphabet $\mathcal{A}$
- For DNA this would be $\mathcal{A} = \{A, C, G, T, -\}$ where $-$ represents gaps (insertion or deletion)
- For proteins it would the amino acid and a gap
- We consider "weighted parsimony" where we weight the cost of a substitution by a matrix $s(a, b)$
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Dynamic Programming Solution

```
tree::cost() {
  cost = [];                                  % initialise array
  if tree.isLeaf() {                         % leaf
    for a∈ 𝒜 {
      cost[a] = ∞
    }
    cost[x] = 0;                              % actual sequence
  } else {
    for a∈ 𝒜 {
      leftCost = left.cost()                  % recursively find cost
      rightCost = right.cost()
      cost[a] = min_b leftCost[b] + S(a,b) % dynamic program
      cost[a] += min_b rightCost[b] + S(a,b)
    }
  }
  return cost                                  % return array
}
```

Dynamic Programming Solution

```
tree::cost() {
  cost = [];                                  % initialise array
  if tree.isLeaf() {                         % leaf
    for a∈ 𝒜 {
      cost[a] = ∞
    }
    cost[x] = 0;                              % actual sequence
  } else {
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```
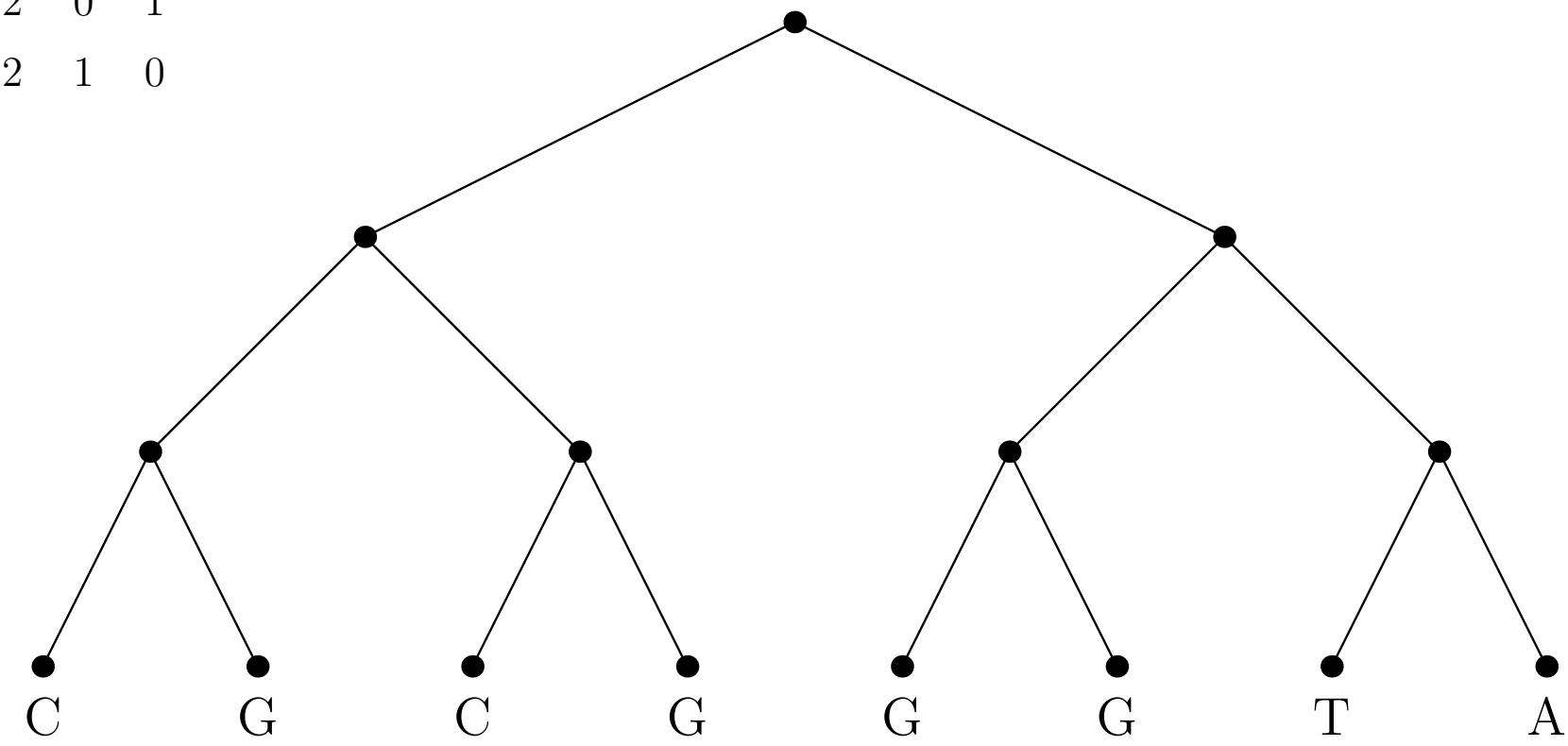

Computing Costs

	A	T	C	G
A	0	1	2	2
T	1	0	2	2
C	2	2	0	1
G	2	2	1	0



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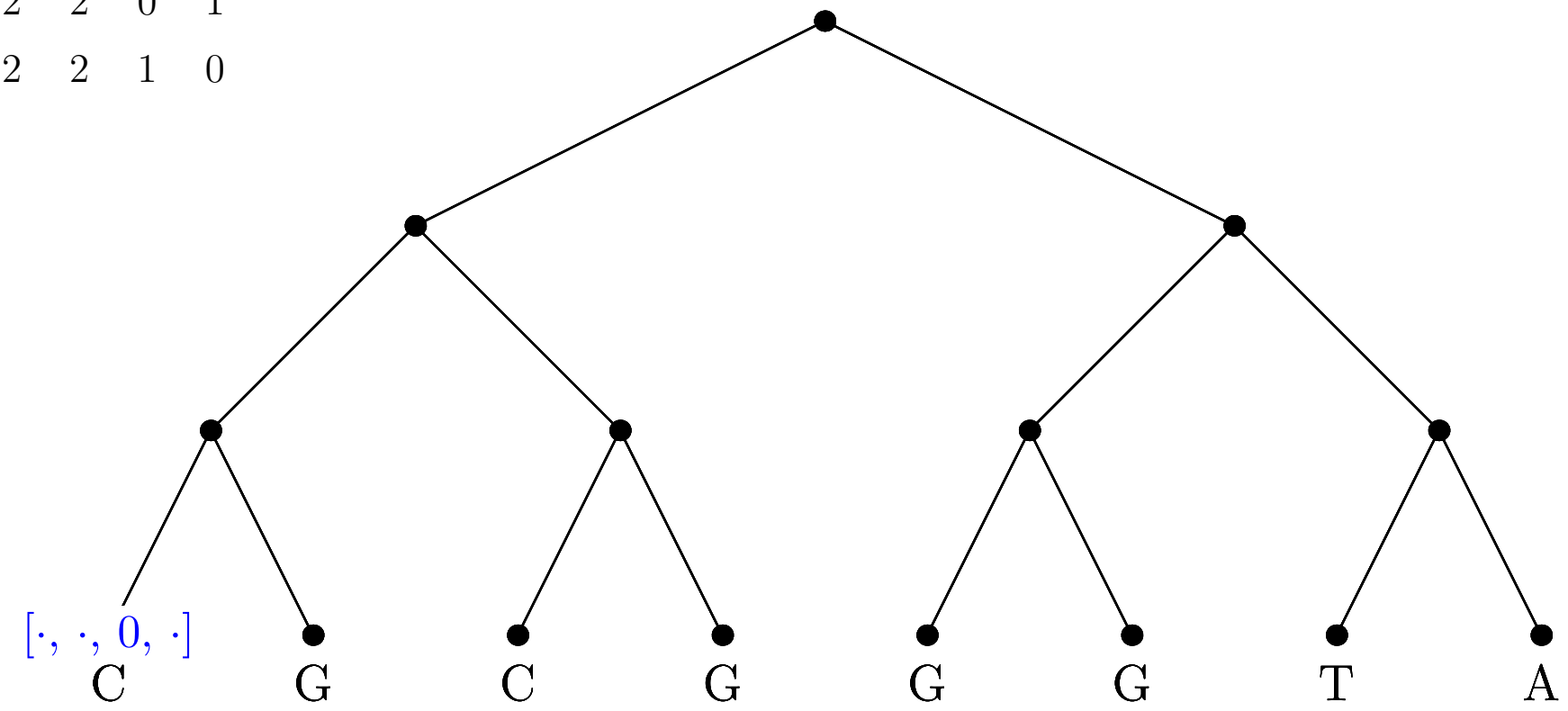

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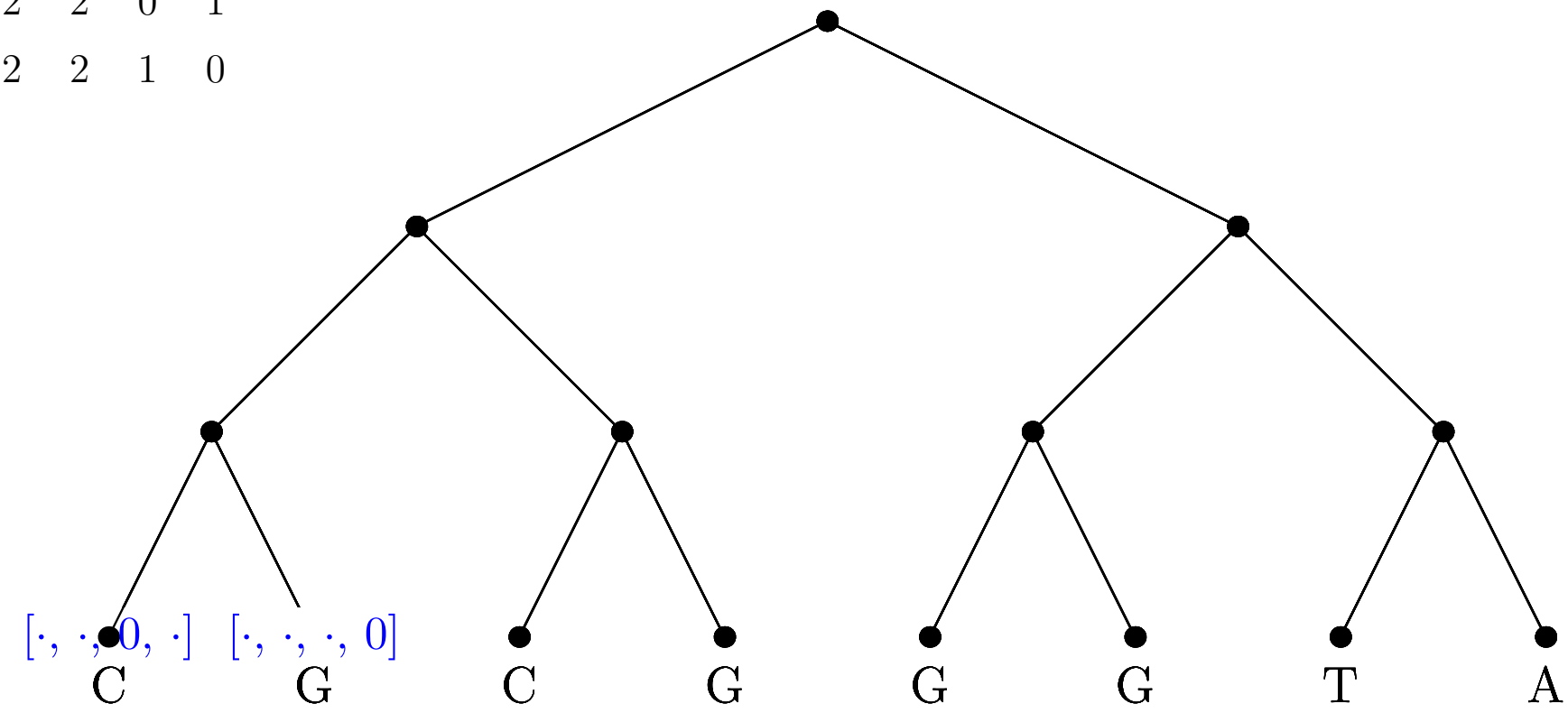

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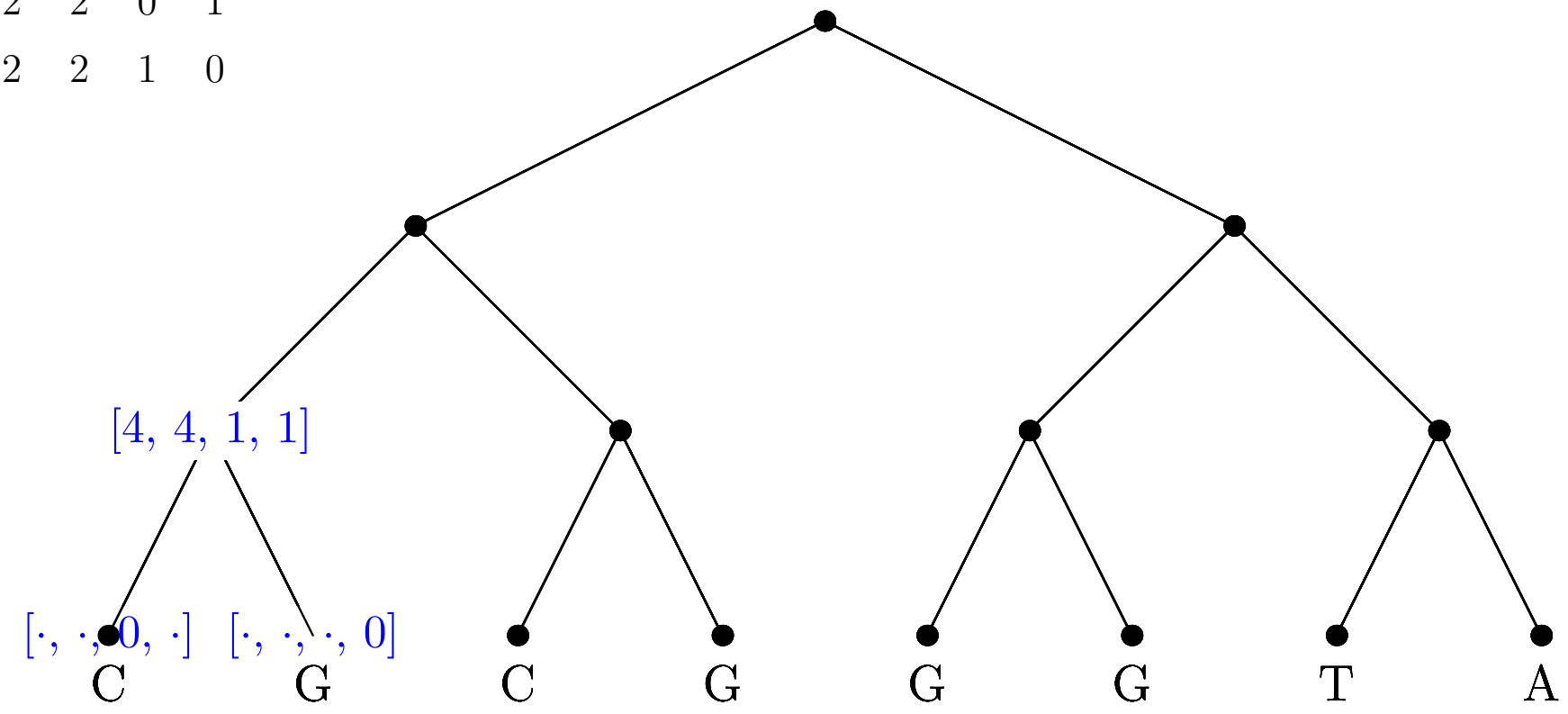

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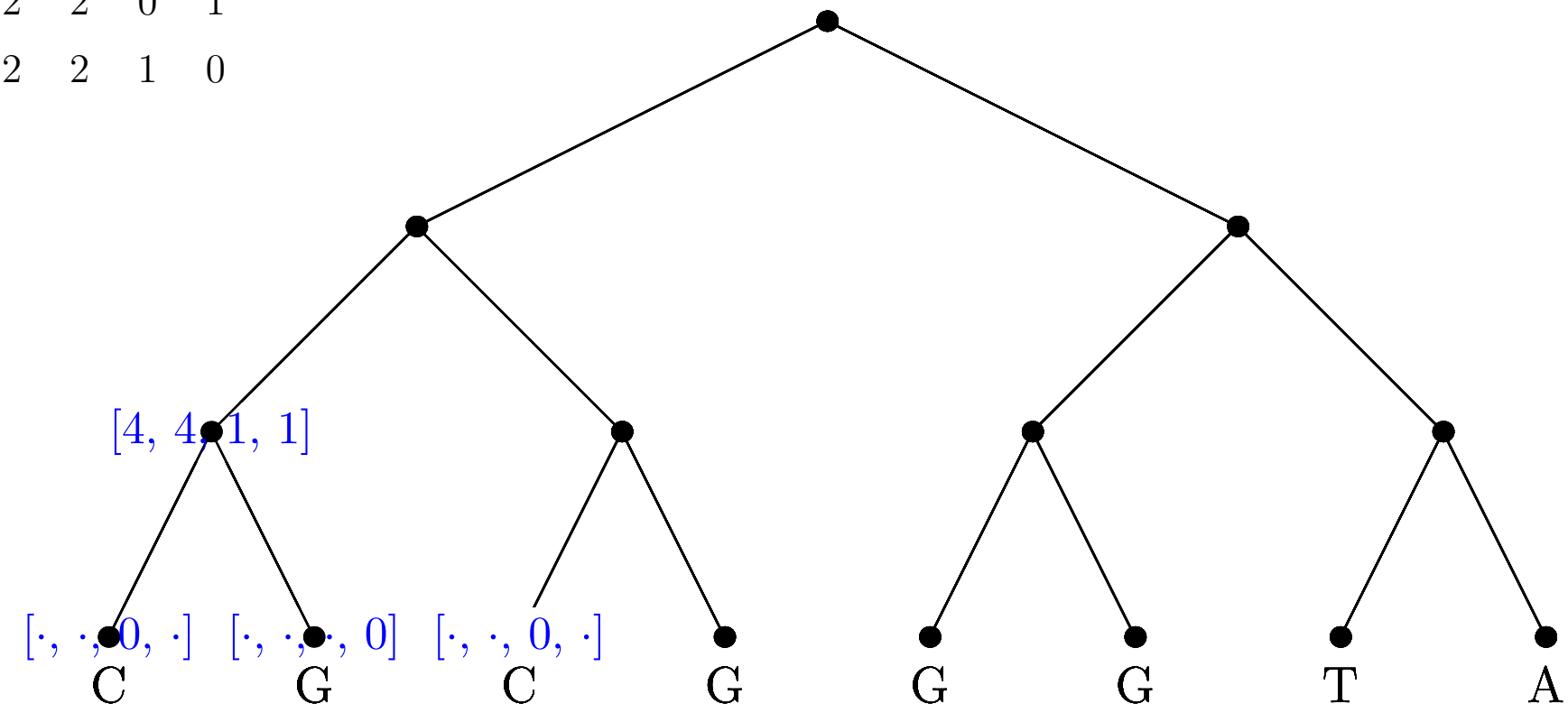

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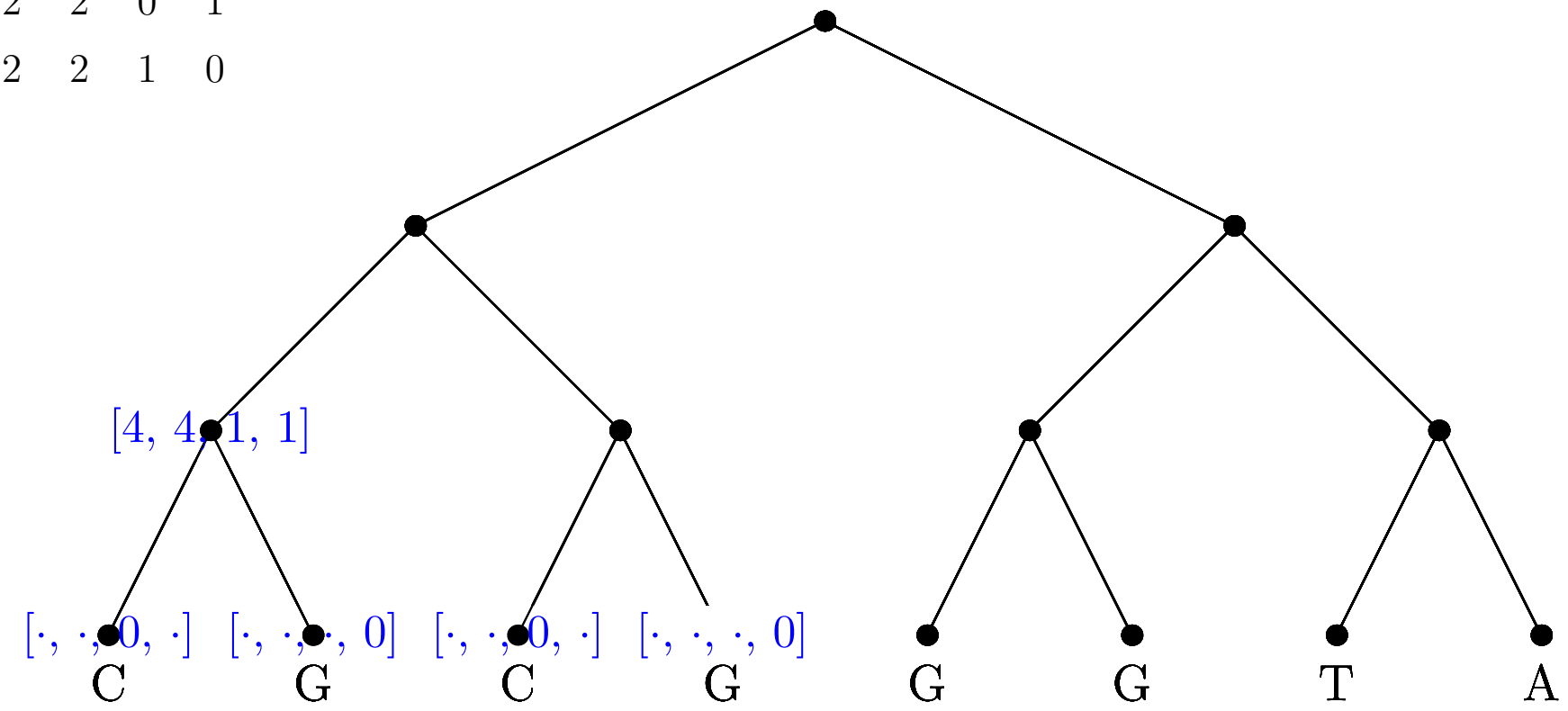

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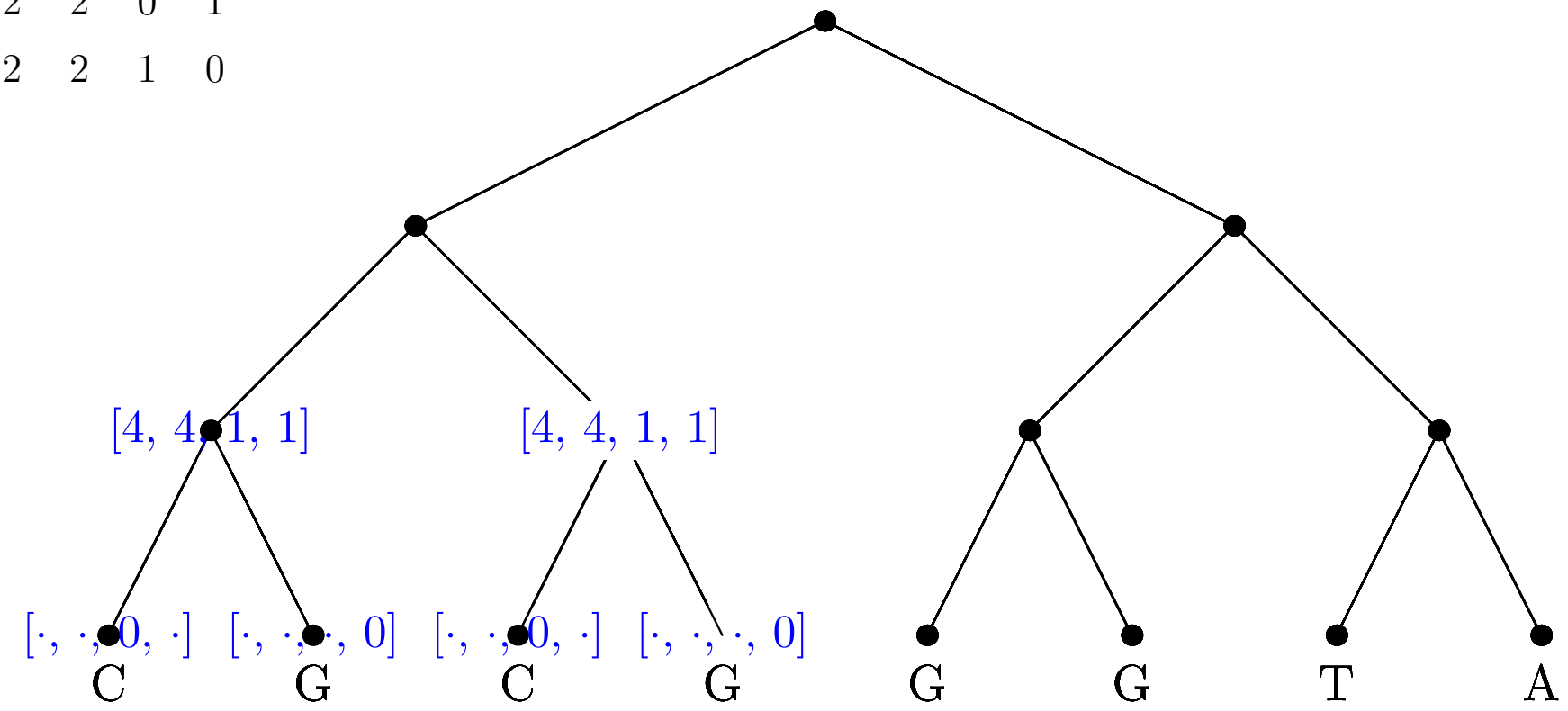

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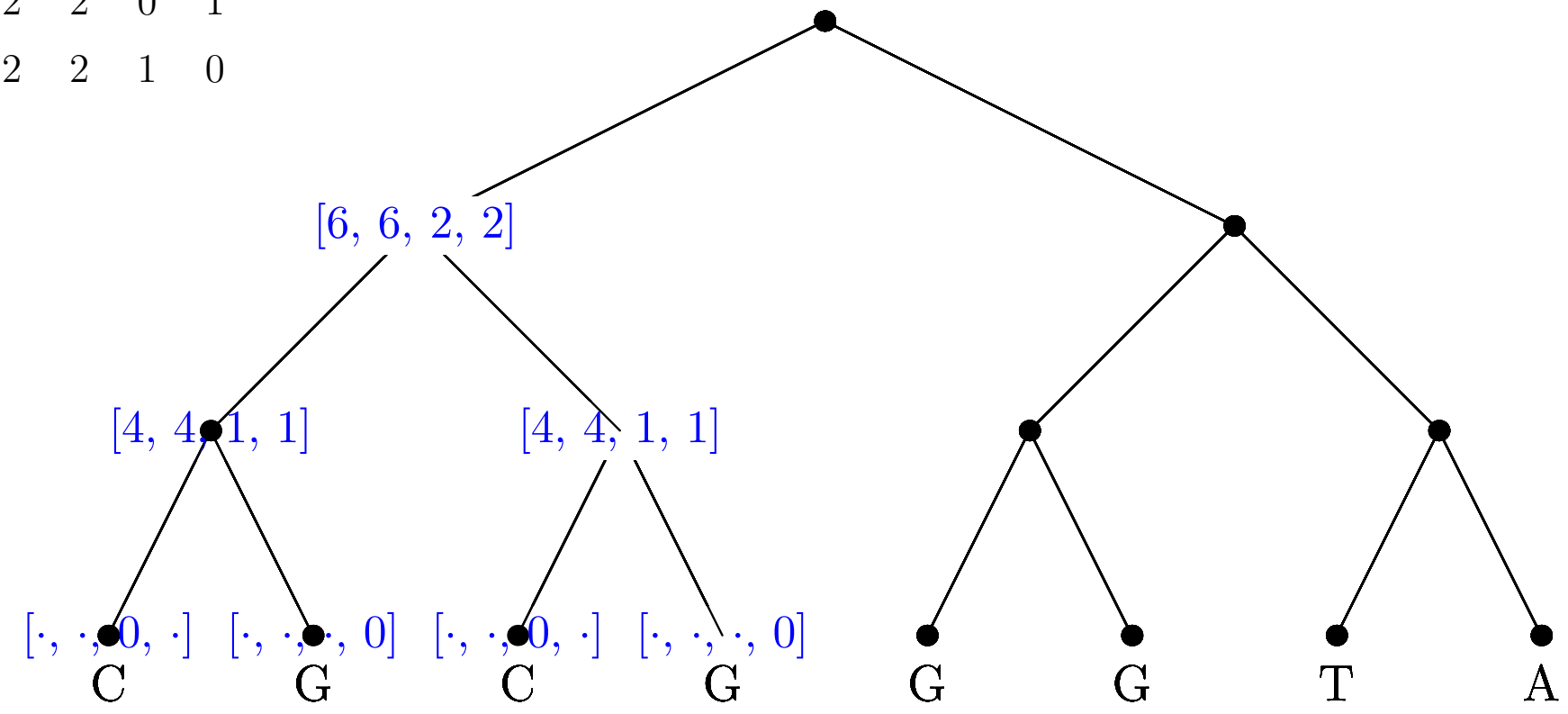

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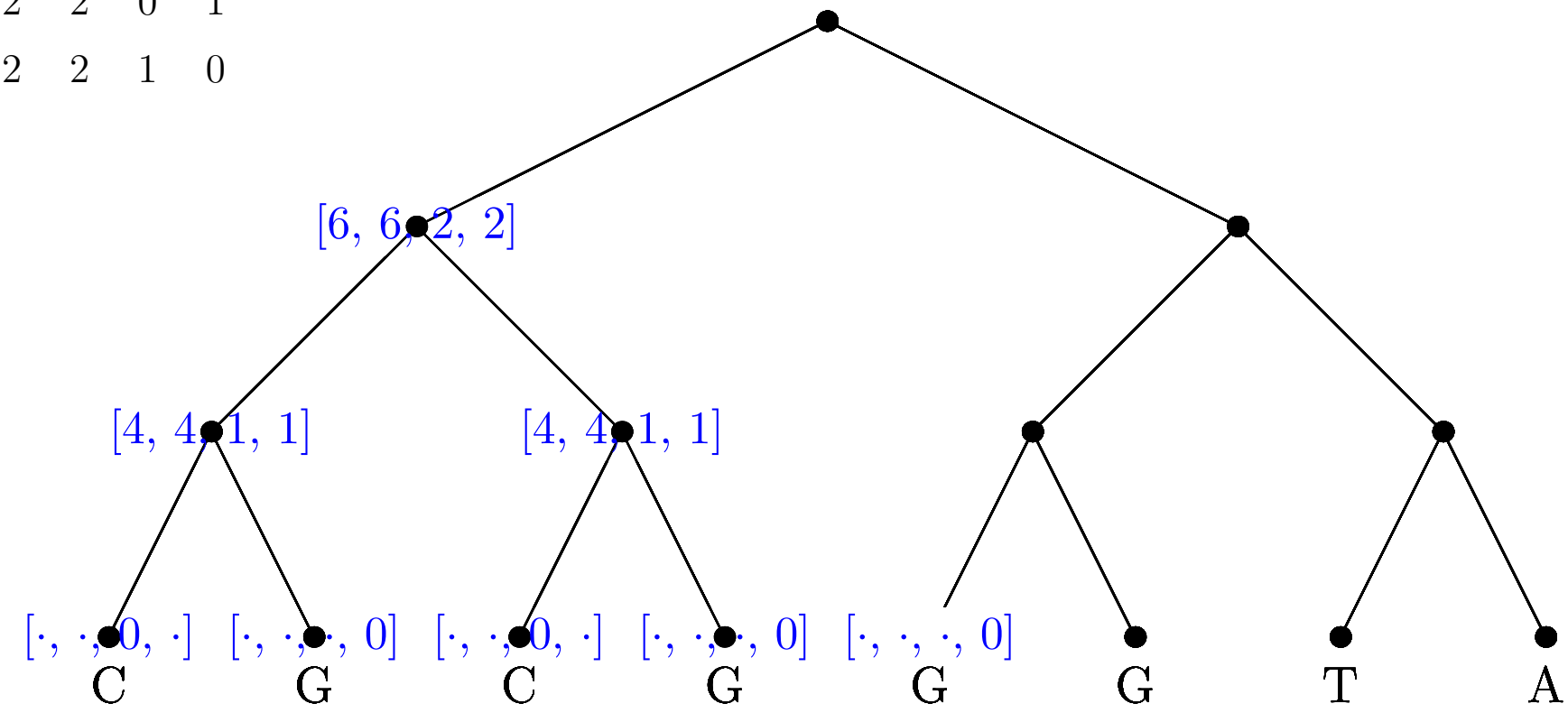

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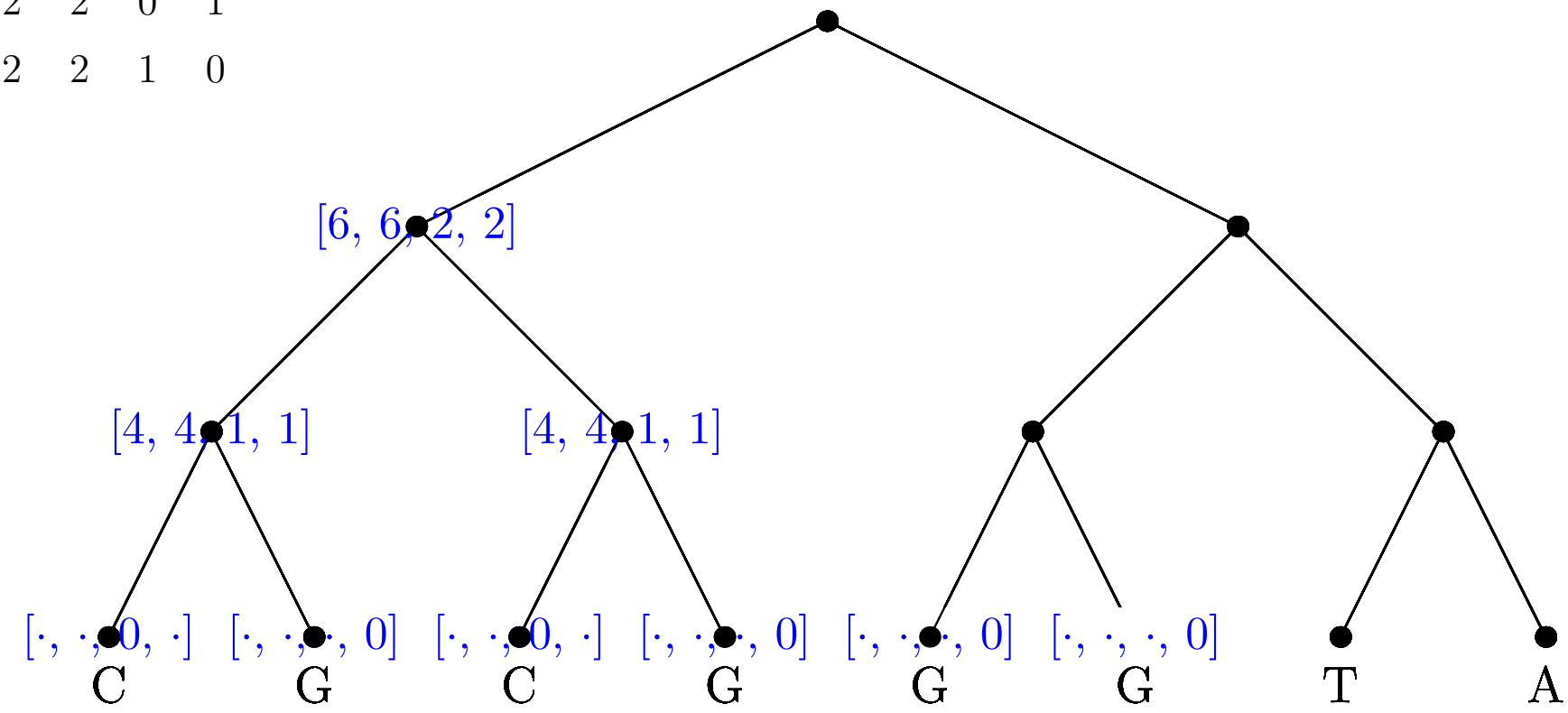

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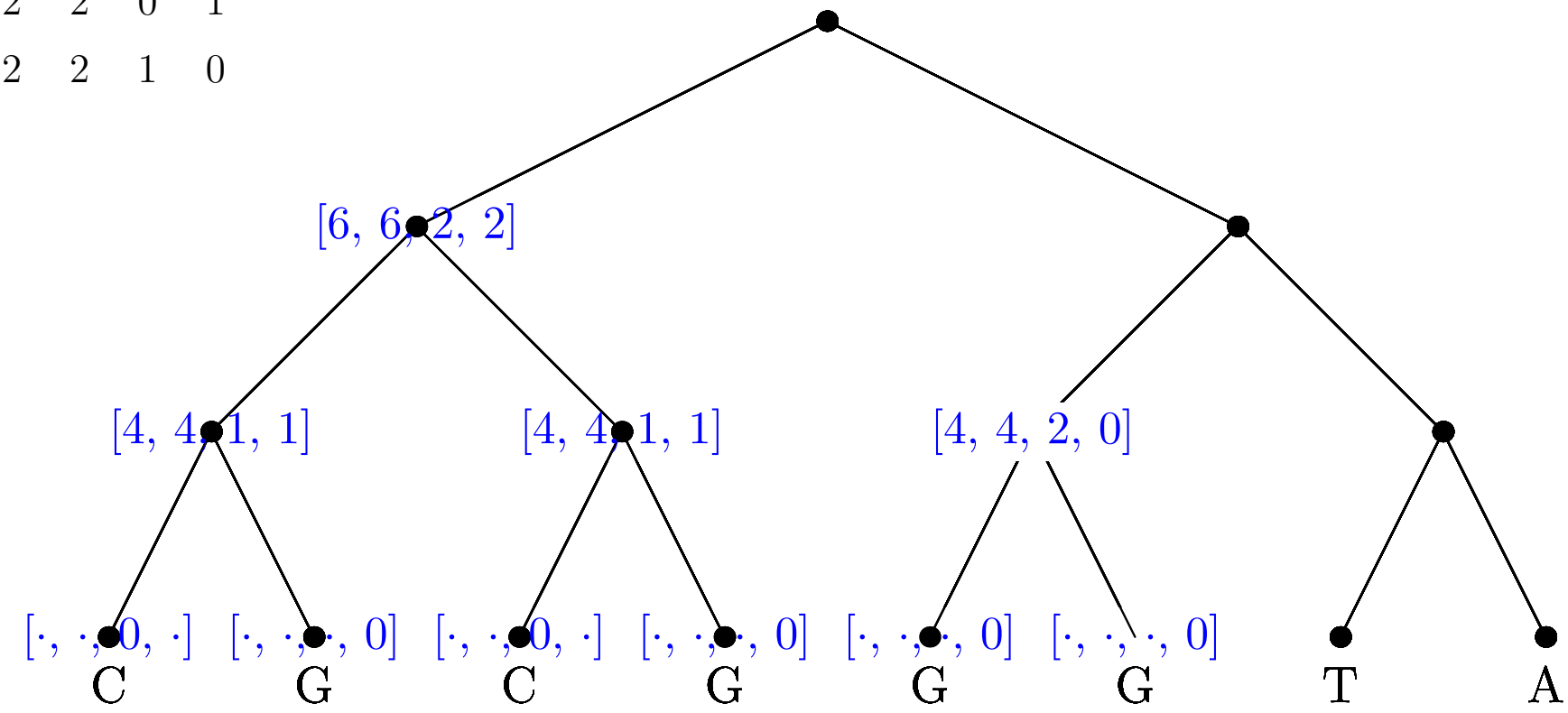

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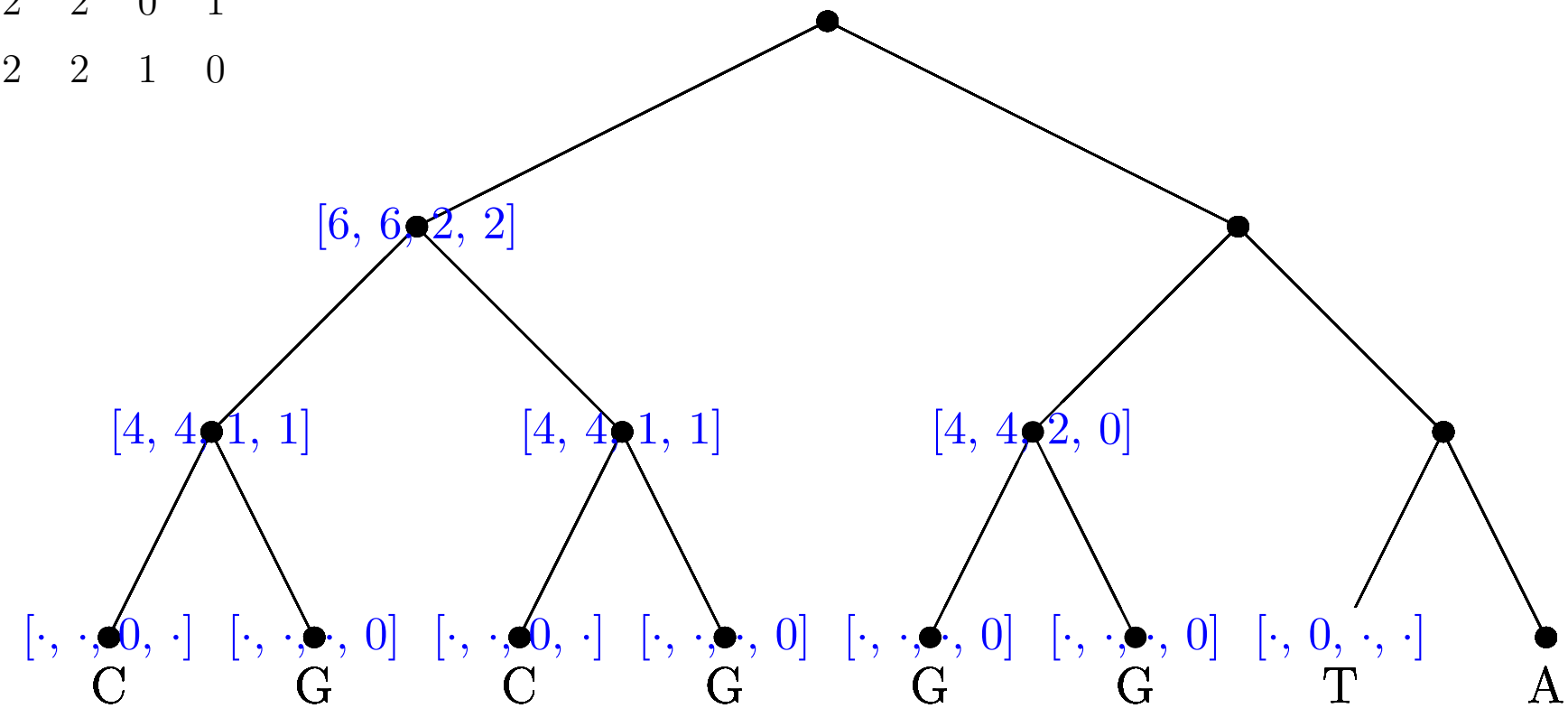

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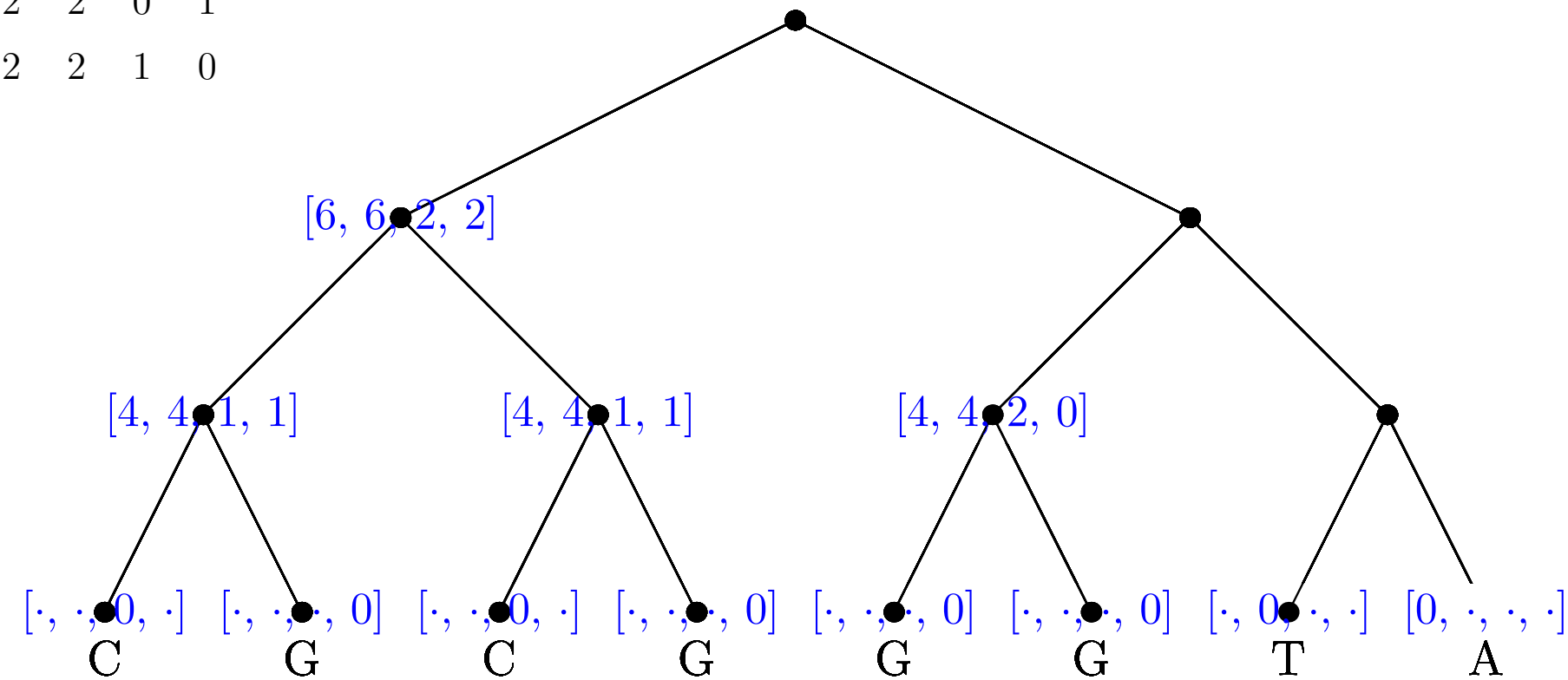

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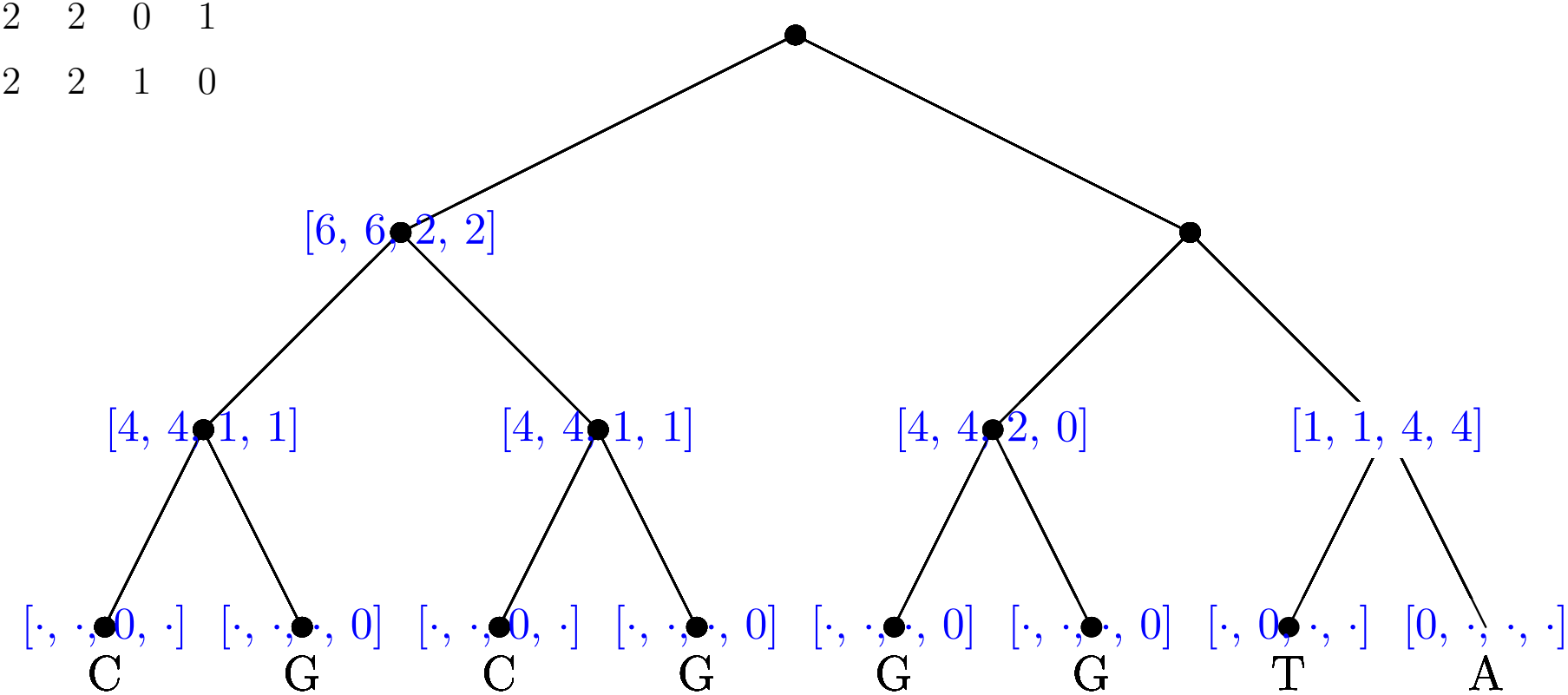

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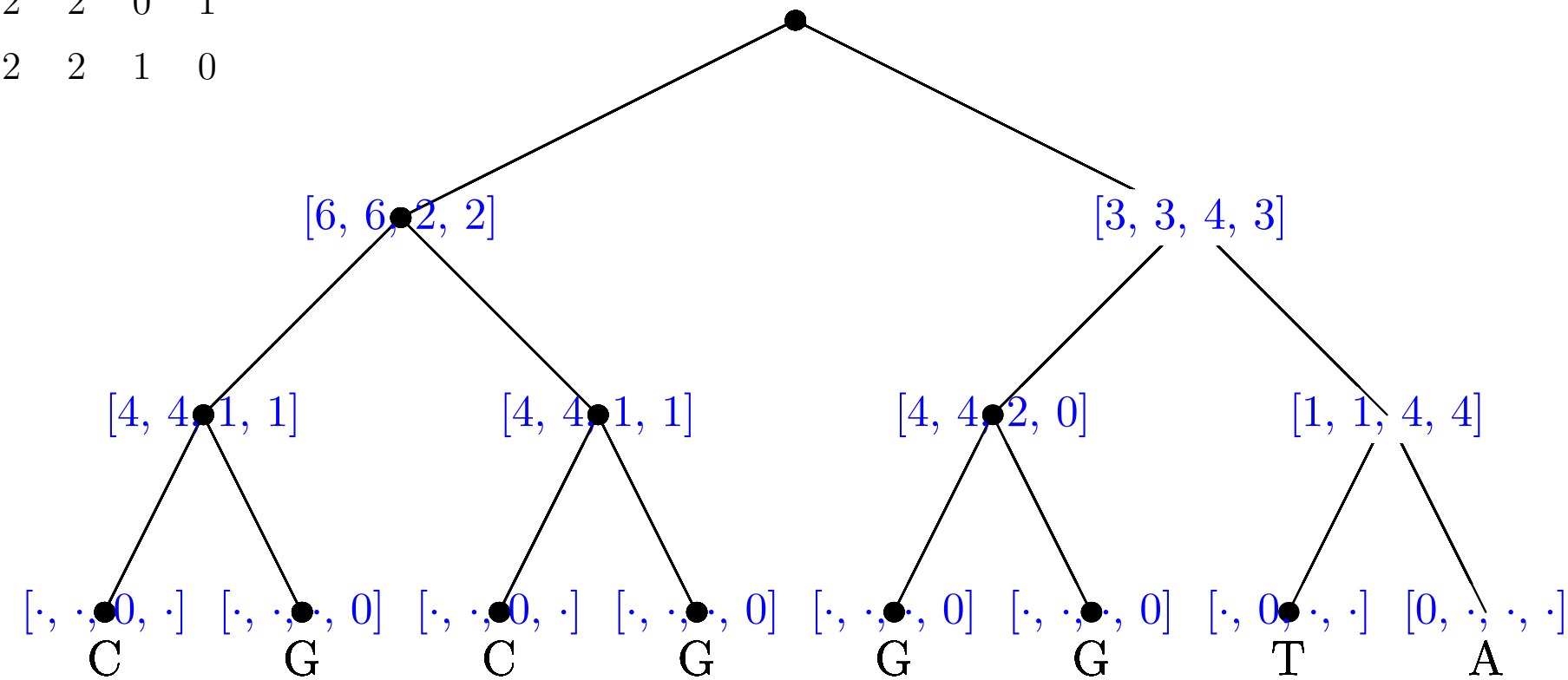

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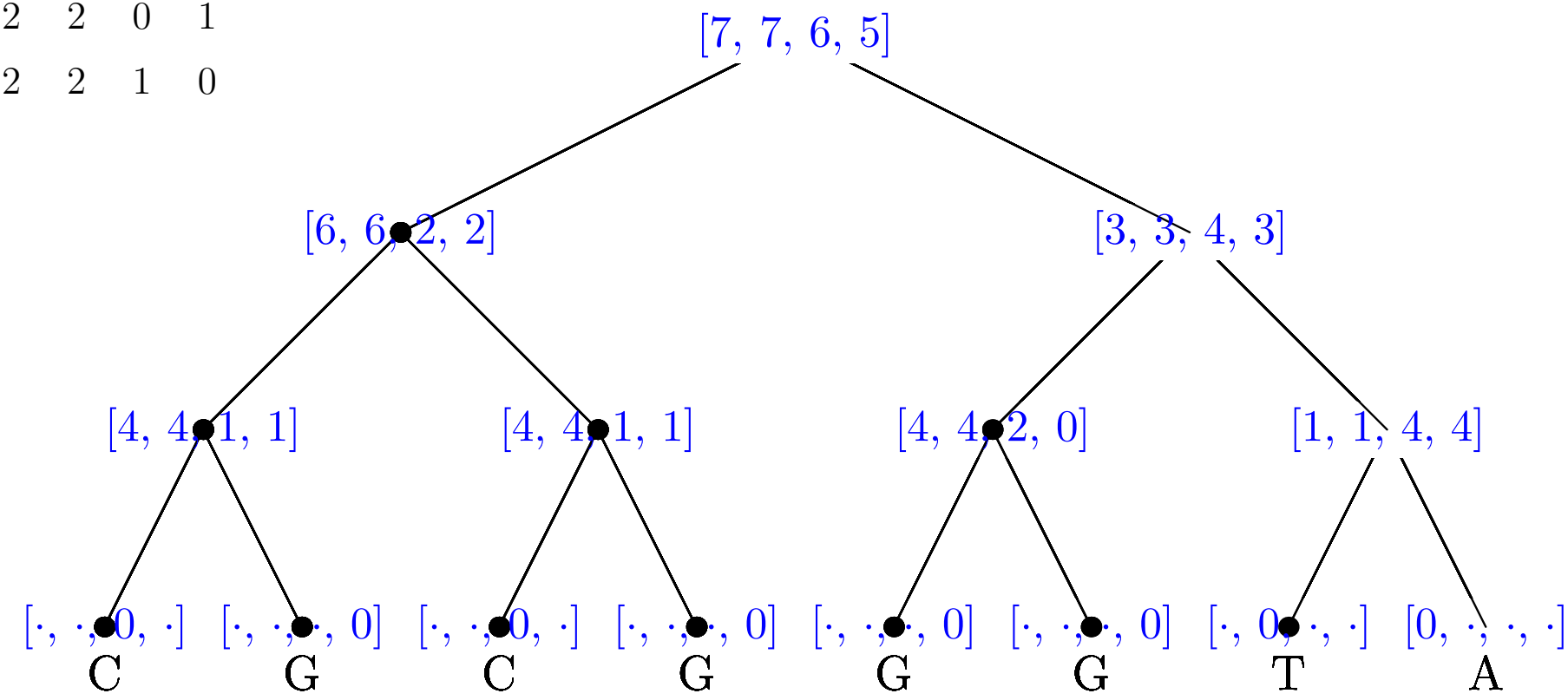

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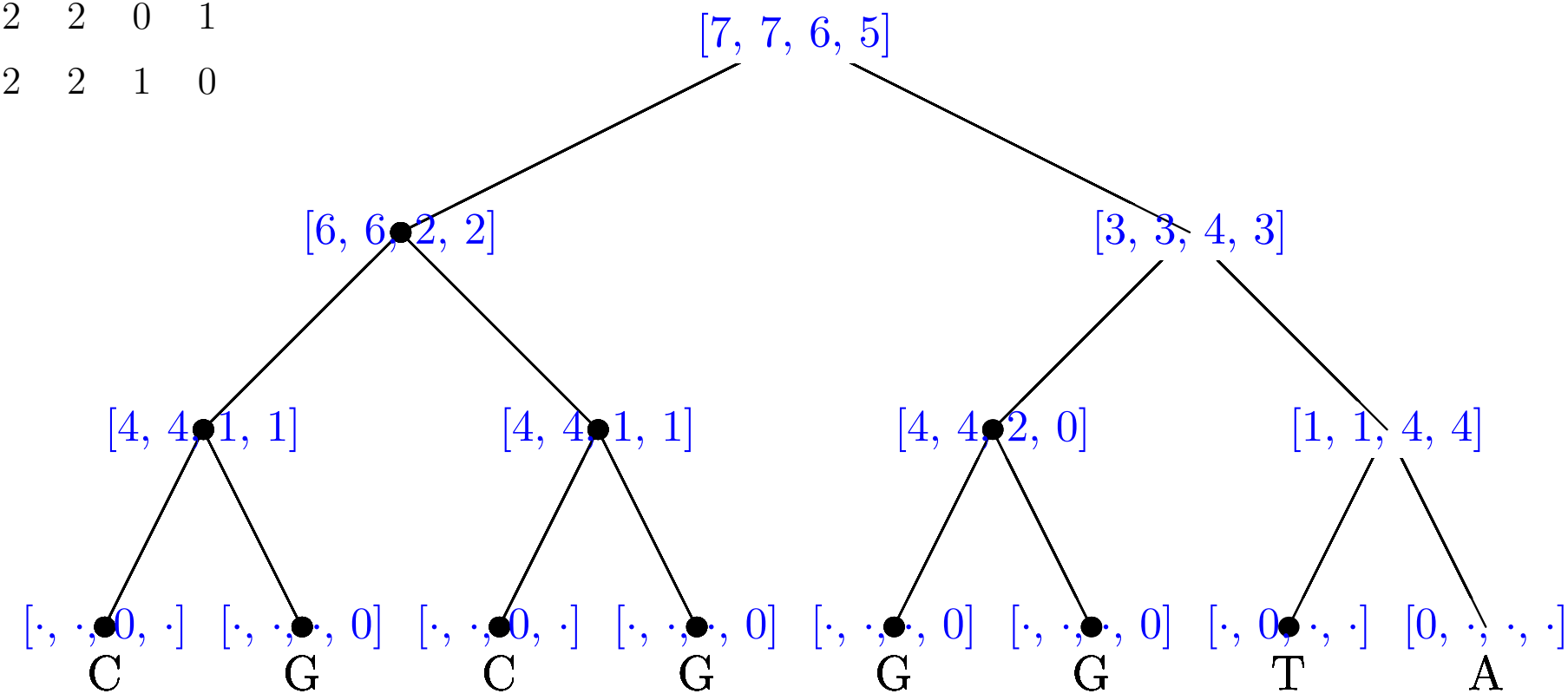

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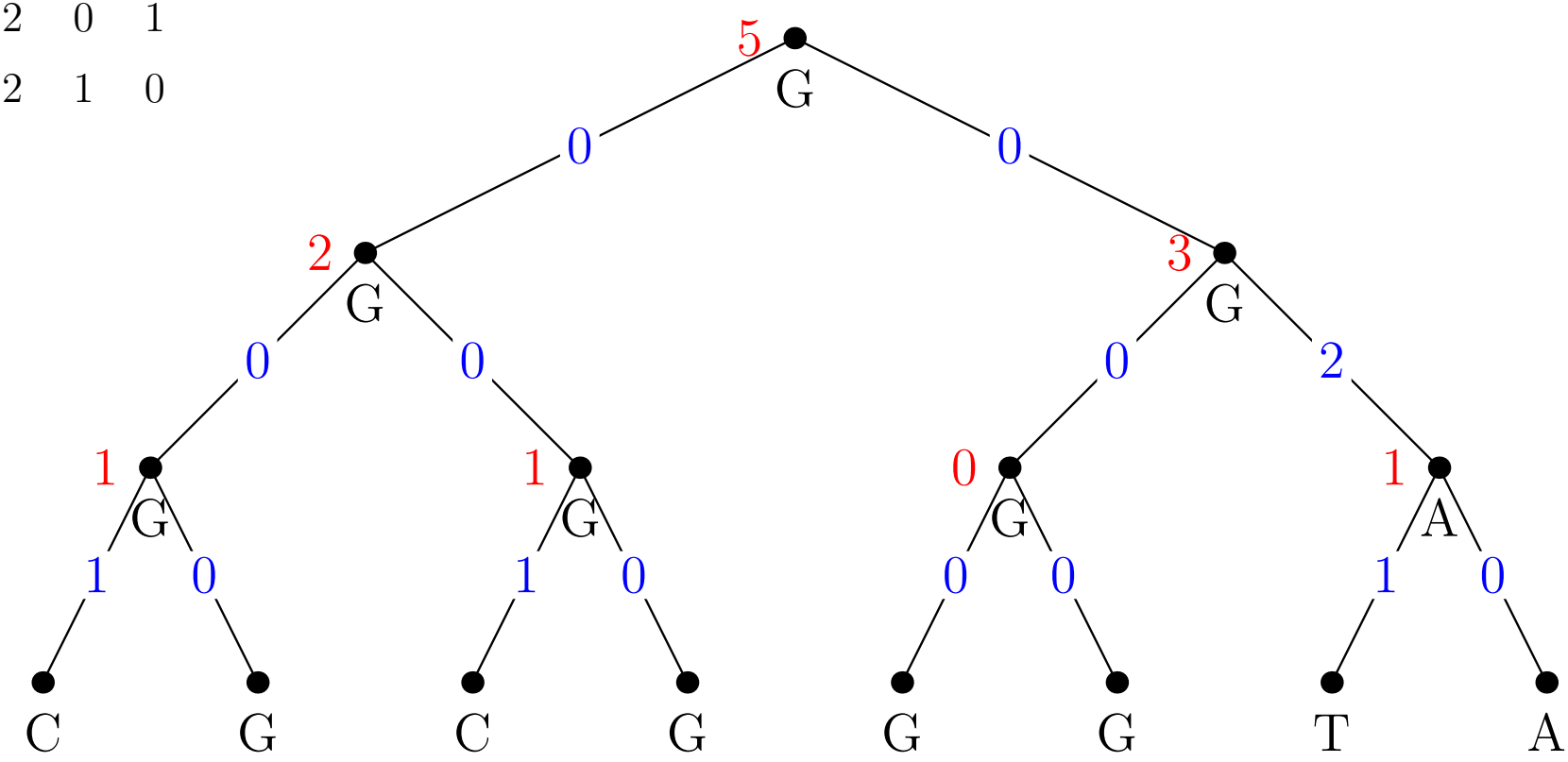
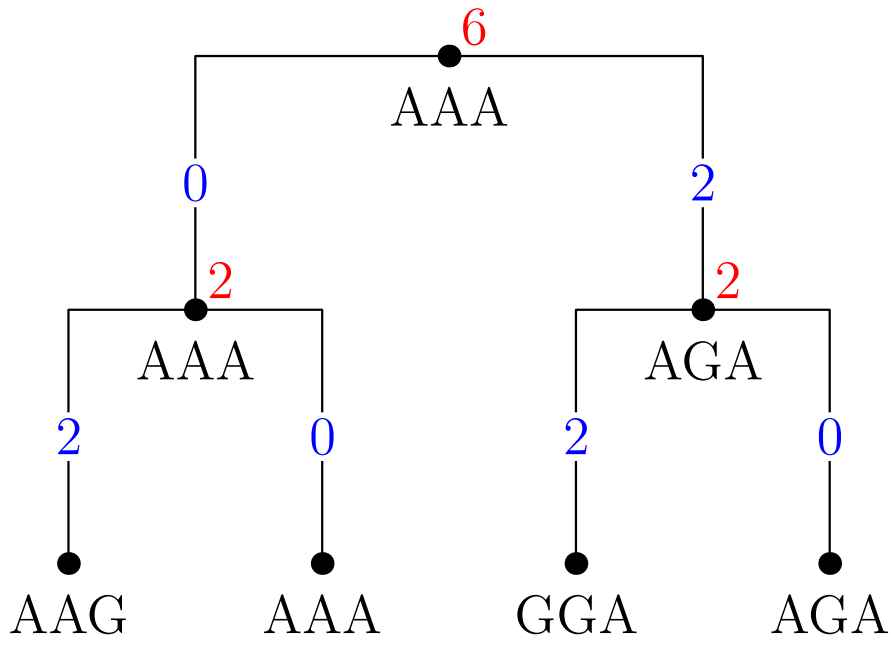


Costs for Sequences

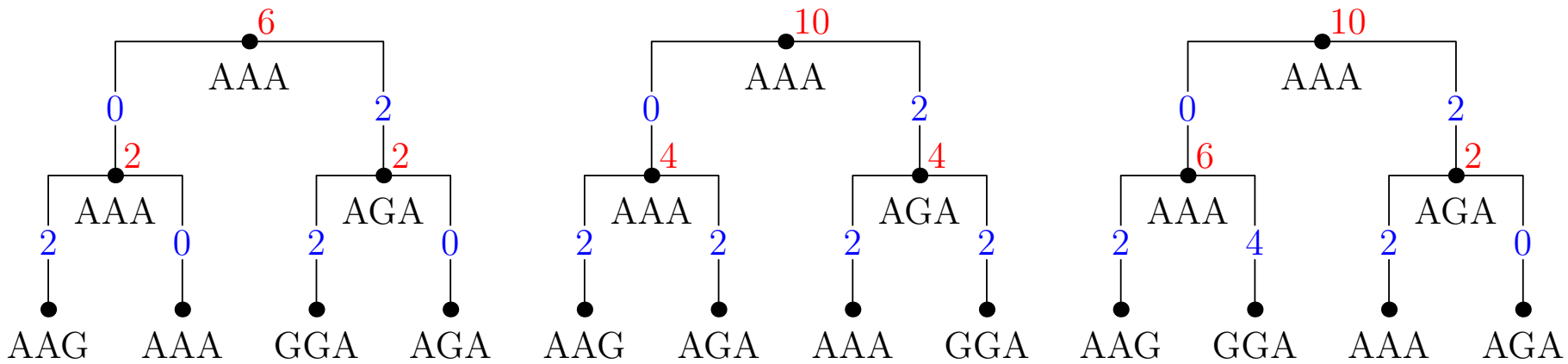
- For sequences we compute the costs at each site

	A	T	C	G
A	0	1	2	2
T	1	0	2	2
C	2	2	0	1
G	2	2	1	0



Many Different Trees

- For a set of sequences there are many different possible trees with different costs



- The parsimony problem is to find the tree with the smallest cost

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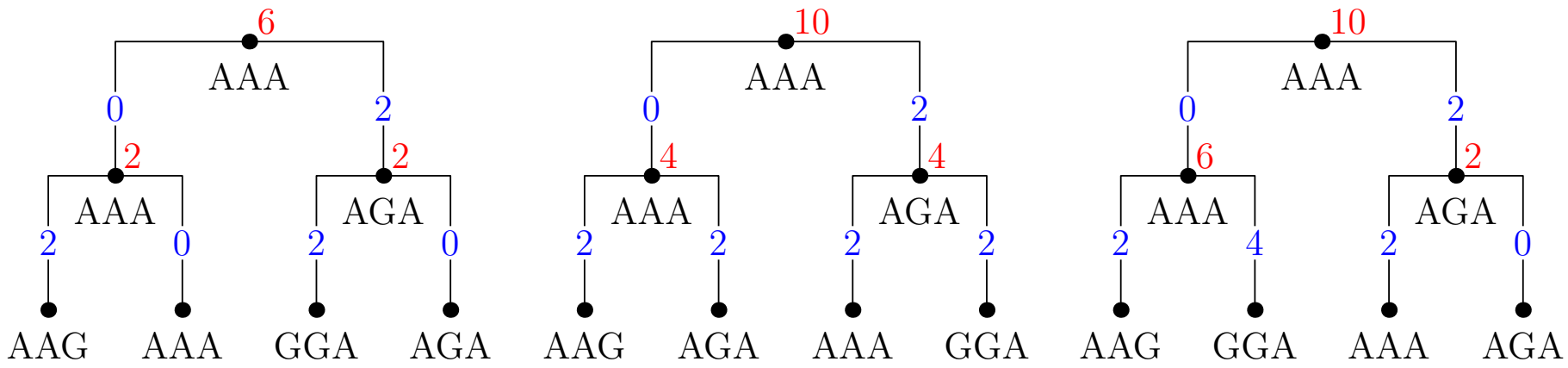

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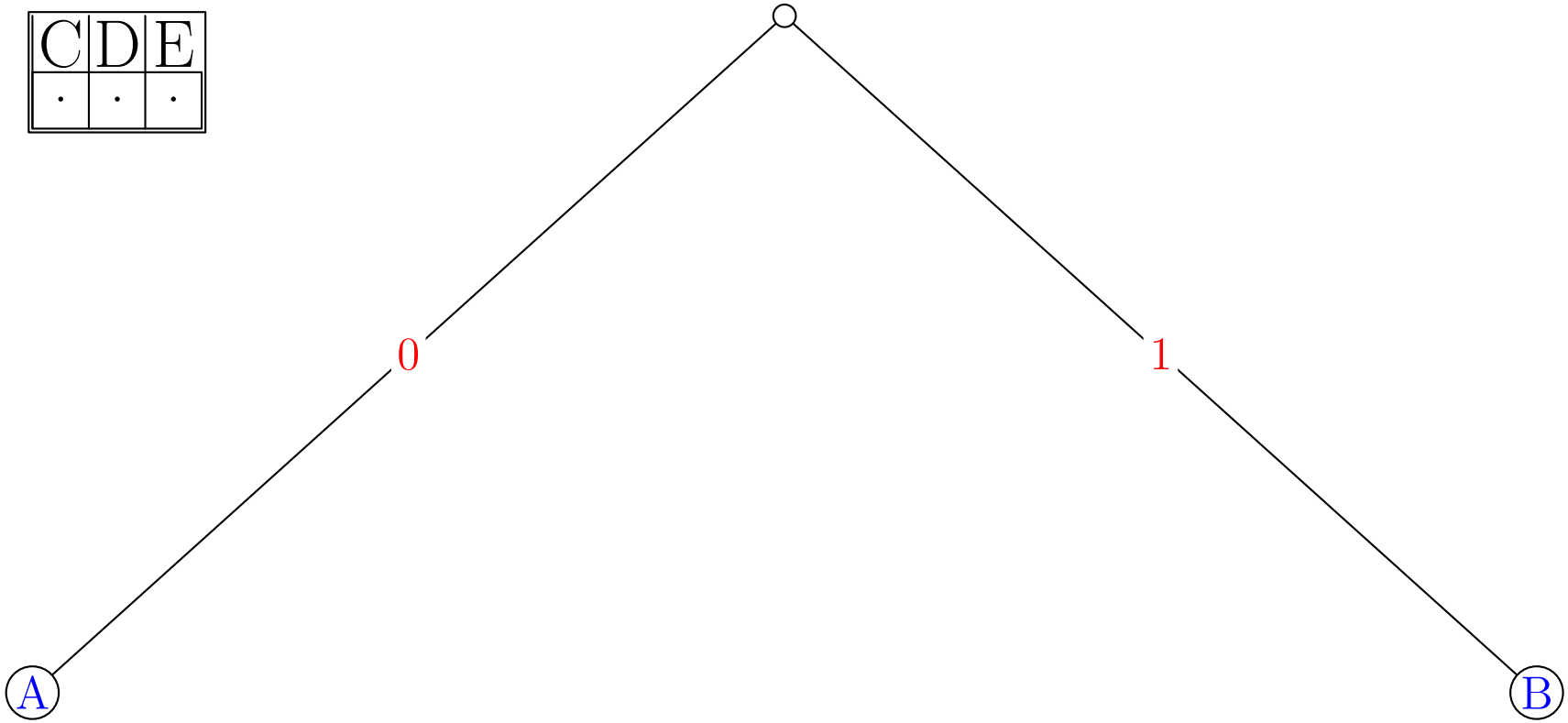
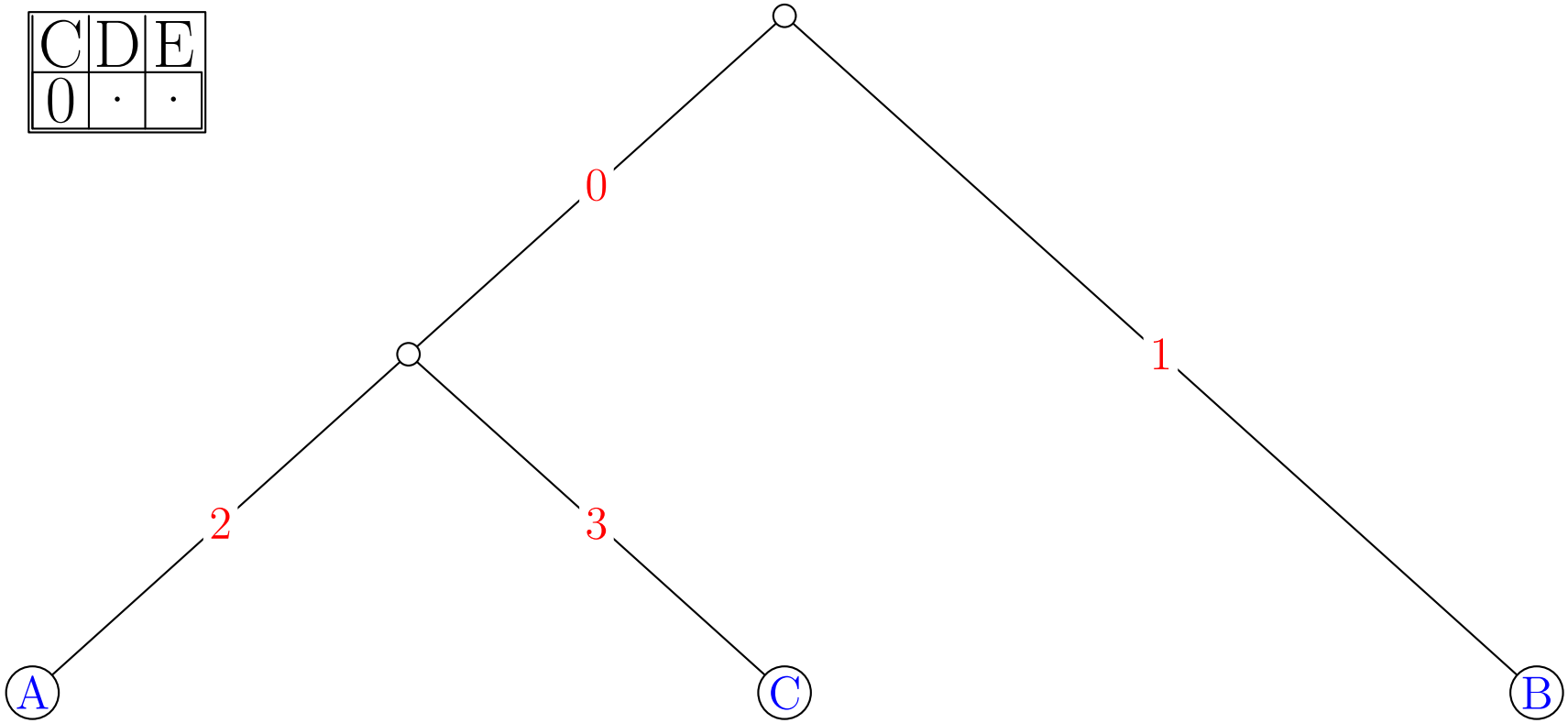
Enumerating Trees



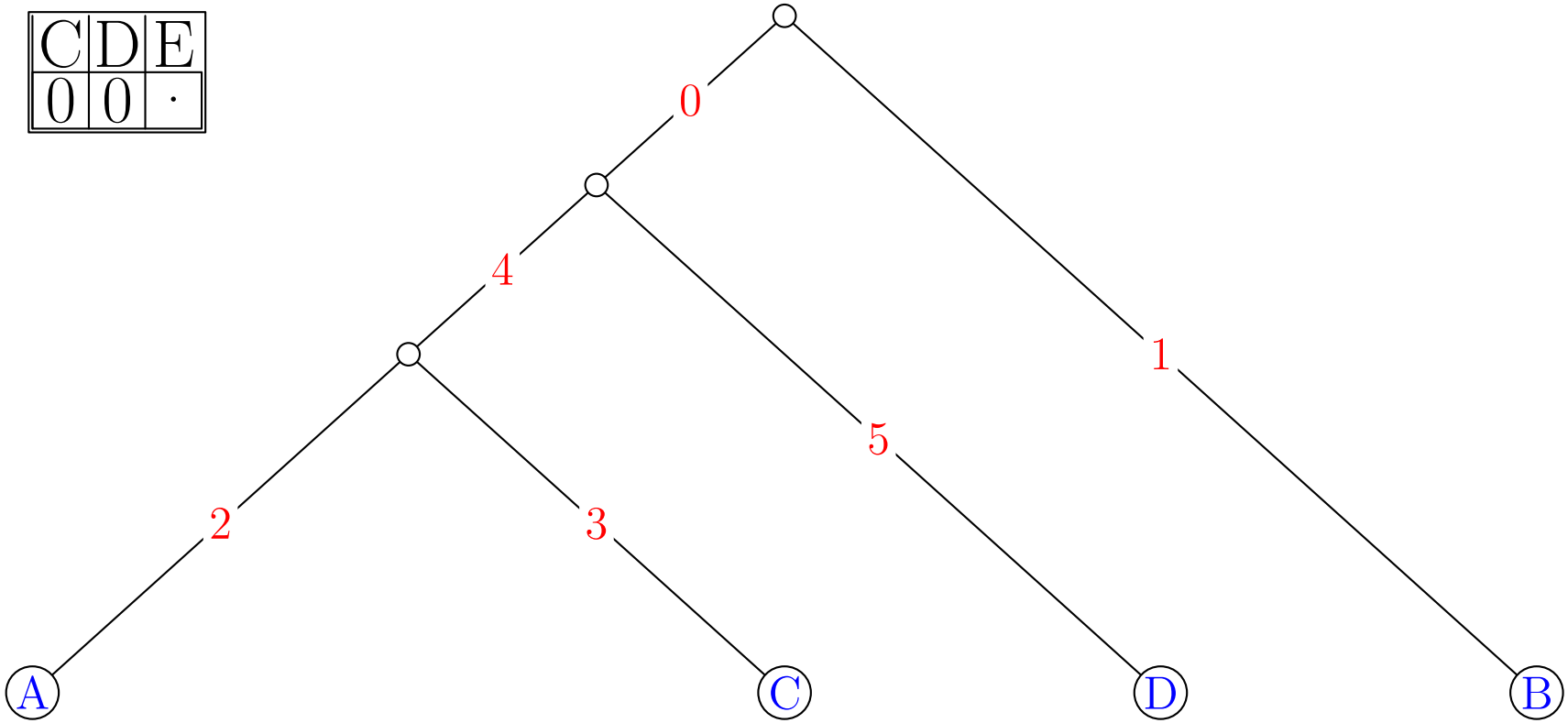
Enumerating Trees

C	D	E
0	·	·



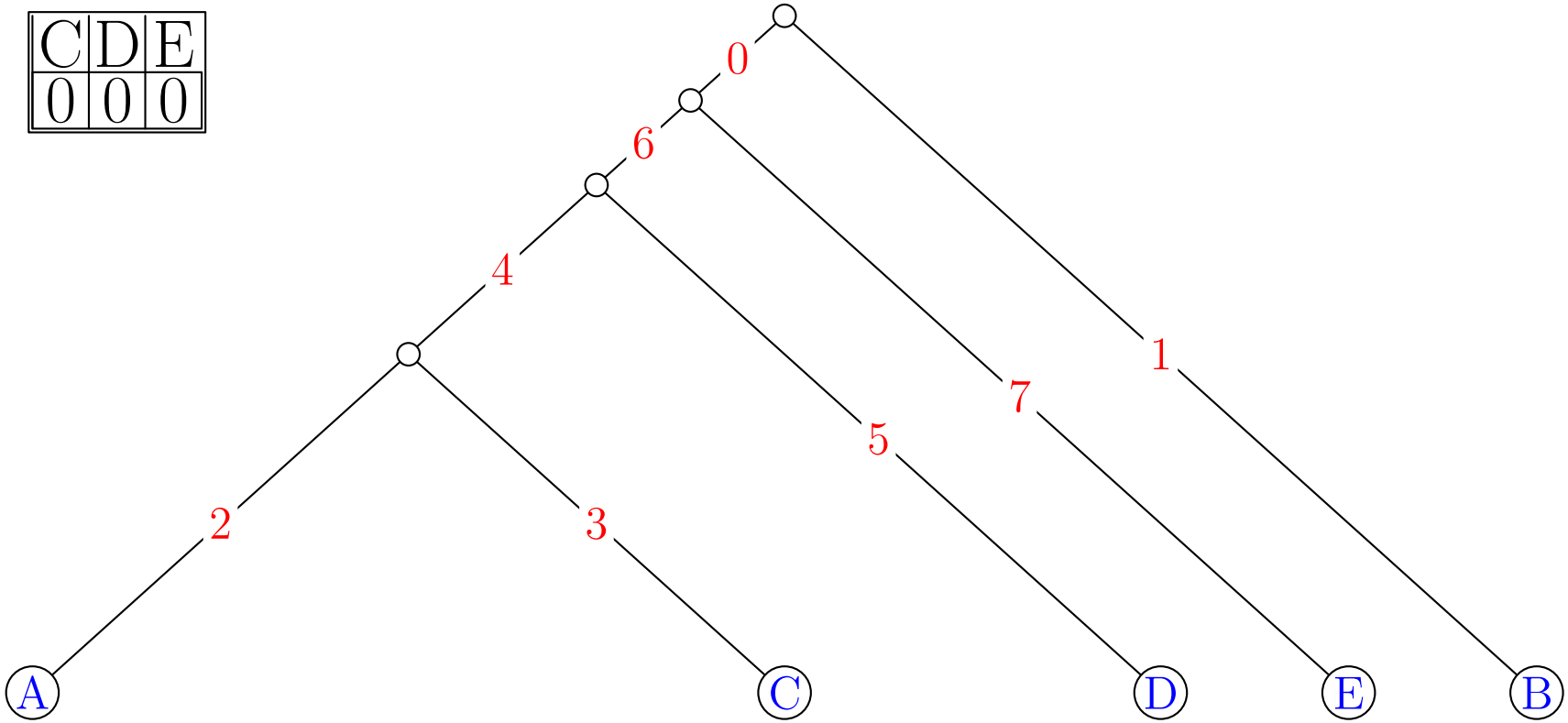
Enumerating Trees

C	D	E
0	0	·



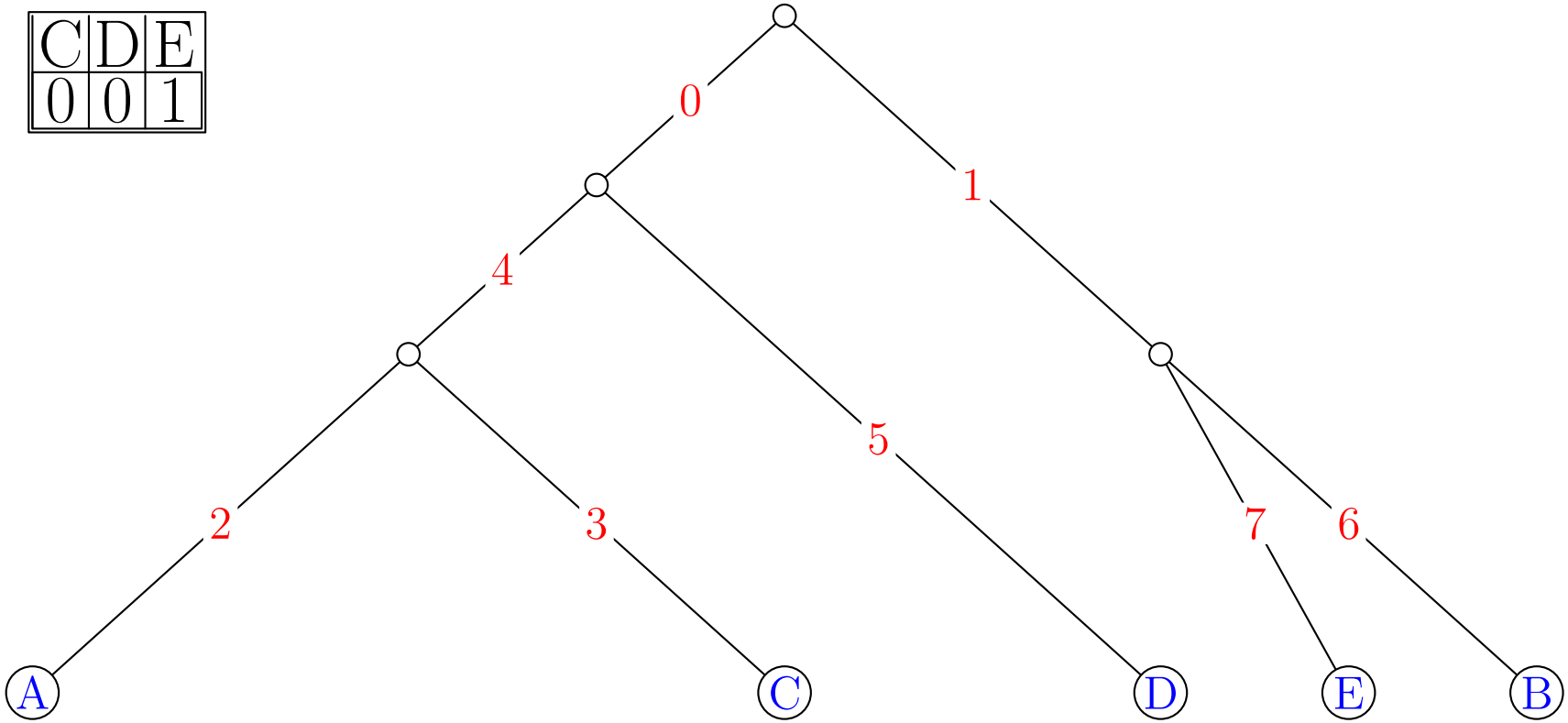
Enumerating Trees

C	D	E
0	0	0



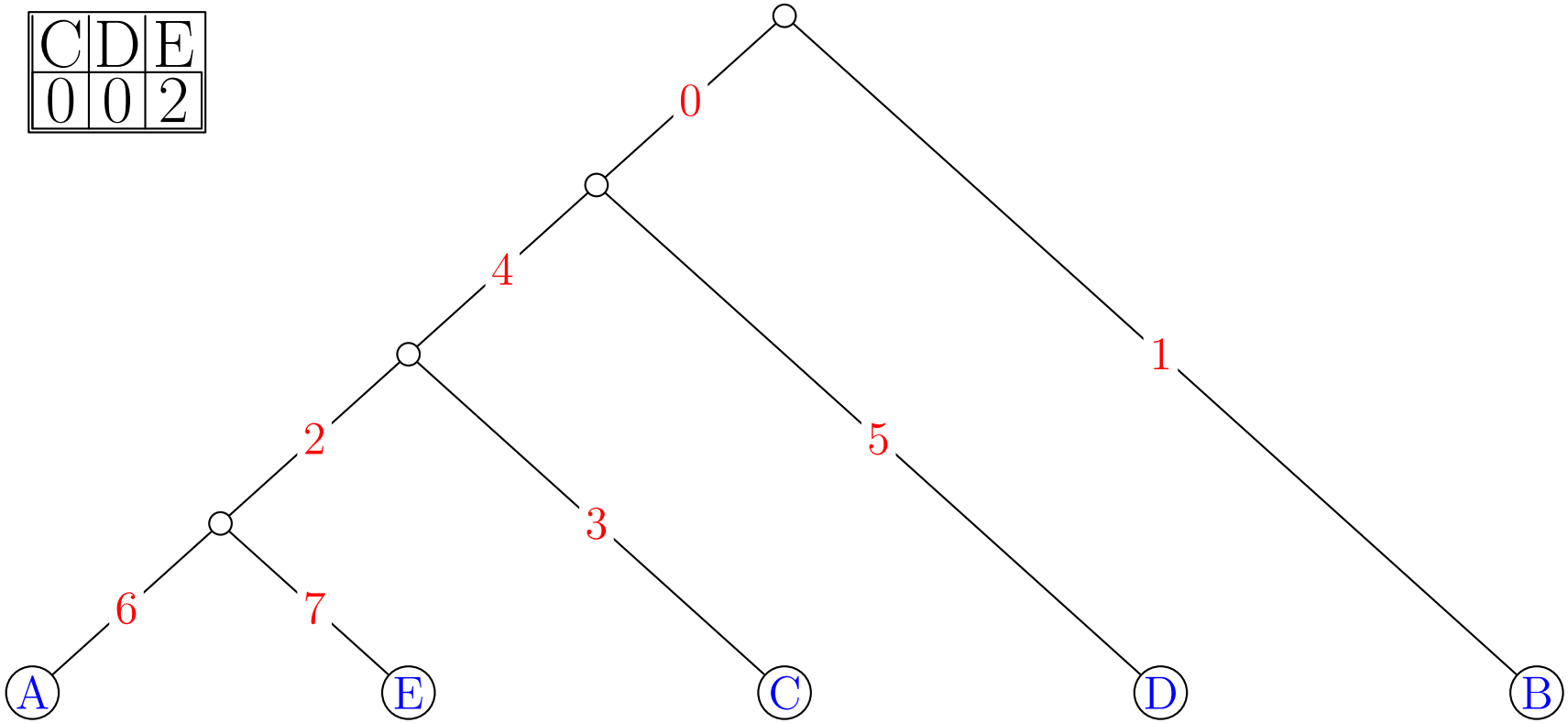
Enumerating Trees

C	D	E
0	0	1



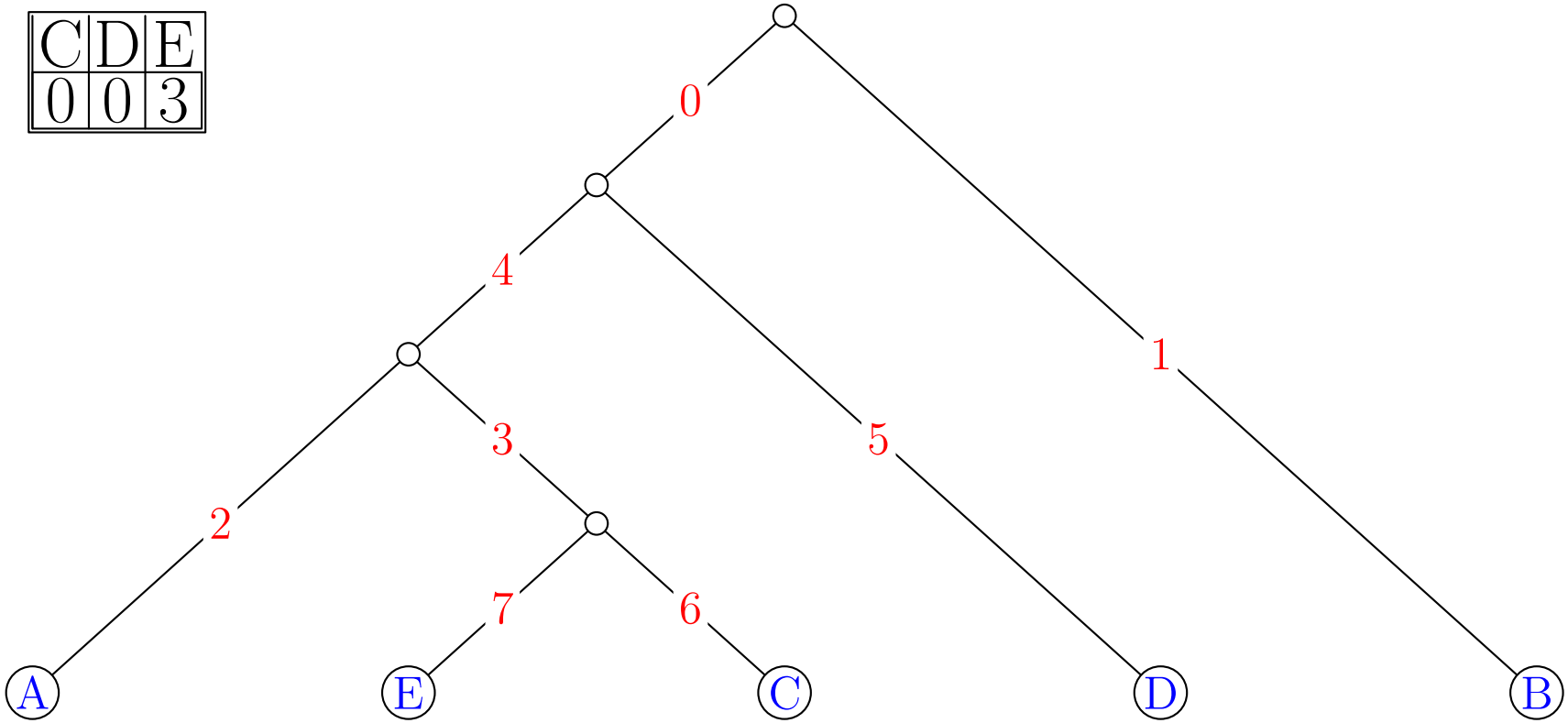
Enumerating Trees

C	D	E
0	0	2



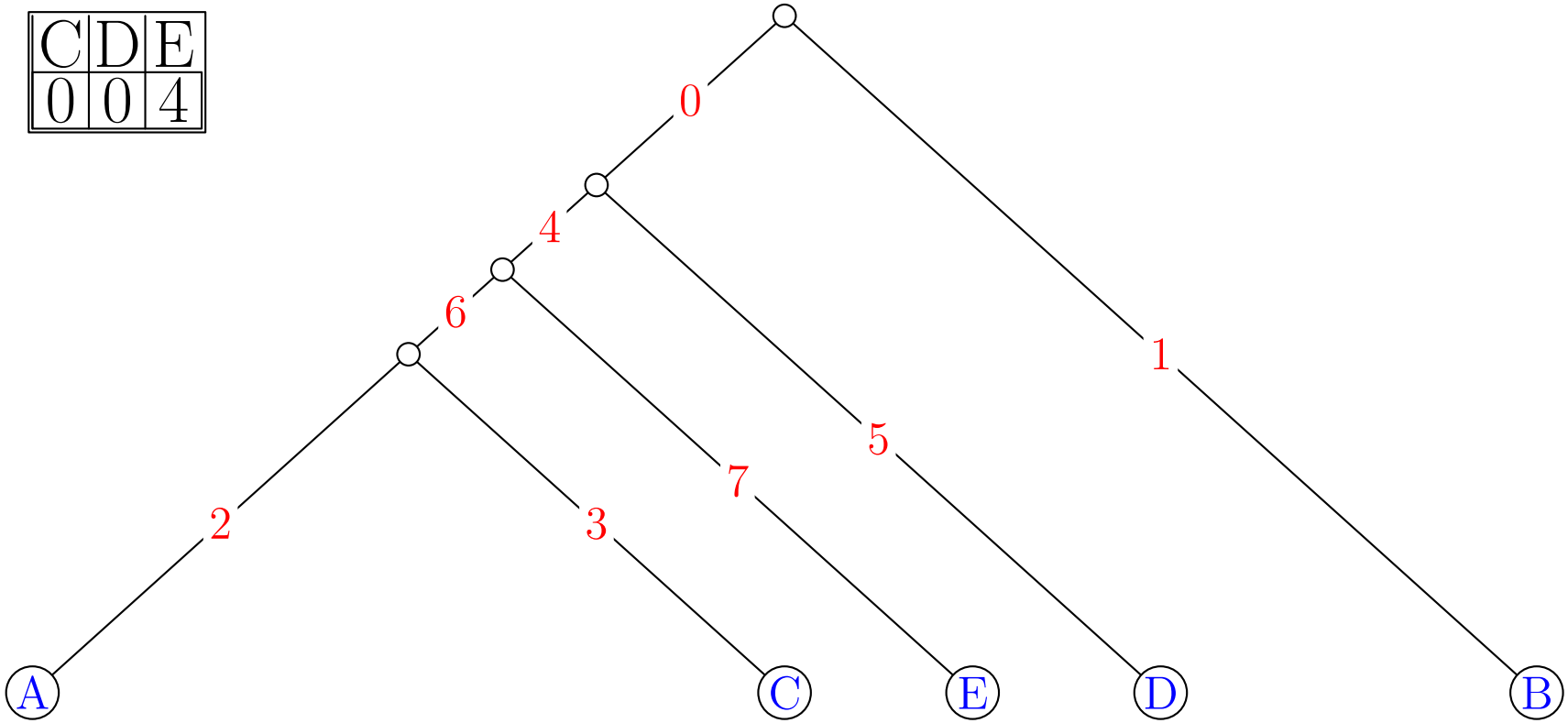
Enumerating Trees

C	D	E
0	0	3



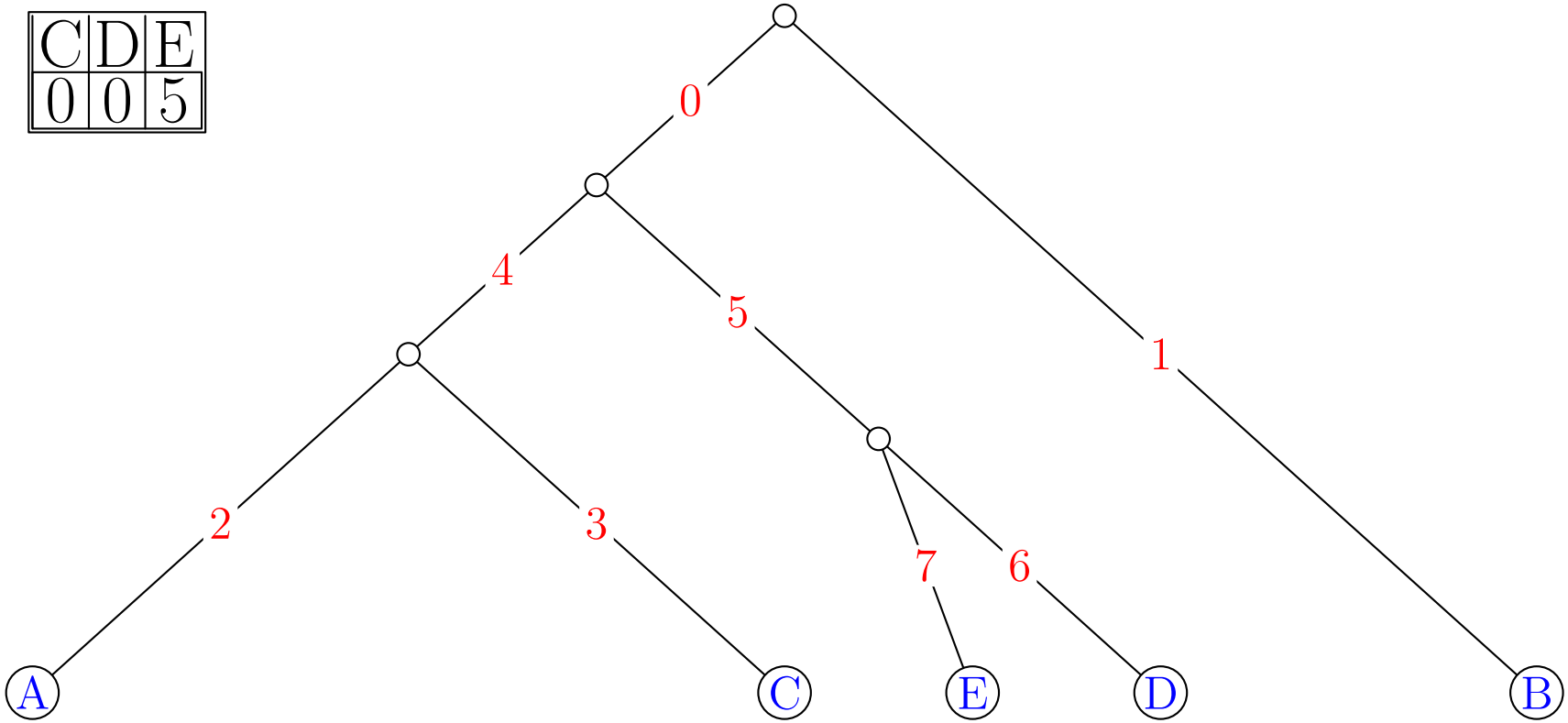
Enumerating Trees

C	D	E
0	0	4



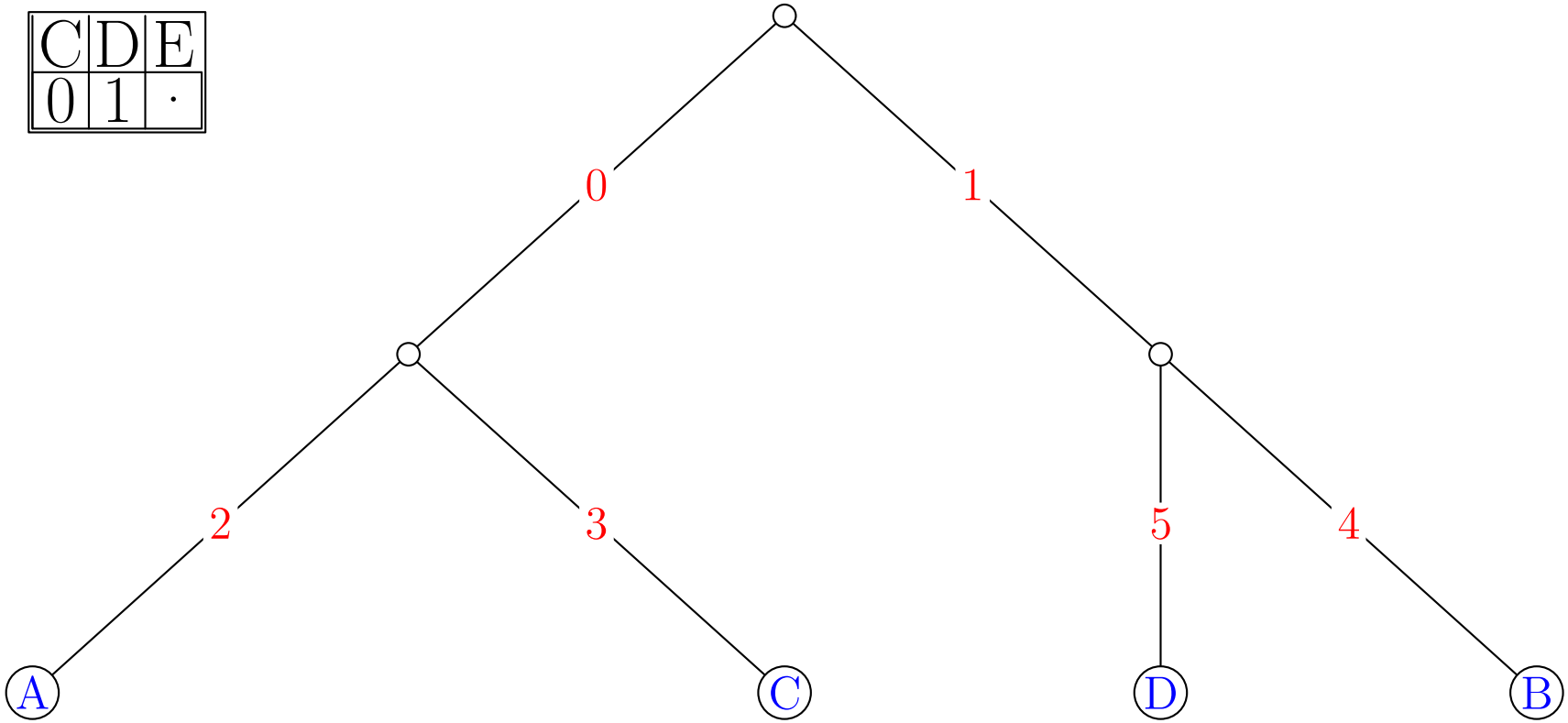
Enumerating Trees

C	D	E
0	0	5



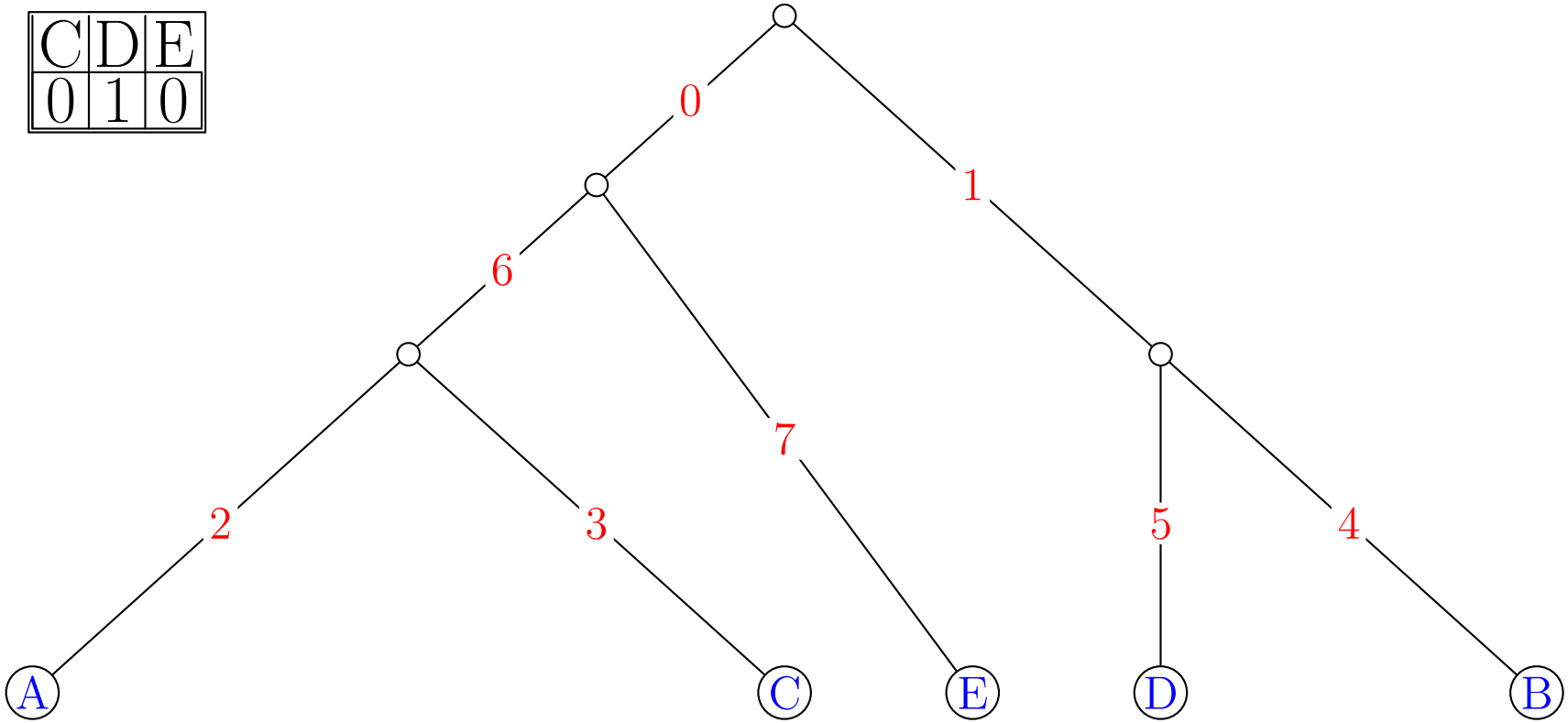
Enumerating Trees

C	D	E
0	1	·



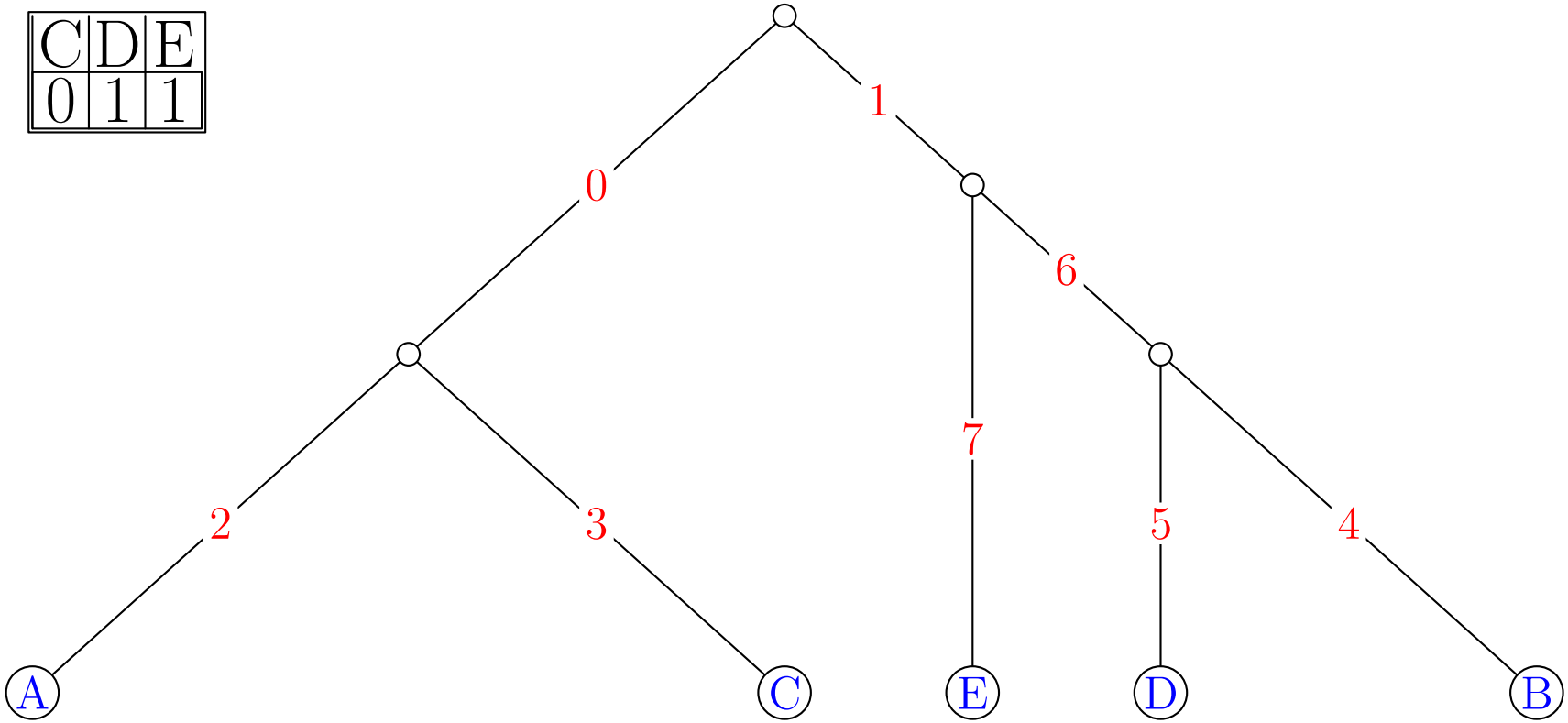
Enumerating Trees

C	D	E
0	1	0



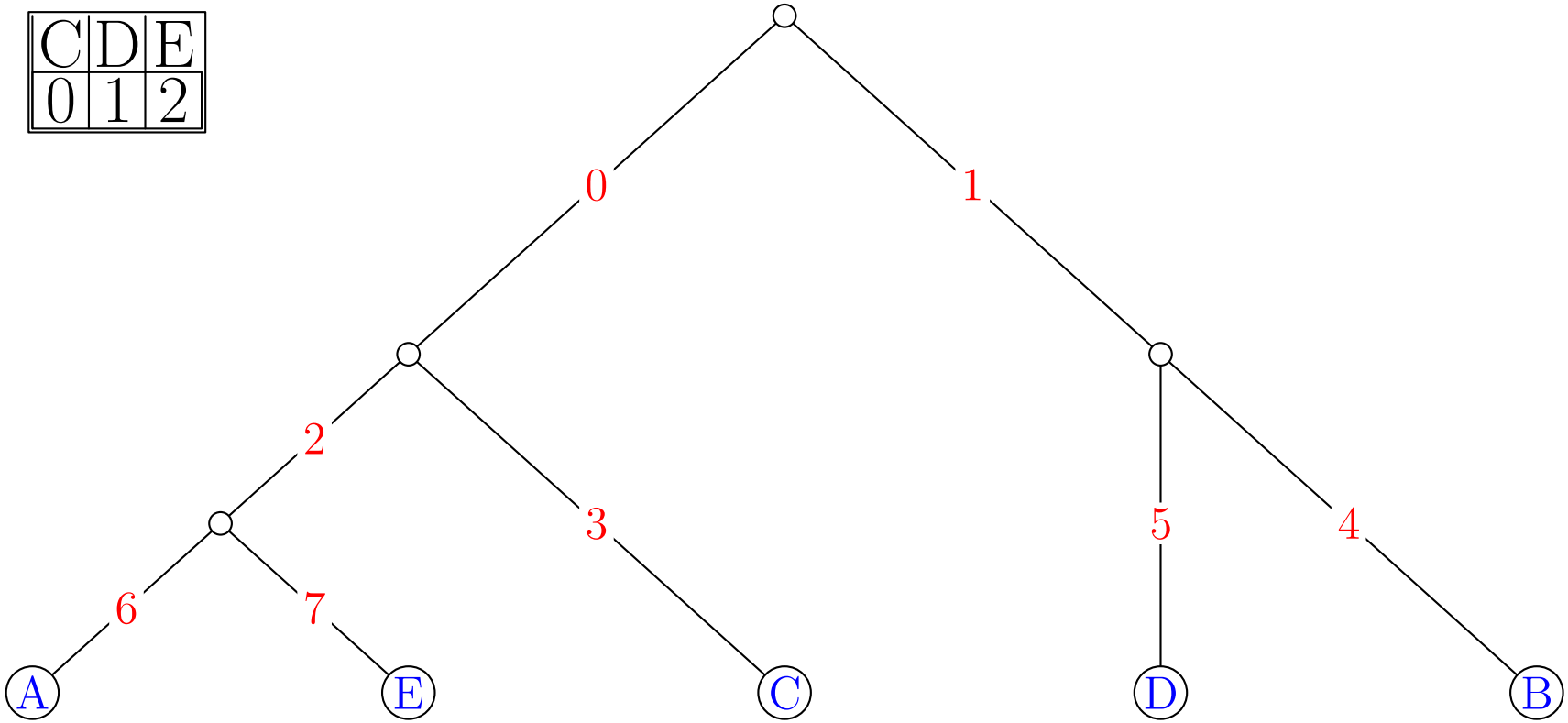
Enumerating Trees

C	D	E
0	1	1



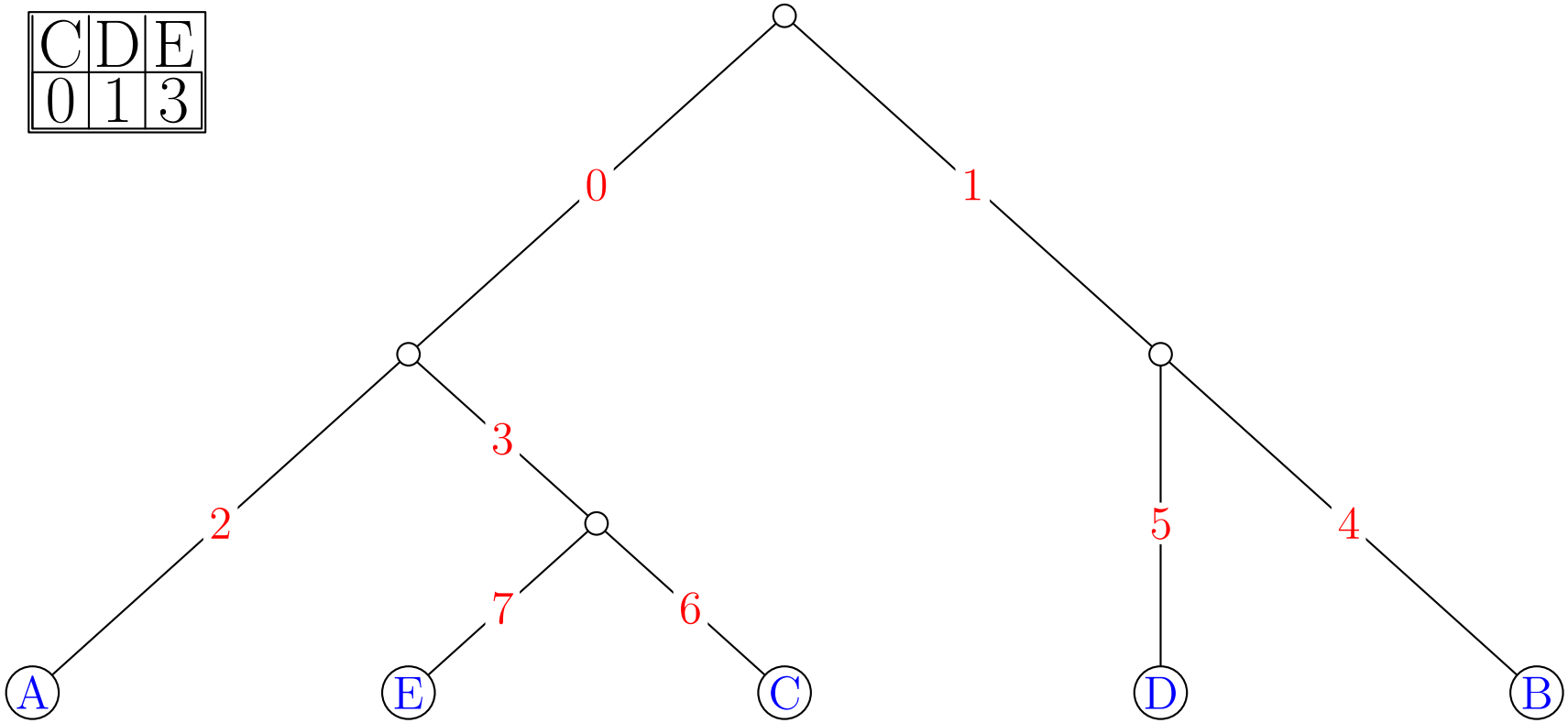
Enumerating Trees

C	D	E
0	1	2



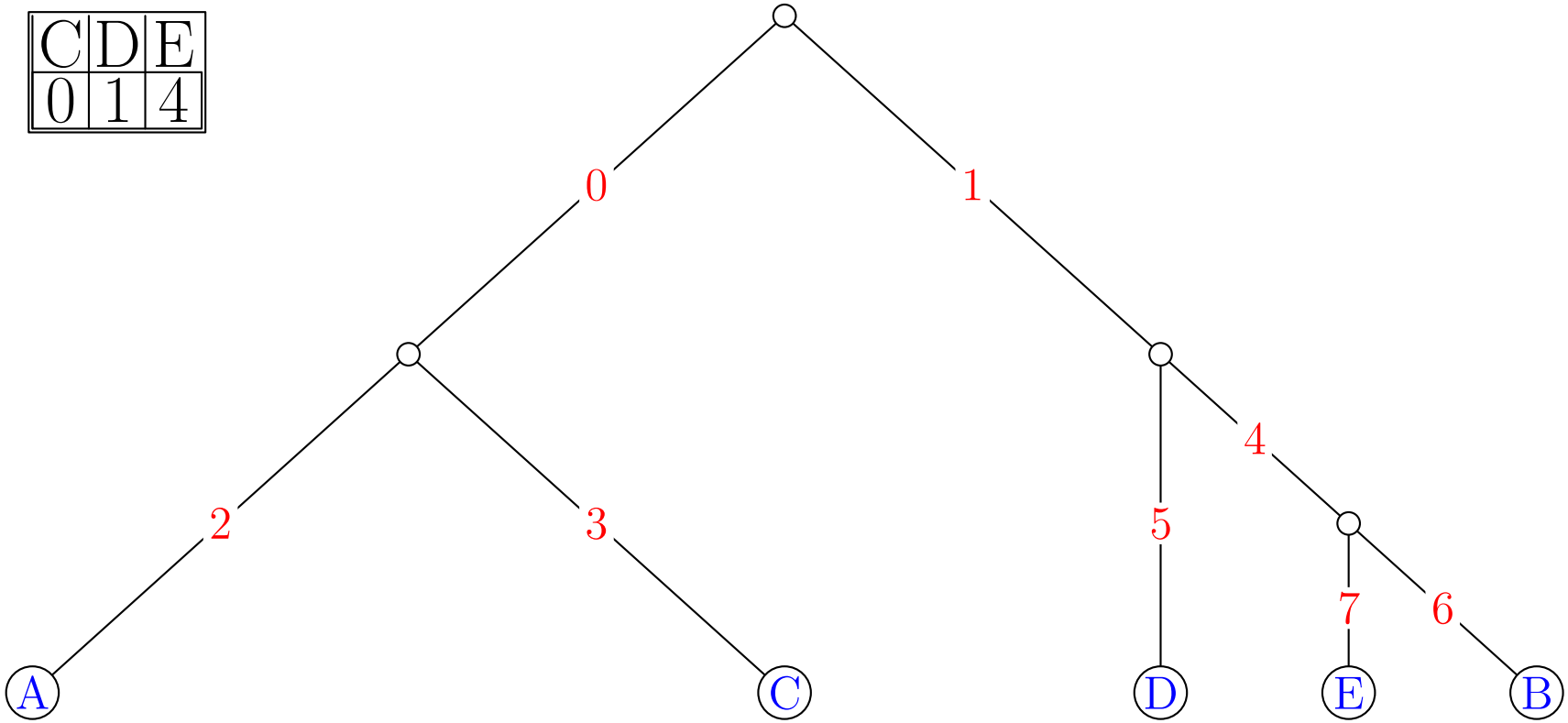
Enumerating Trees

C	D	E
0	1	3



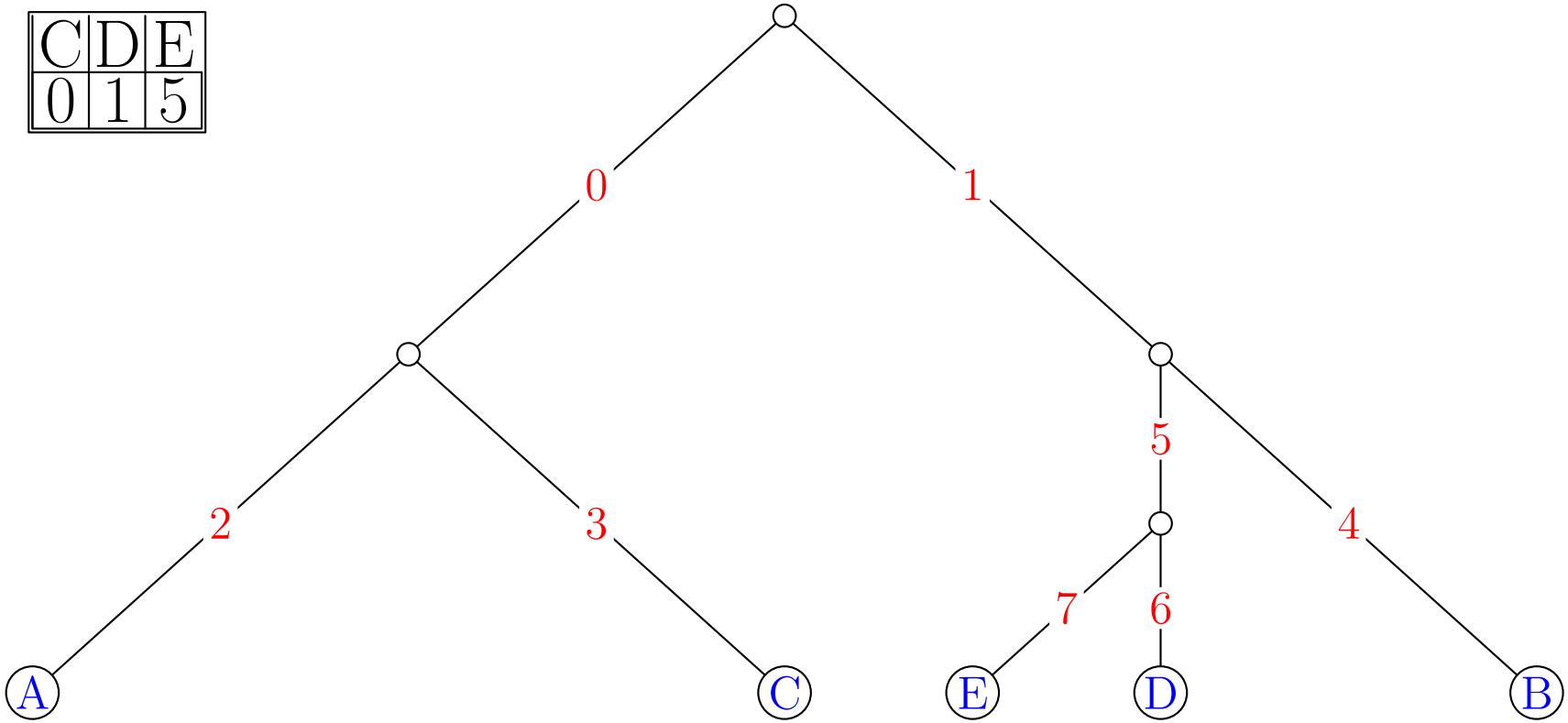
Enumerating Trees

C	D	E
0	1	4



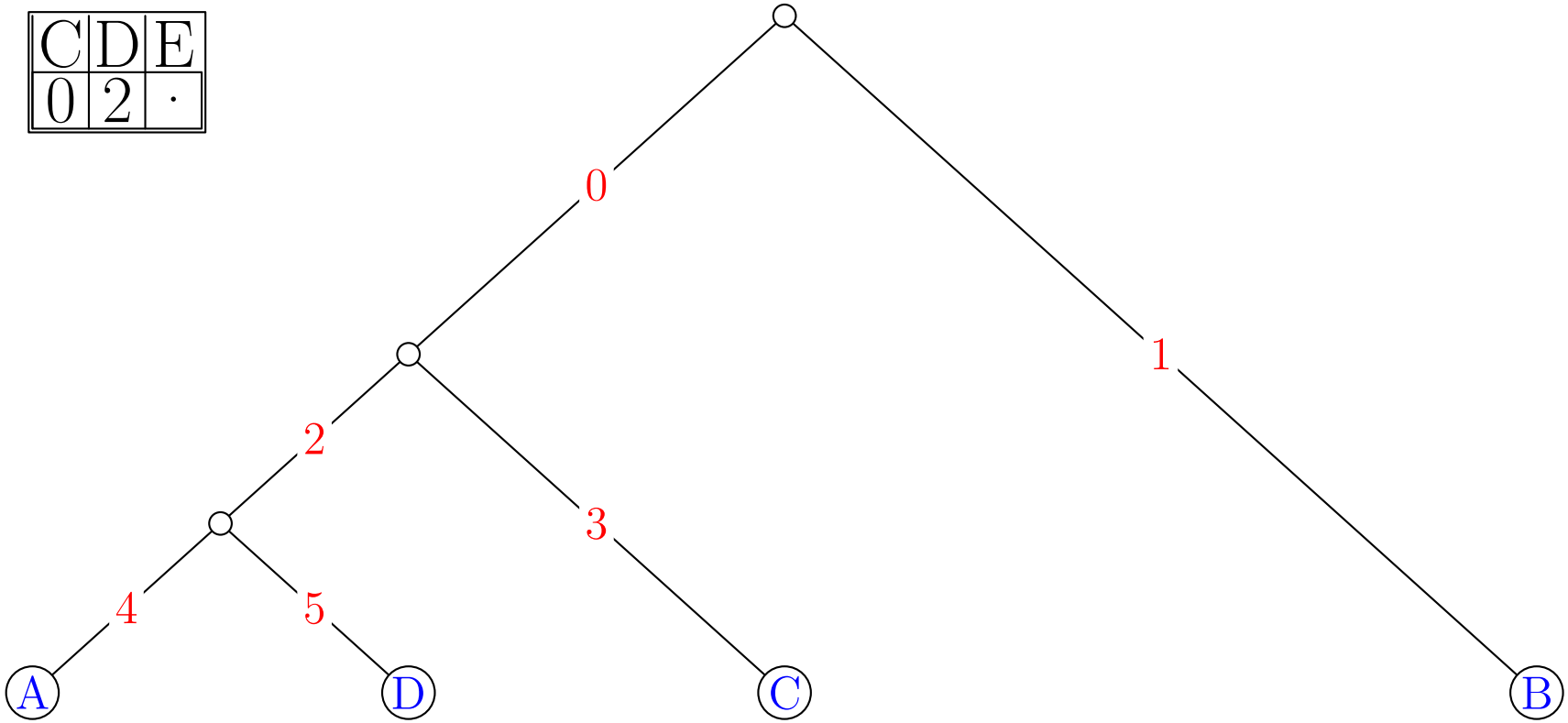
Enumerating Trees

C	D	E
0	1	5



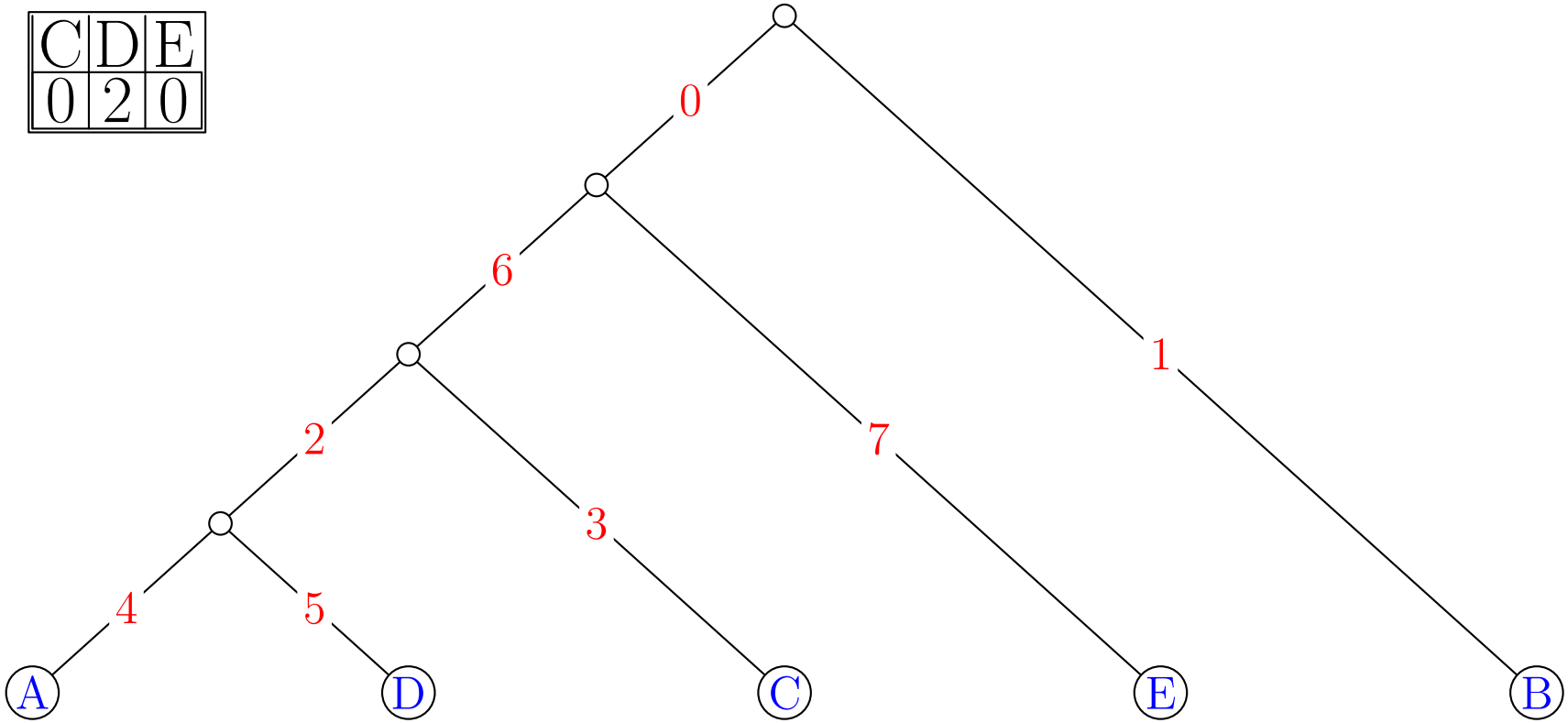
Enumerating Trees

C	D	E
0	2	·



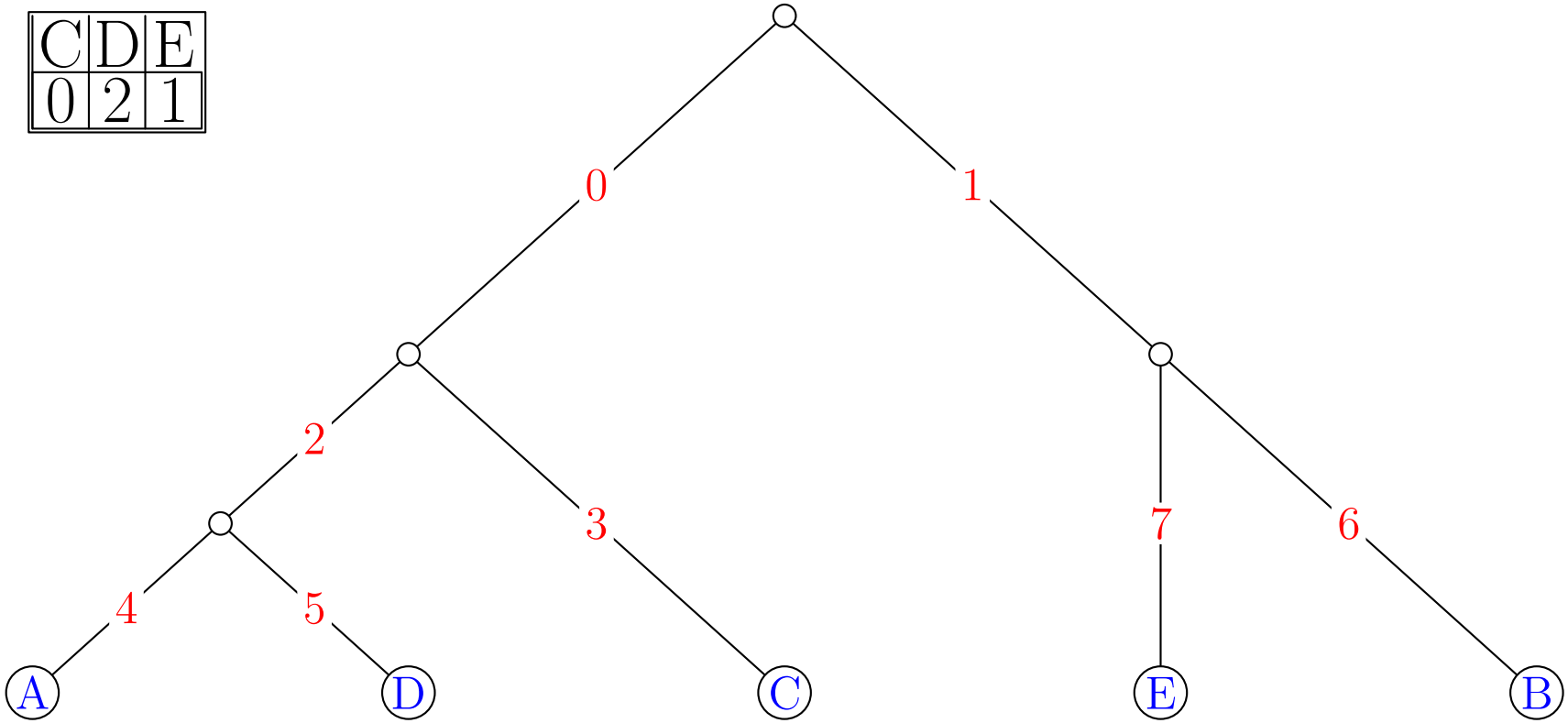
Enumerating Trees

C	D	E
0	2	0



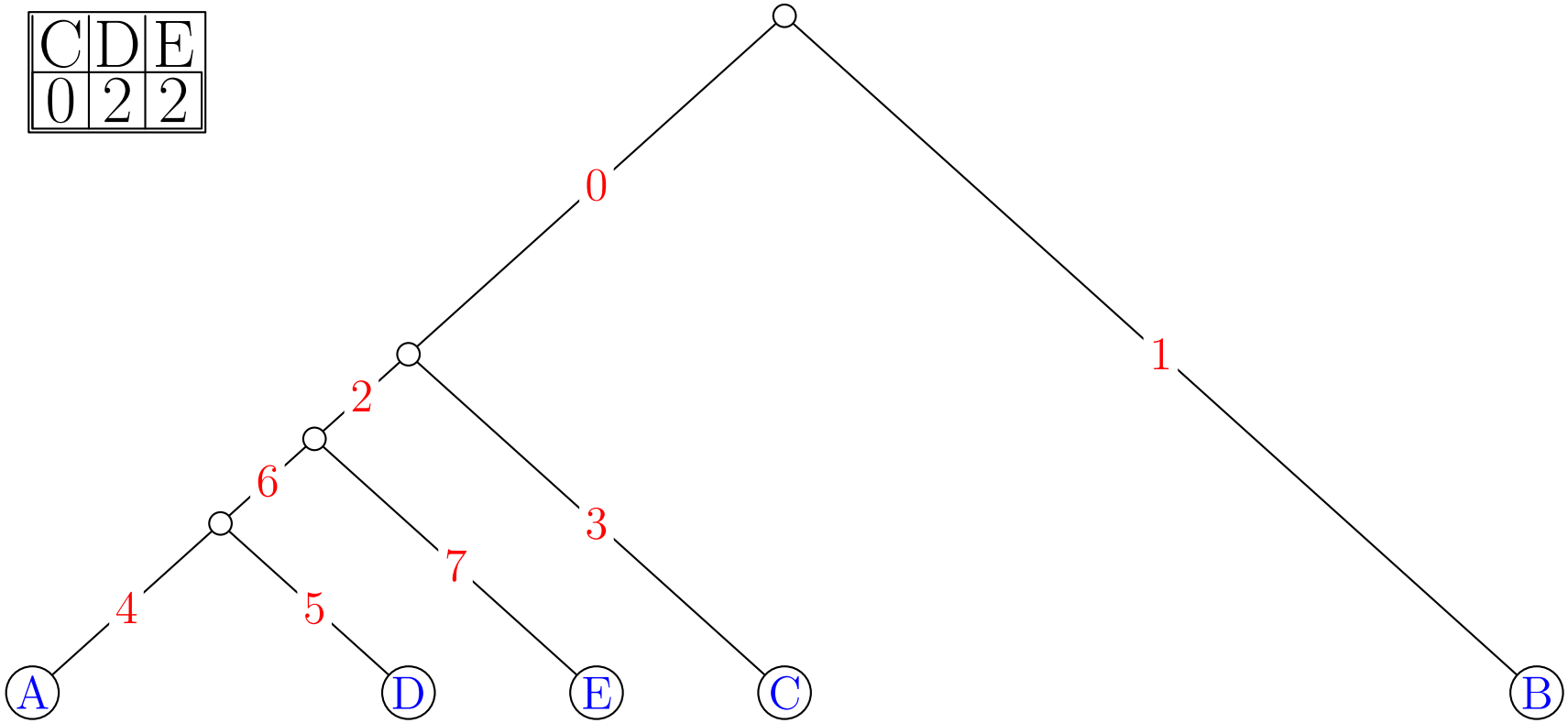
Enumerating Trees

C	D	E
0	2	1



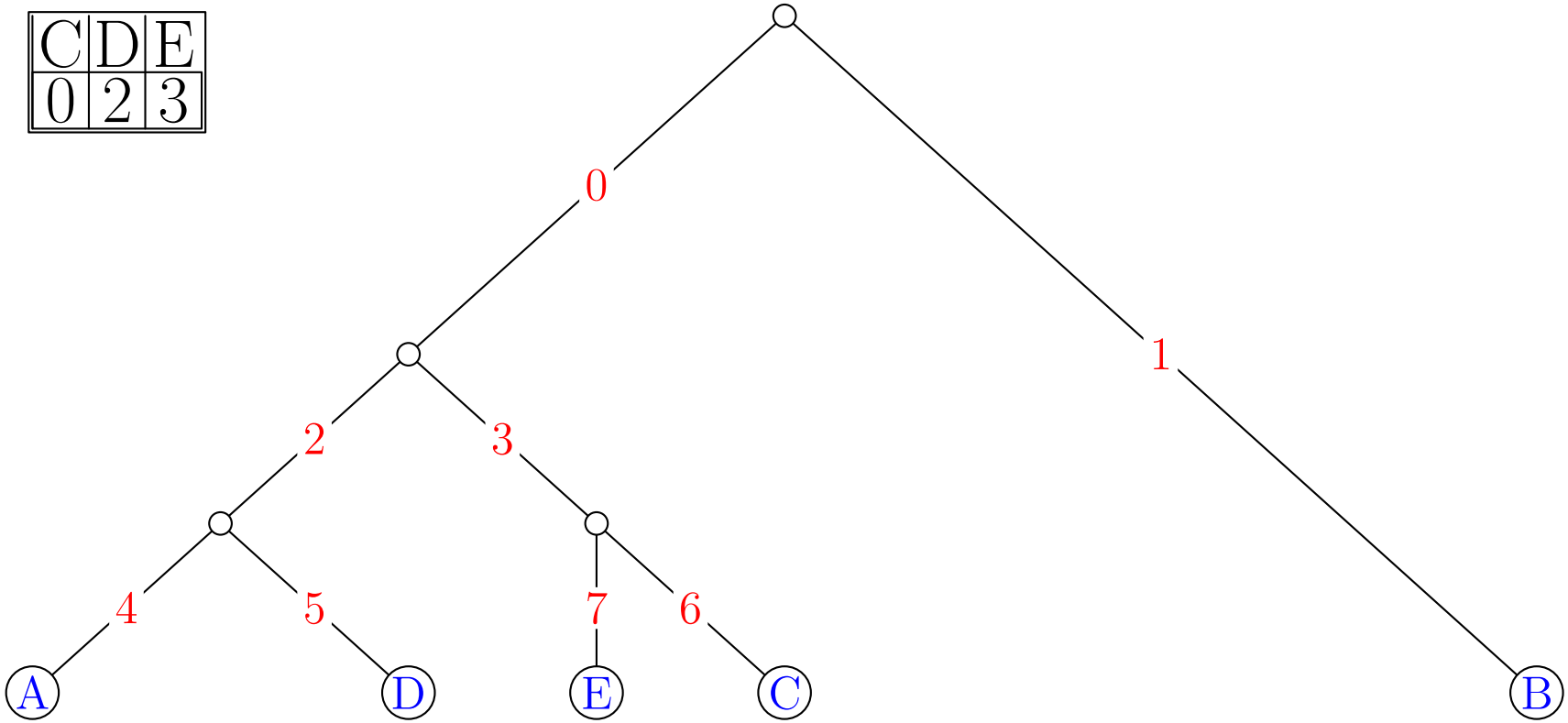
Enumerating Trees

C	D	E
0	2	2



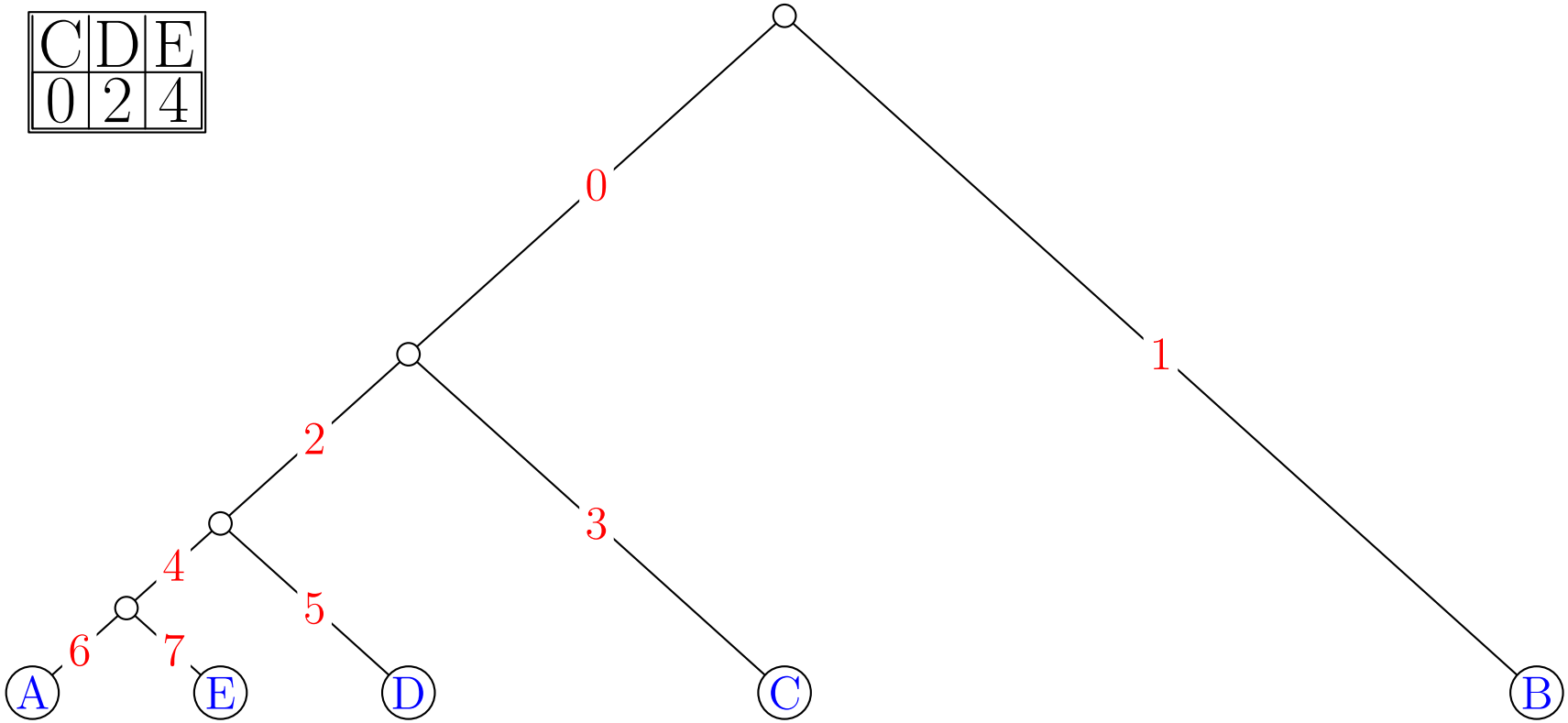
Enumerating Trees

C	D	E
0	2	3



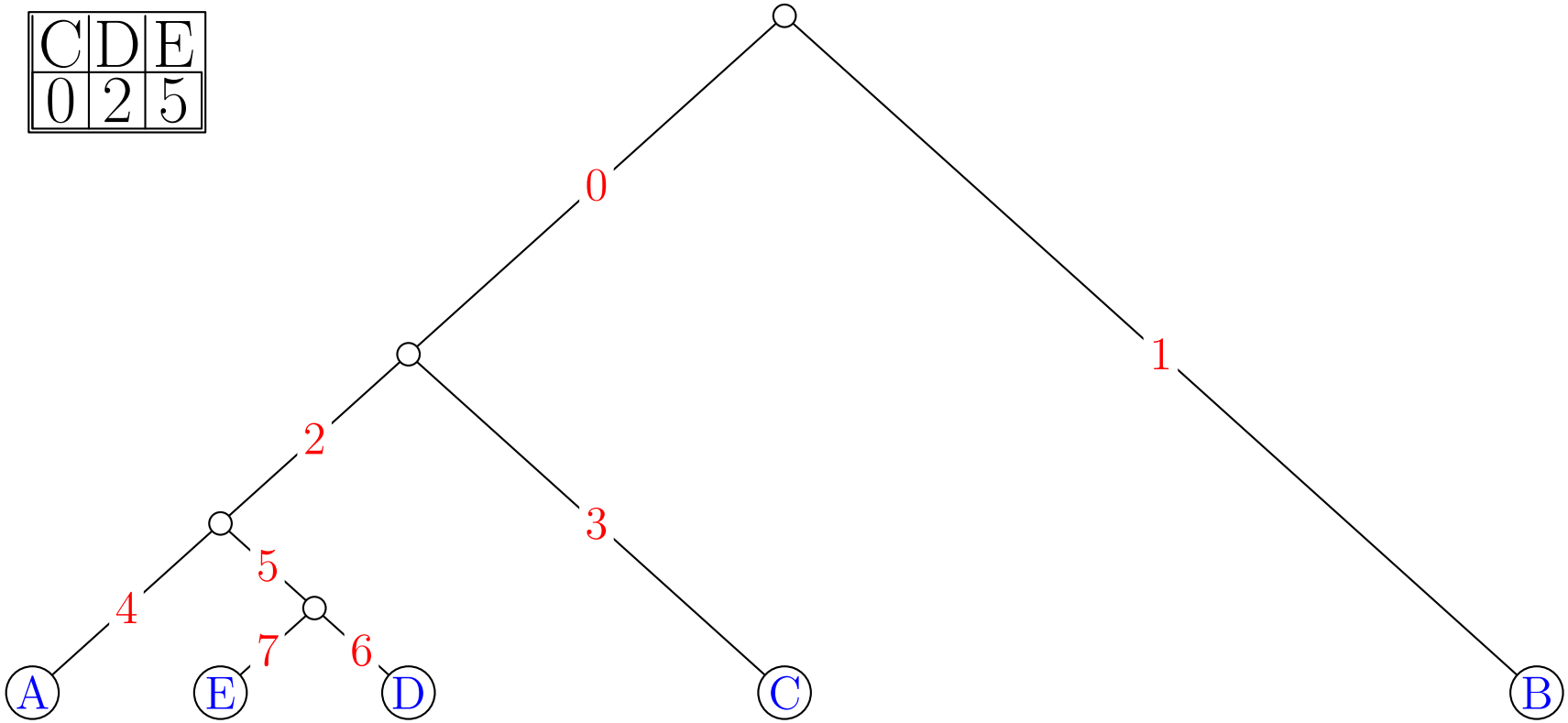
Enumerating Trees

C	D	E
0	2	4



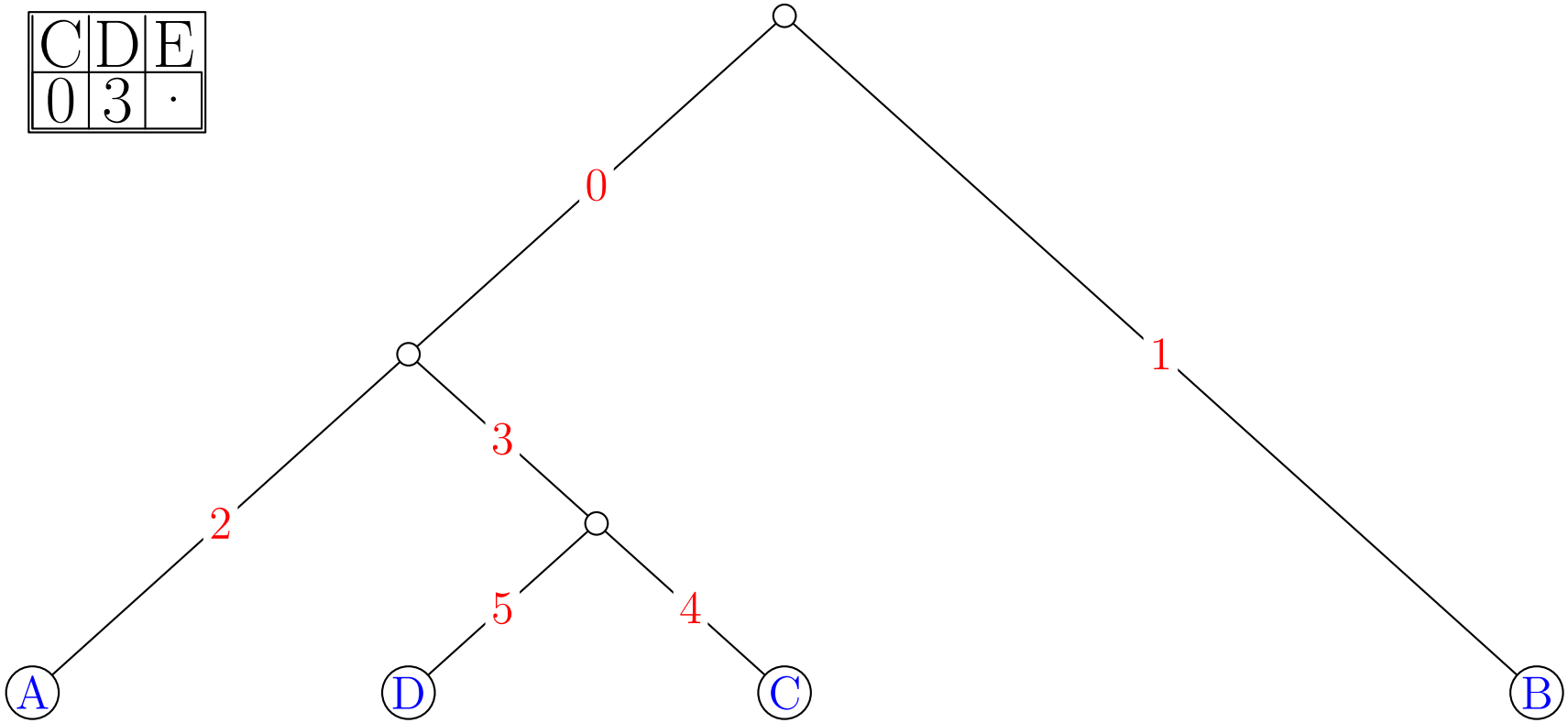
Enumerating Trees

C	D	E
0	2	5



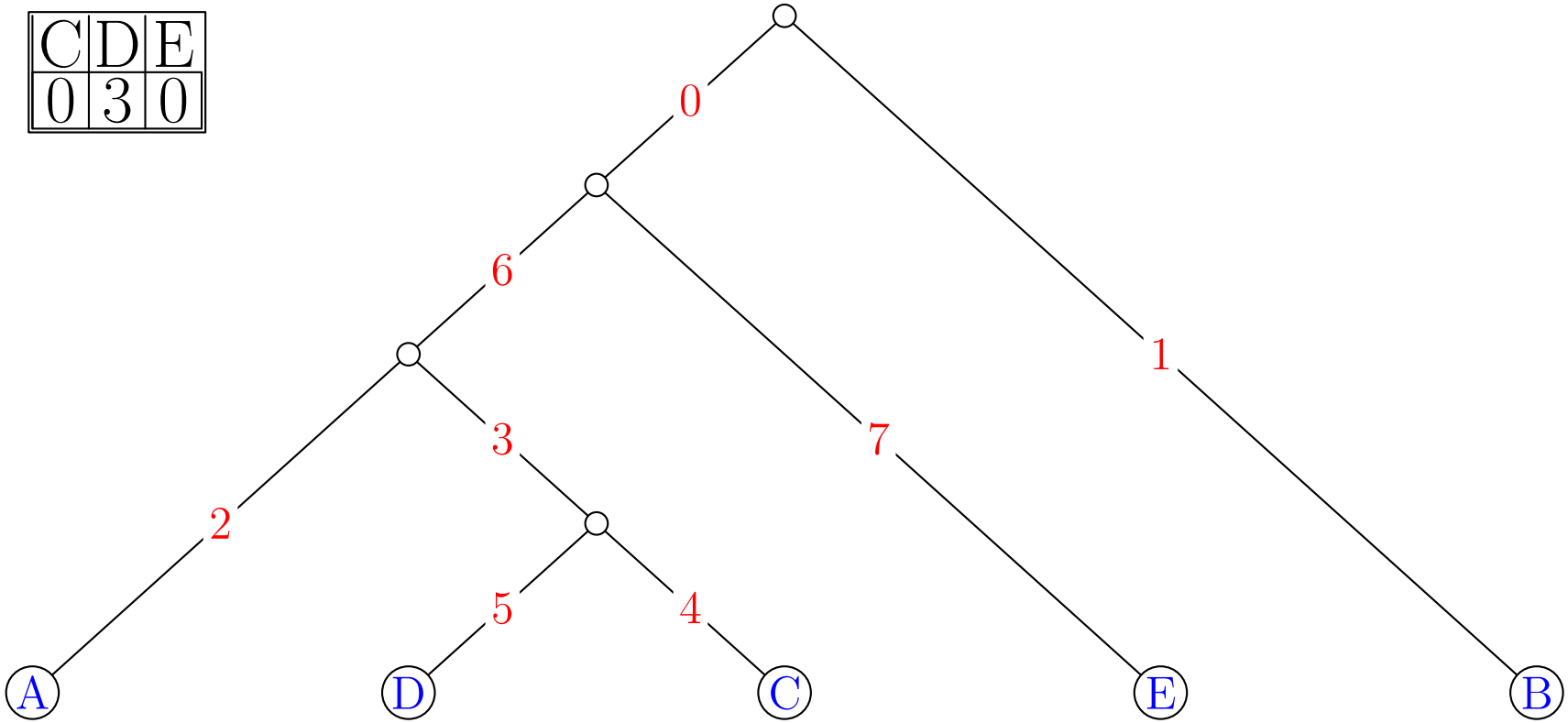
Enumerating Trees

C	D	E
0	3	·



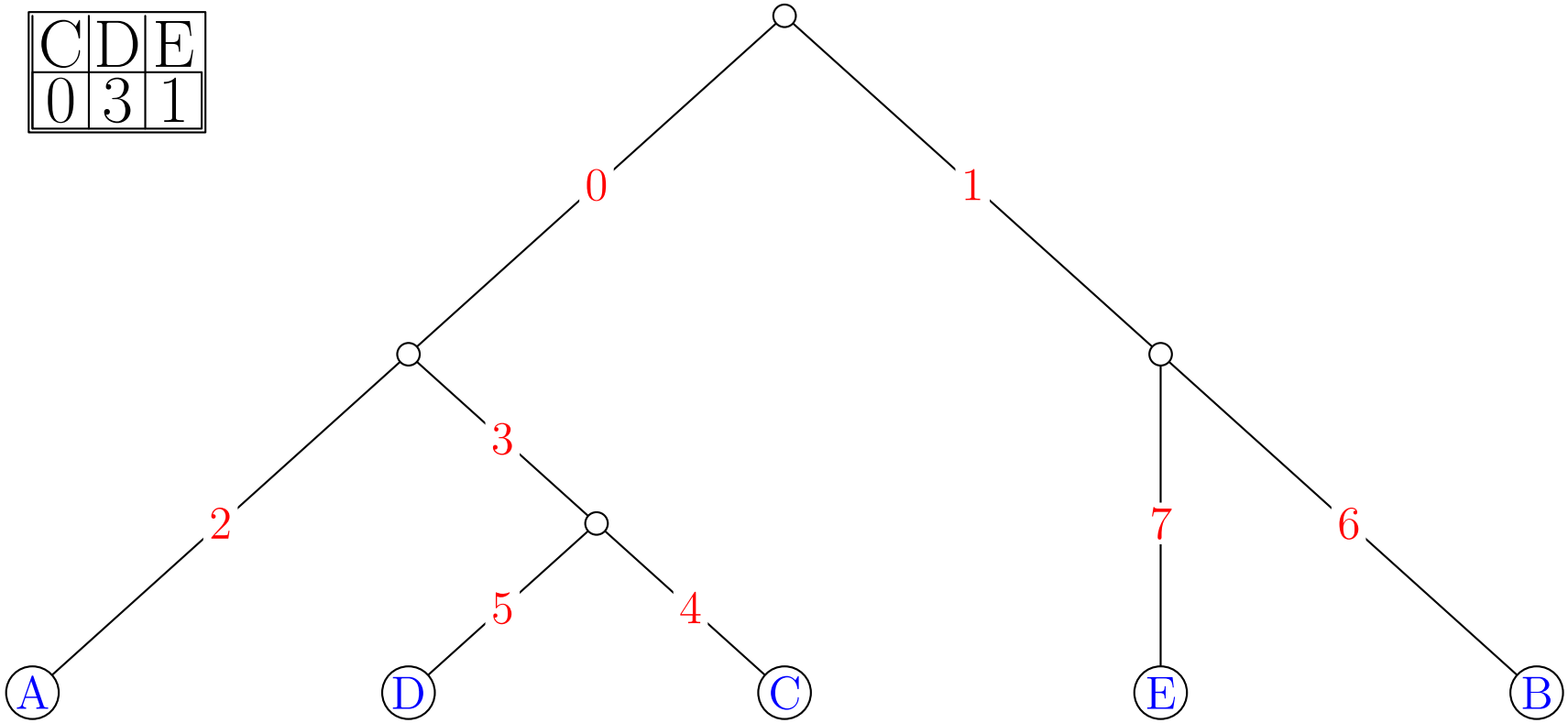
Enumerating Trees

C	D	E
0	3	0



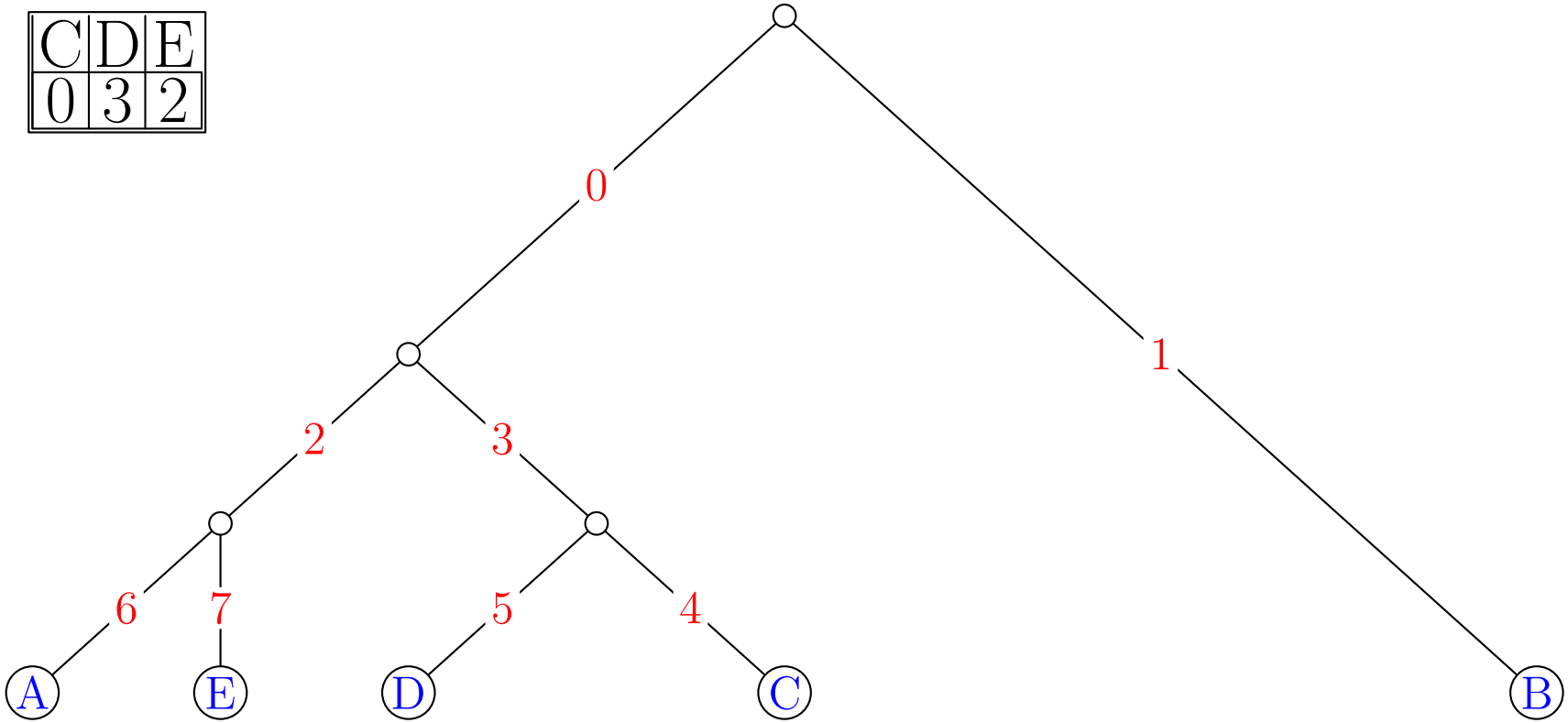
Enumerating Trees

C	D	E
0	3	1



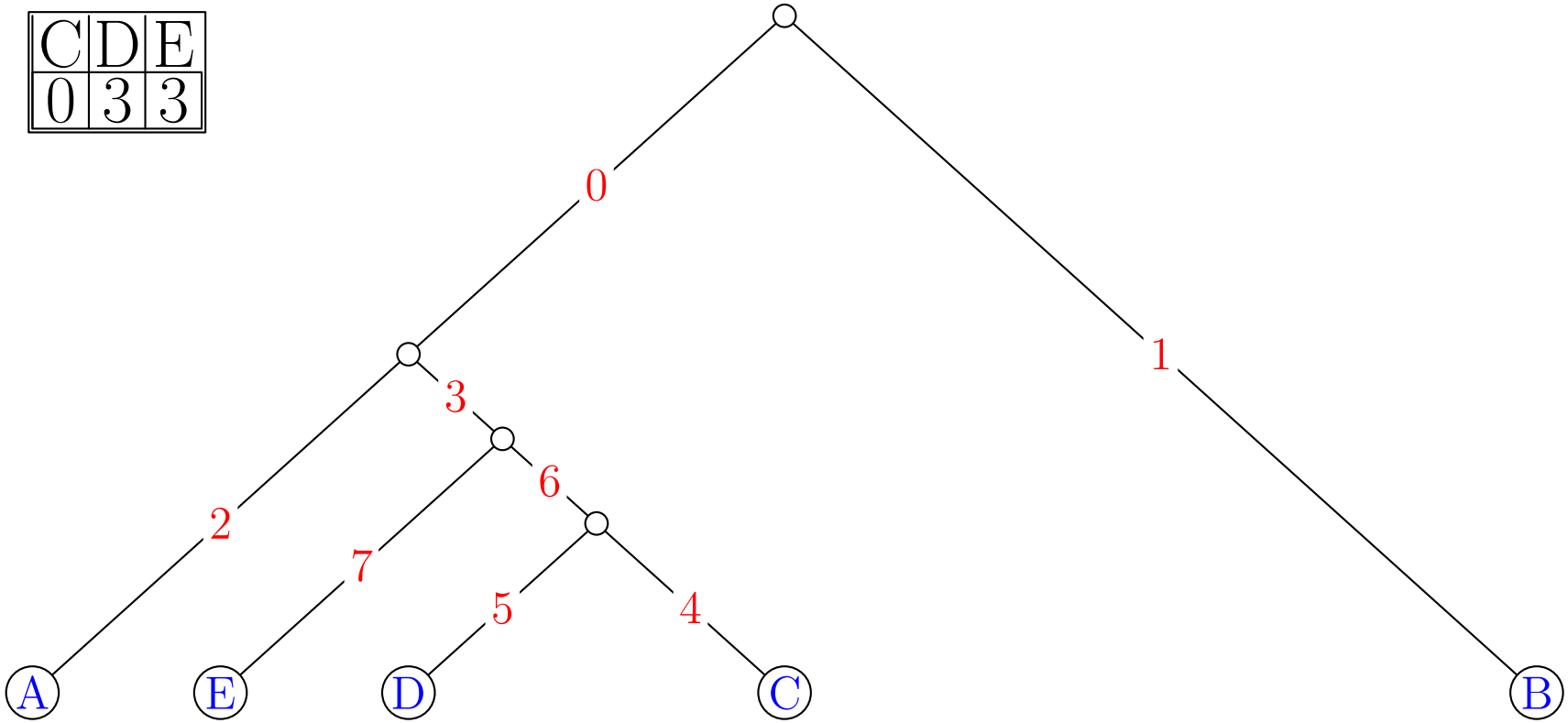
Enumerating Trees

C	D	E
0	3	2



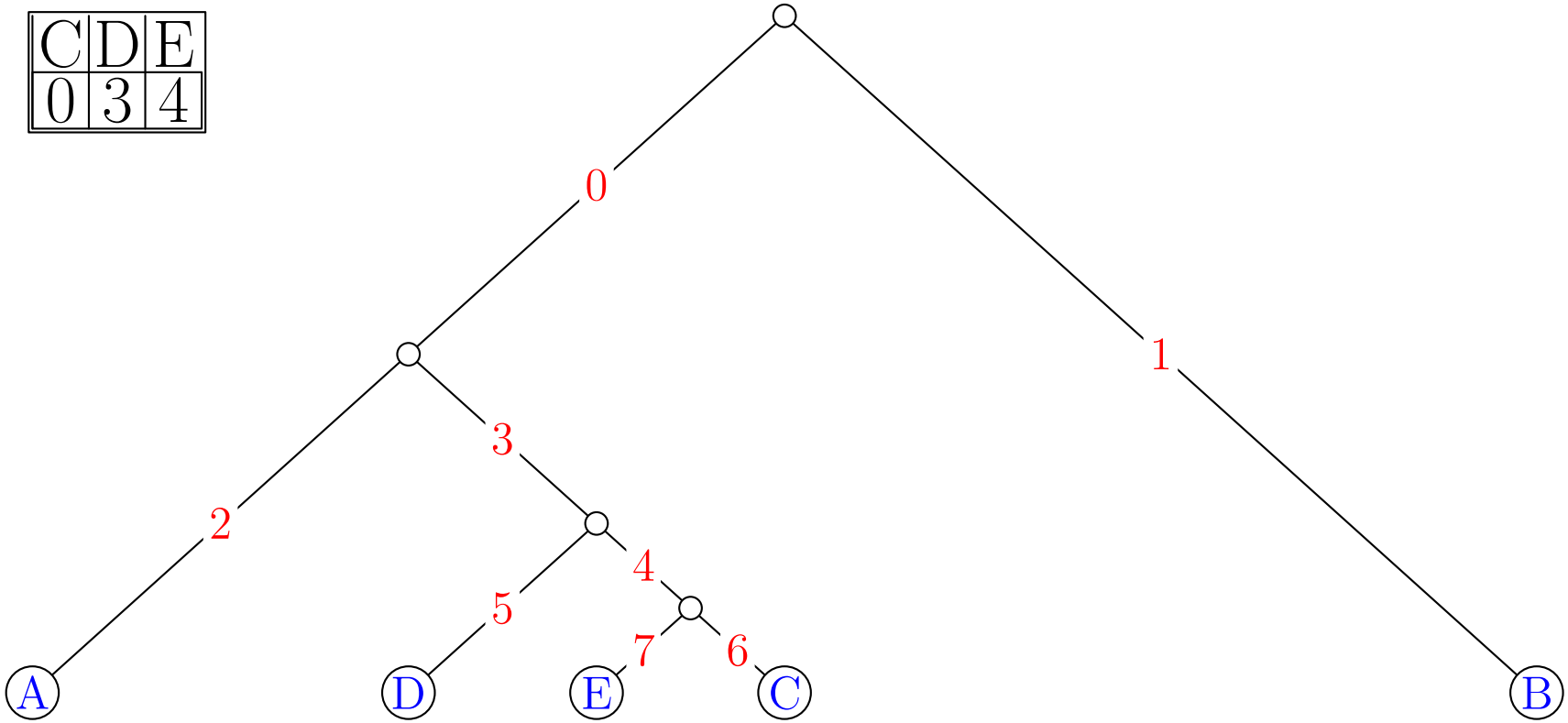
Enumerating Trees

C	D	E
0	3	3



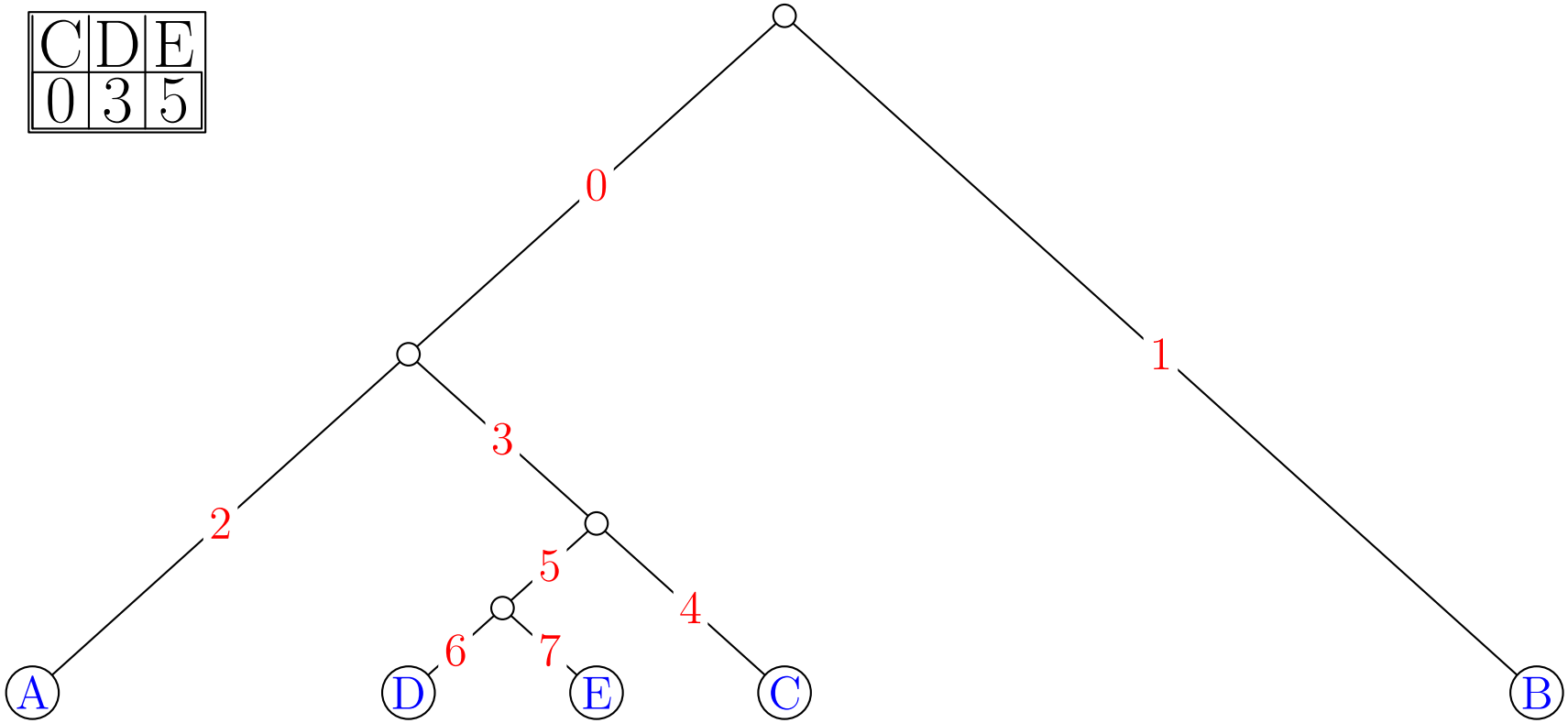
Enumerating Trees

C	D	E
0	3	4



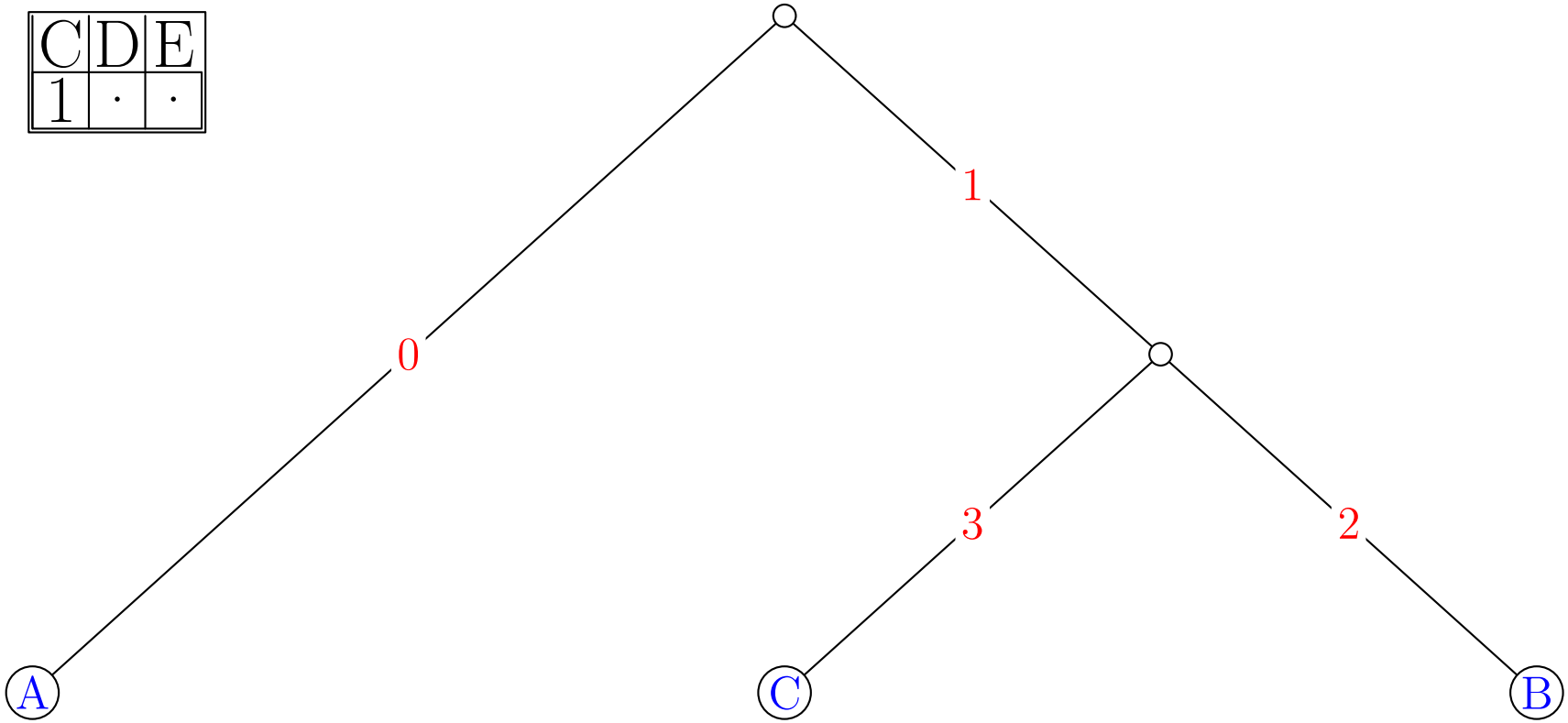
Enumerating Trees

C	D	E
0	3	5



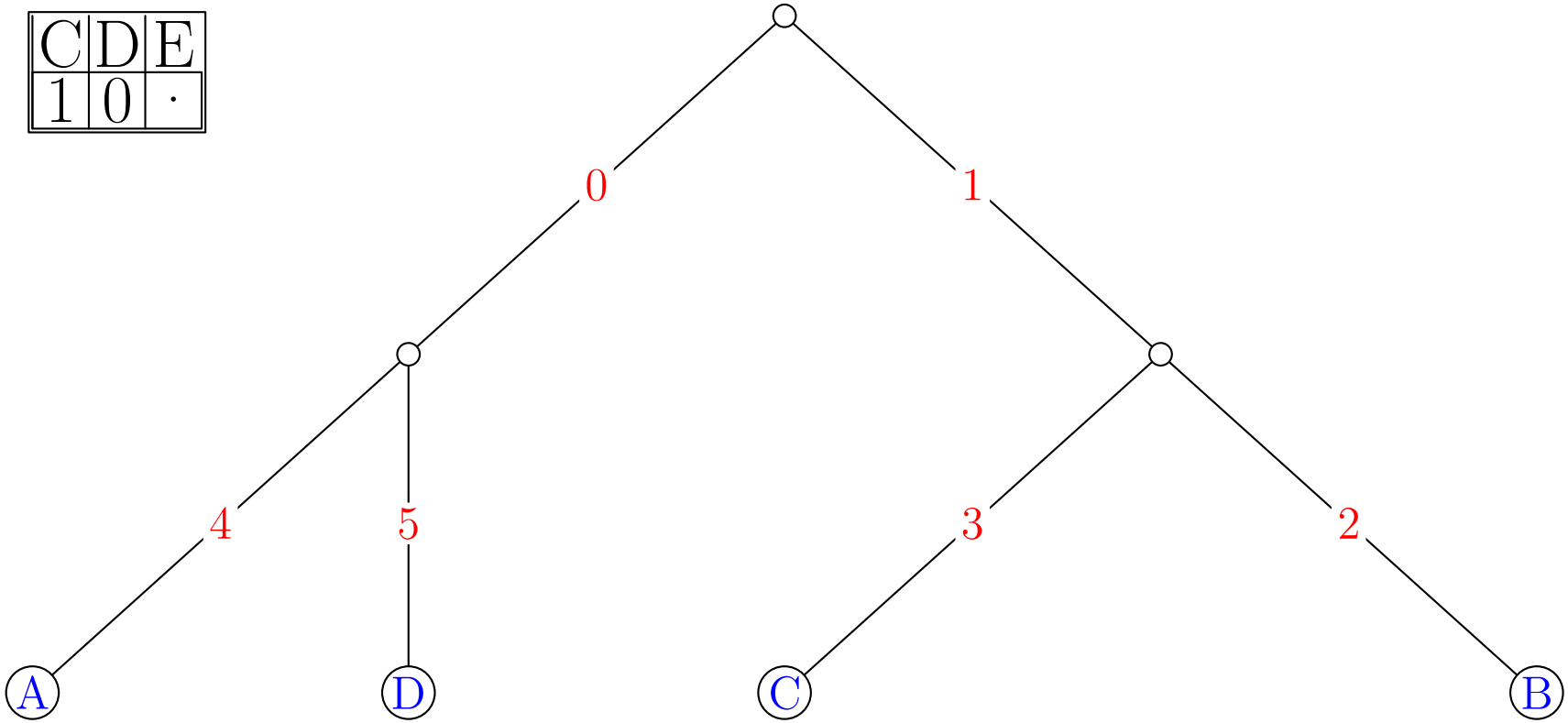
Enumerating Trees

C	D	E
1	·	·



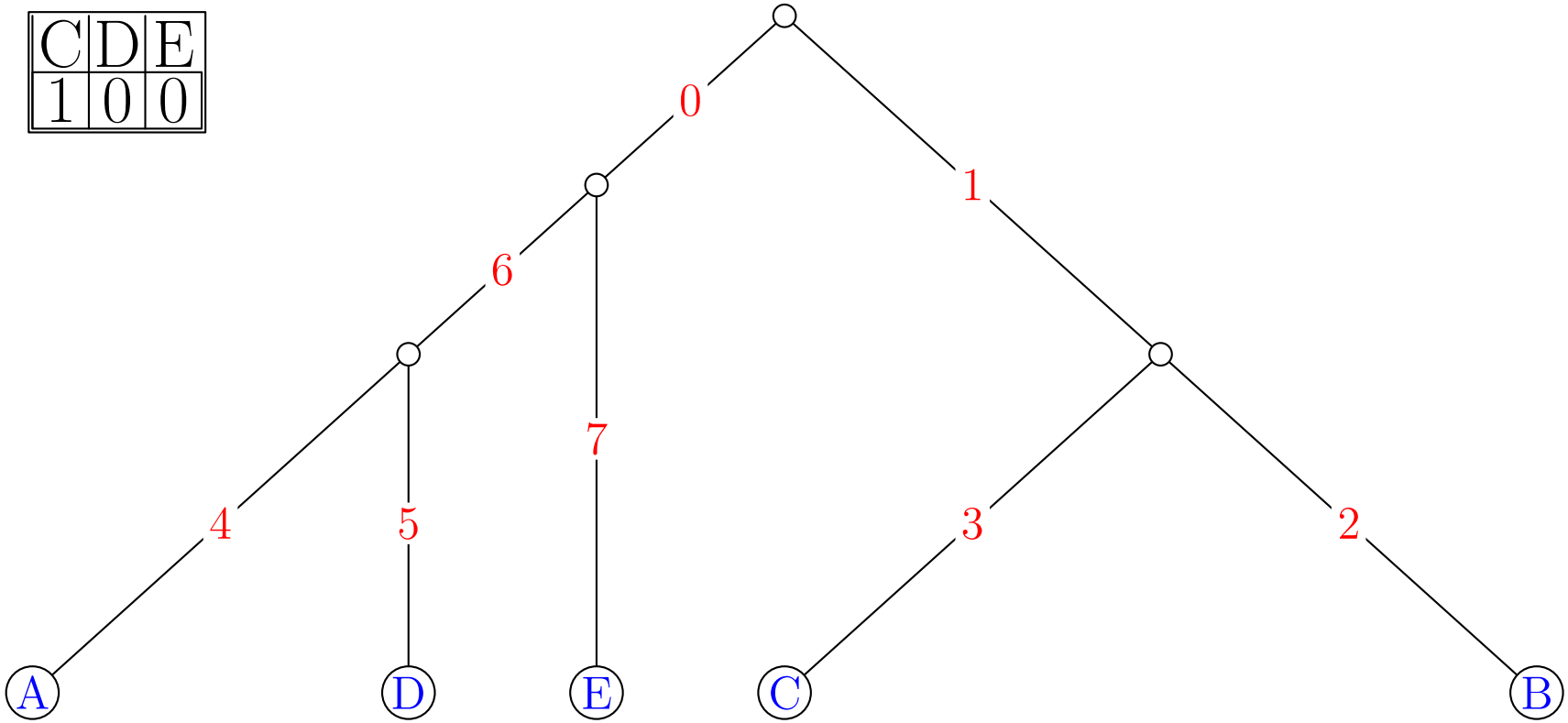
Enumerating Trees

C	D	E
1	0	·



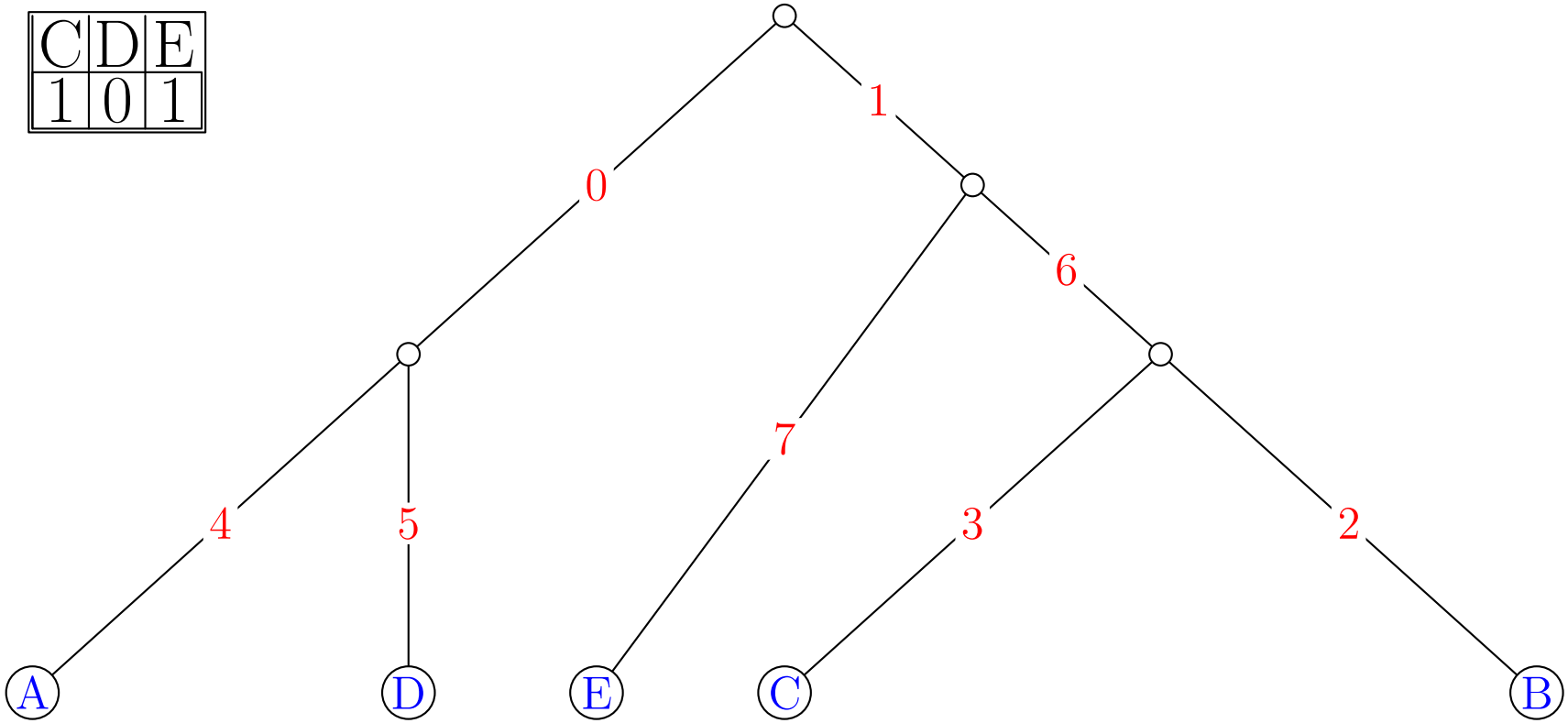
Enumerating Trees

C	D	E
1	0	0



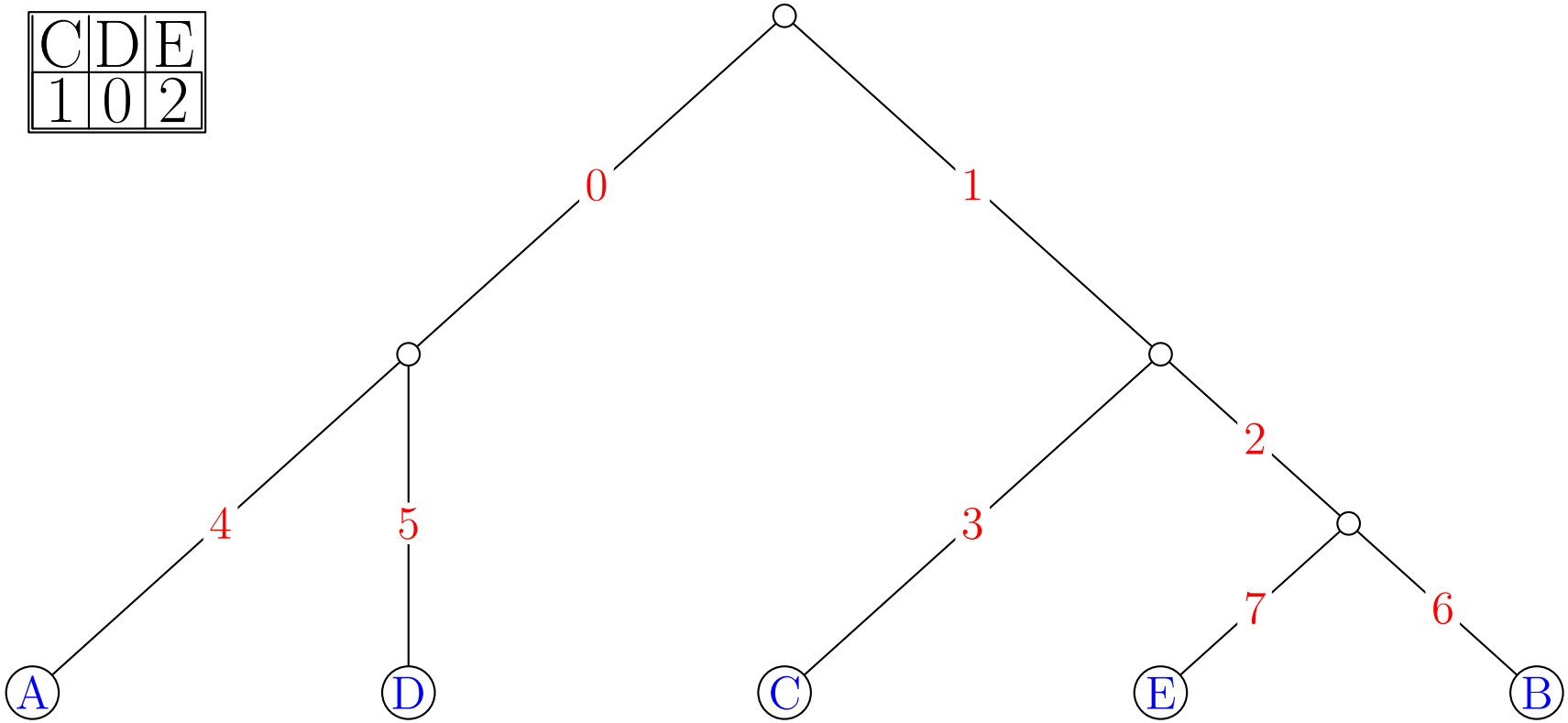
Enumerating Trees

C	D	E
1	0	1



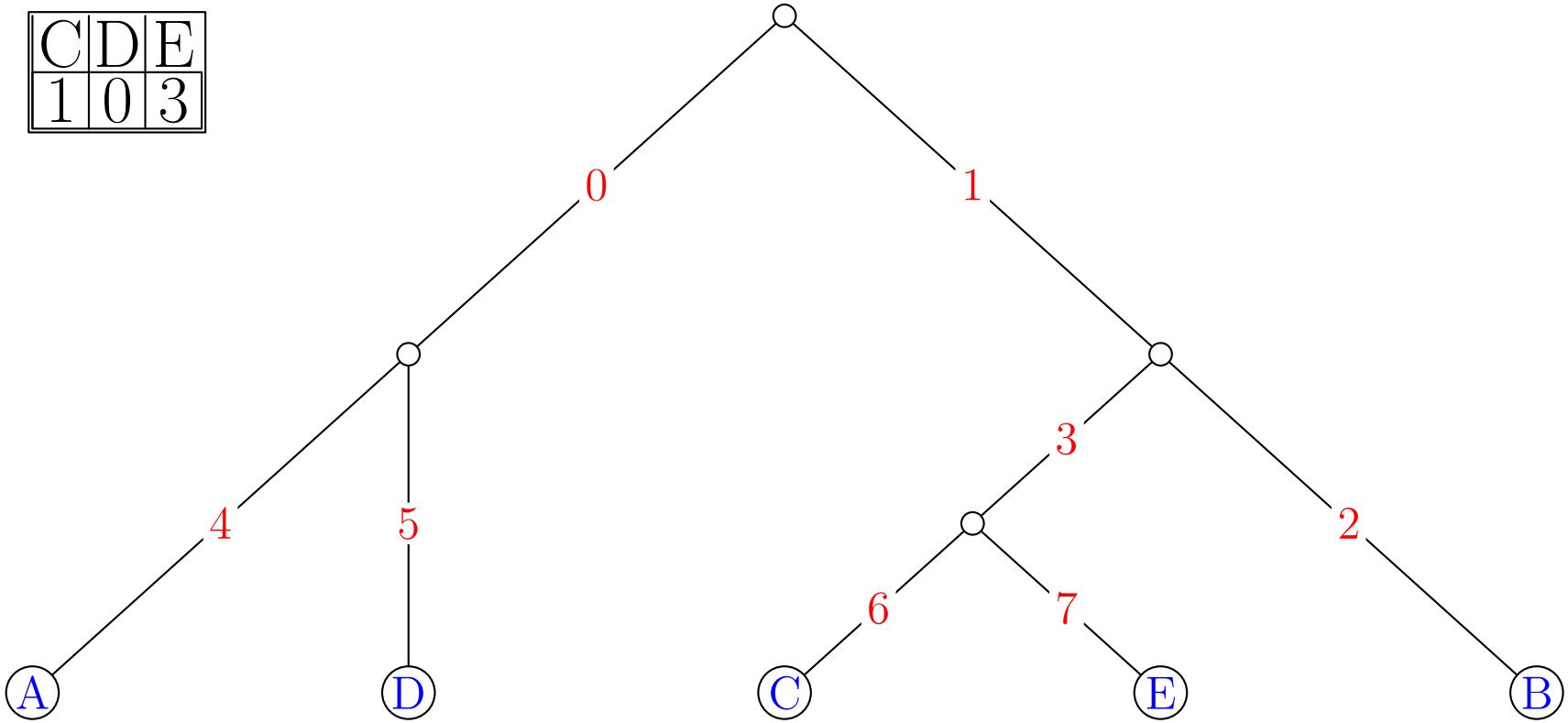
Enumerating Trees

C	D	E
1	0	2

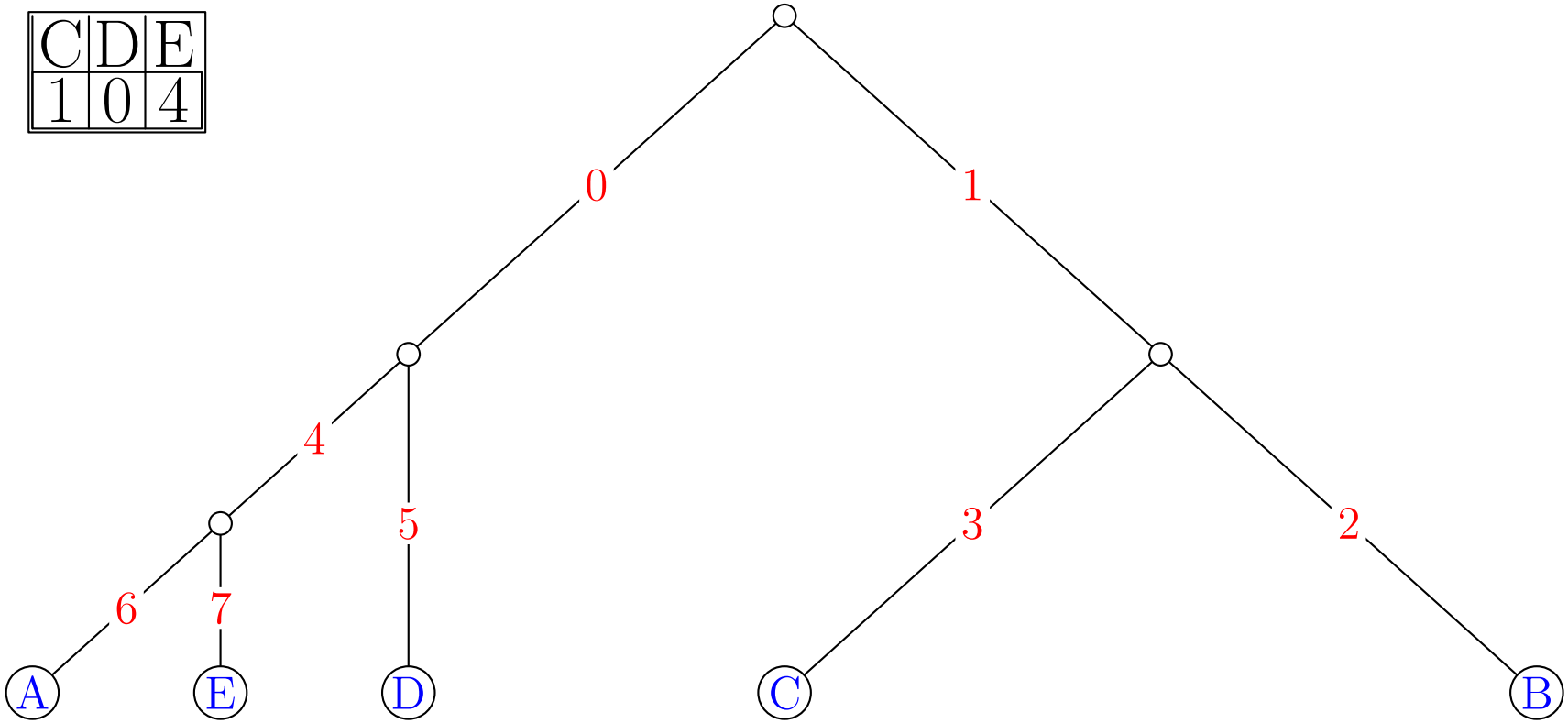


Enumerating Trees

C	D	E
1	0	3

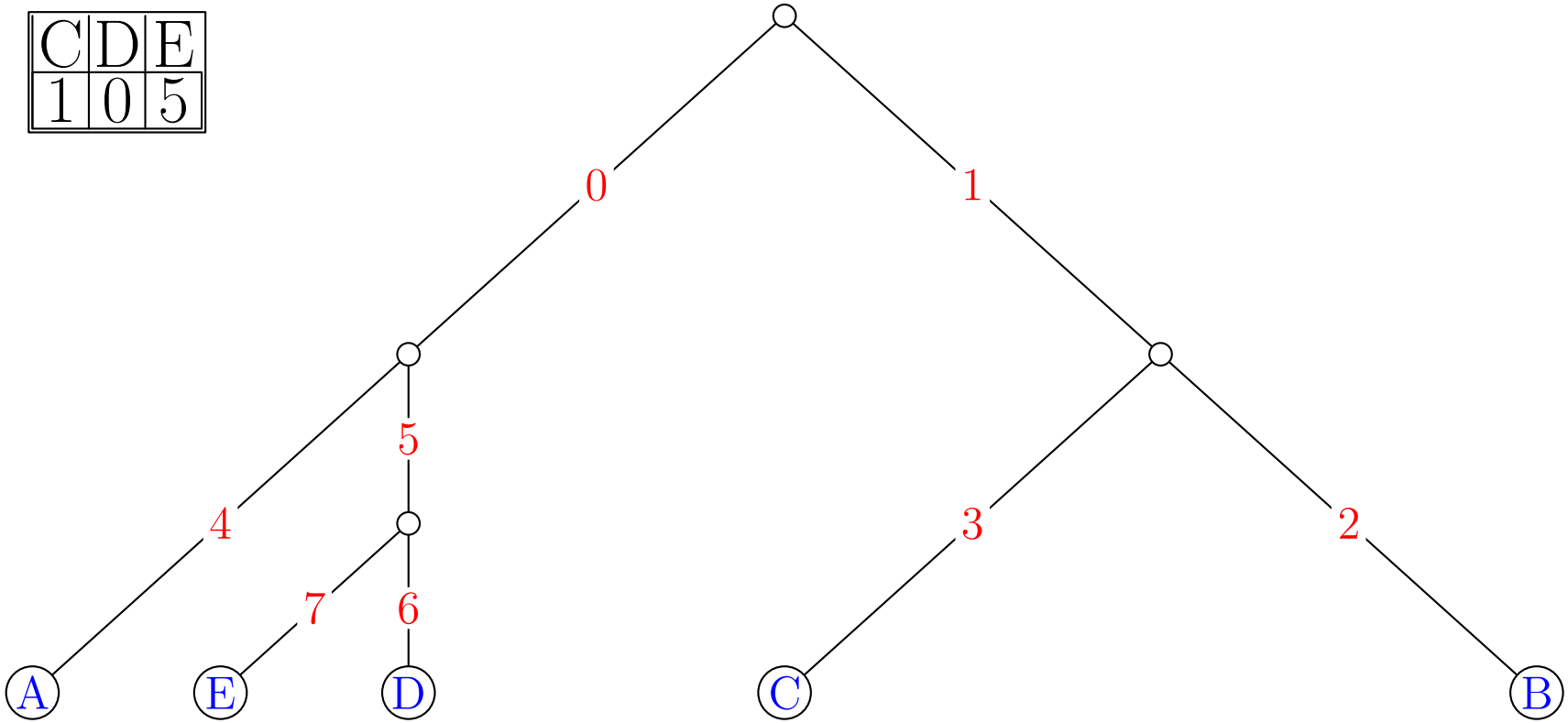


Enumerating Trees



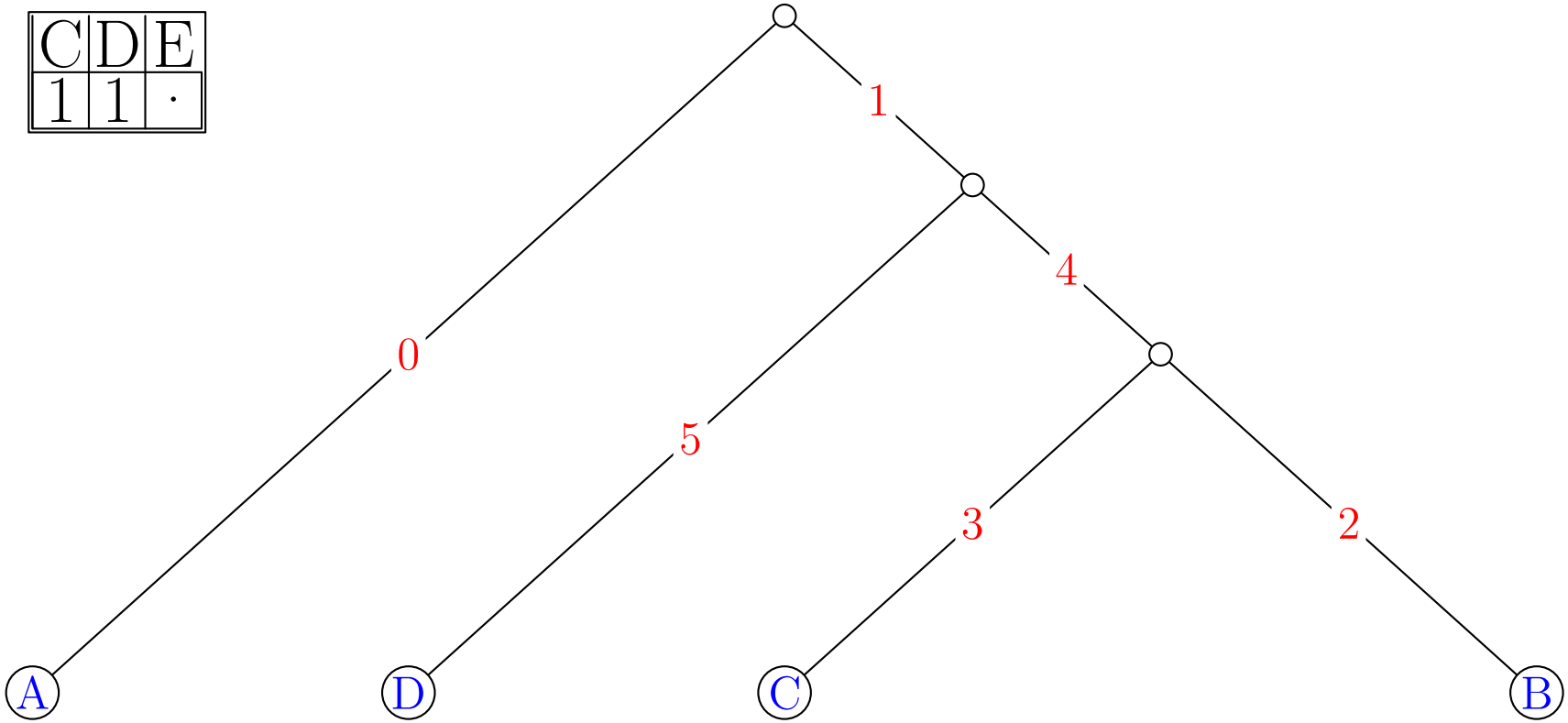
Enumerating Trees

C	D	E
1	0	5



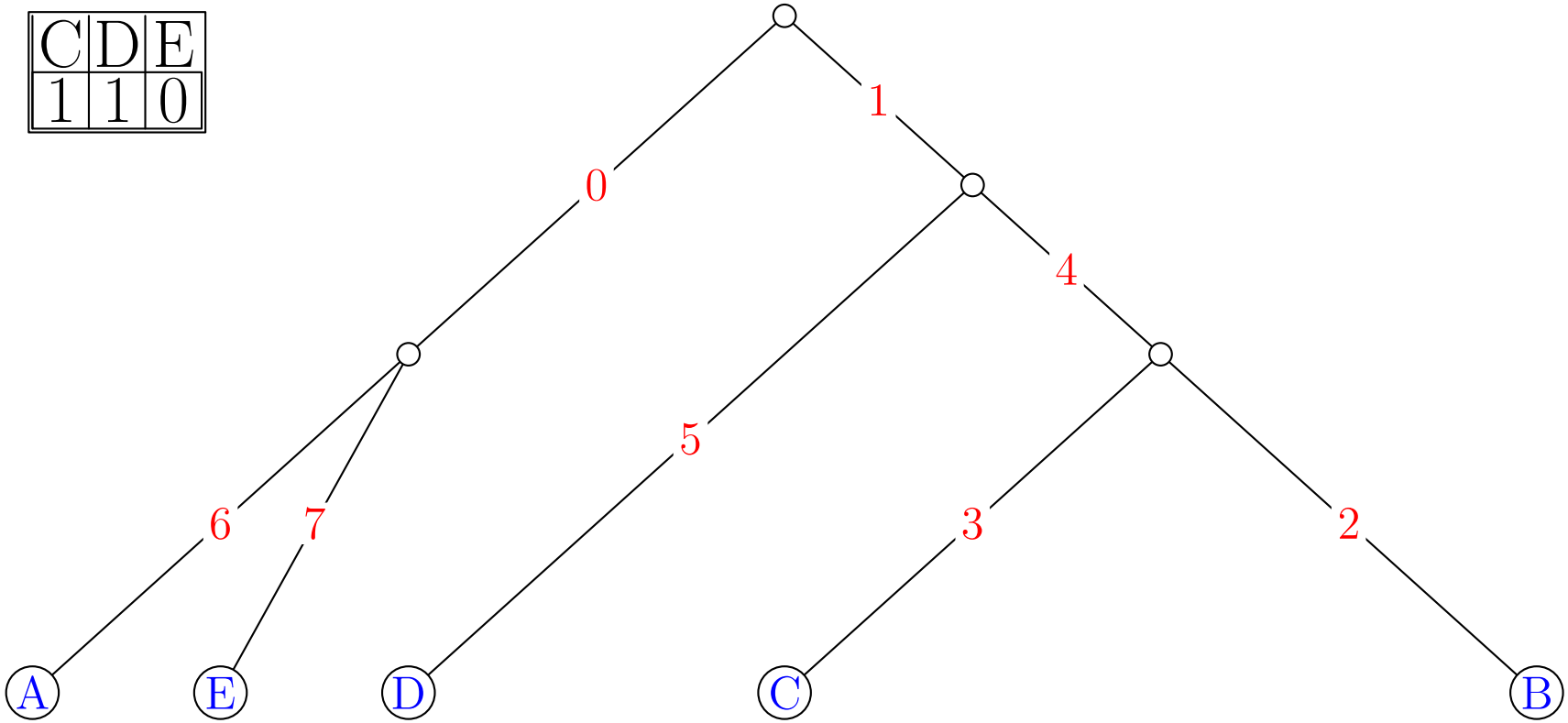
Enumerating Trees

C	D	E
1	1	·



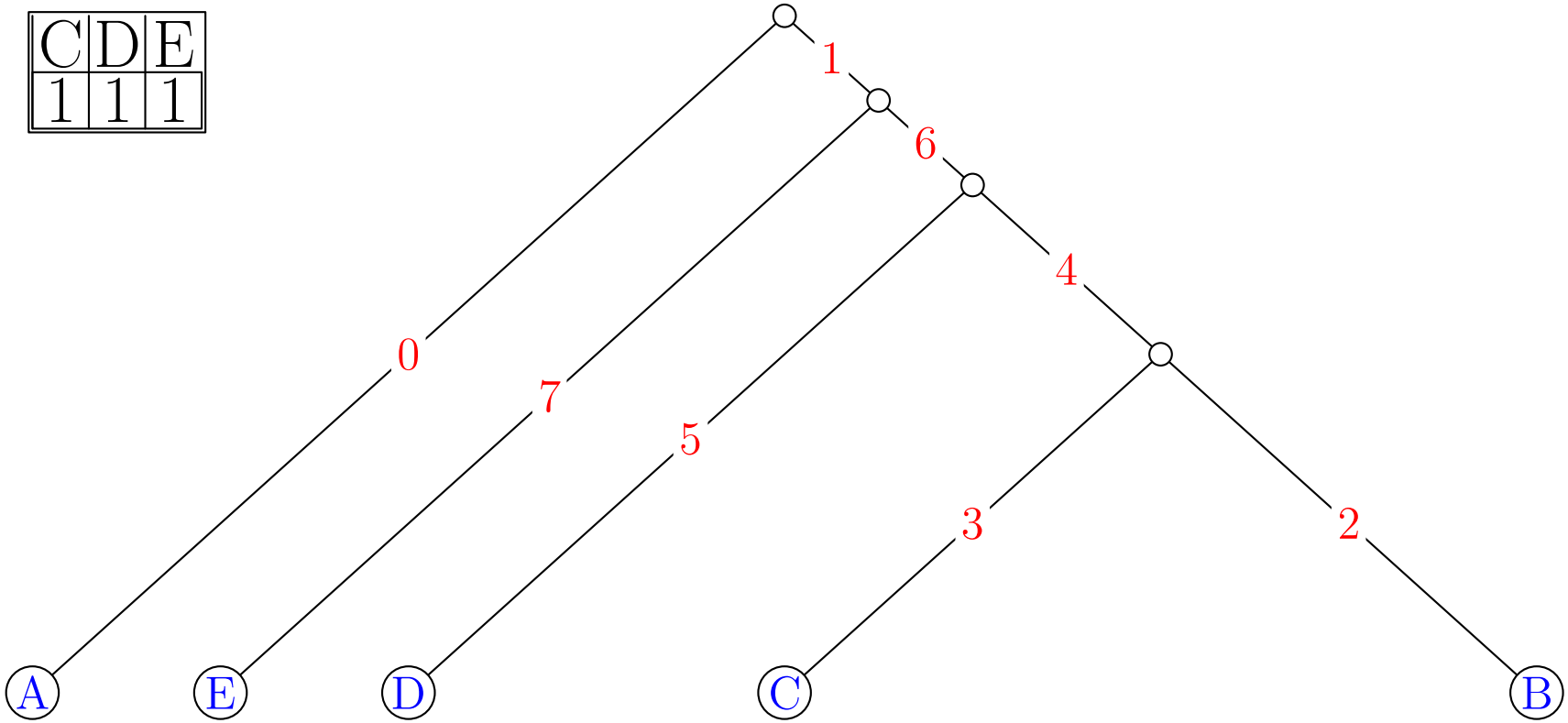
Enumerating Trees

C	D	E
1	1	0



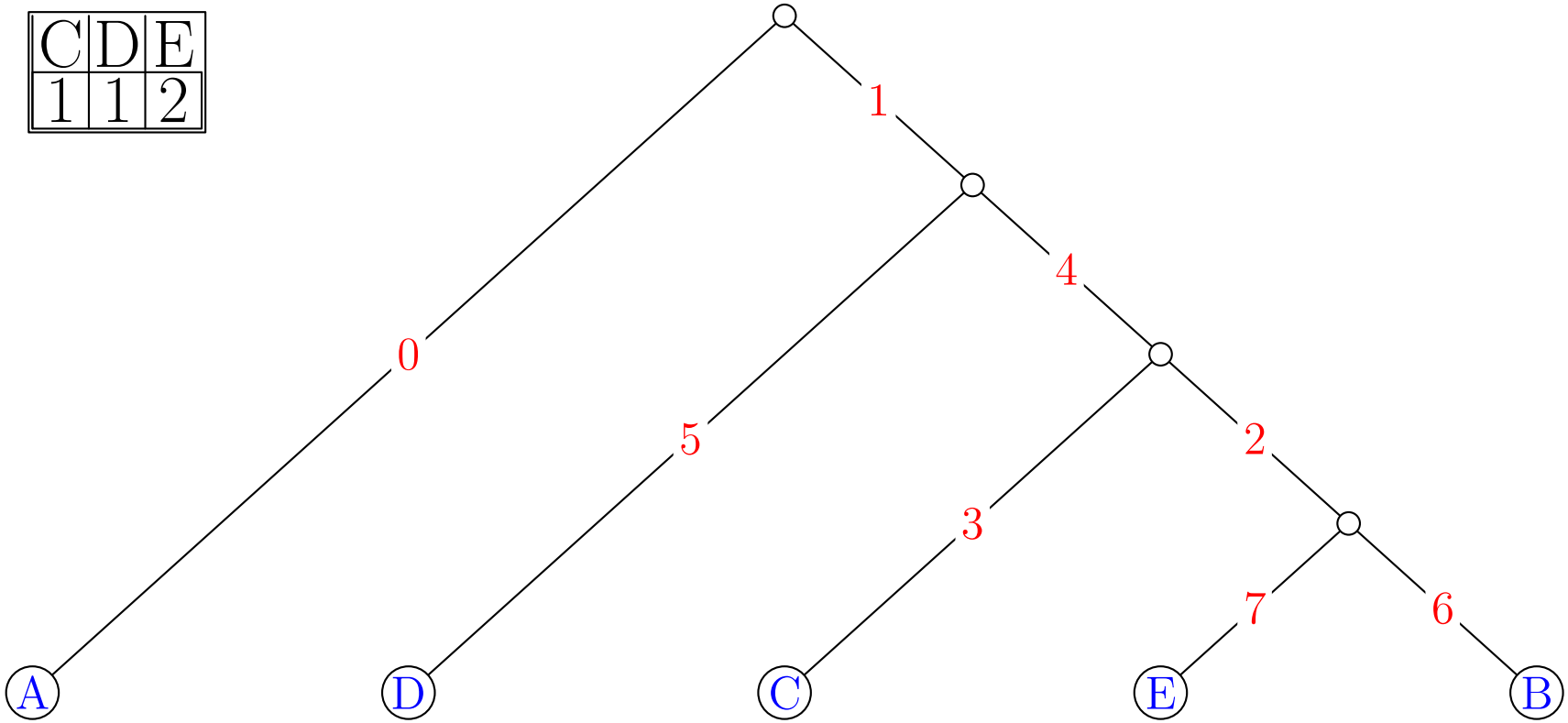
Enumerating Trees

C	D	E
1	1	1



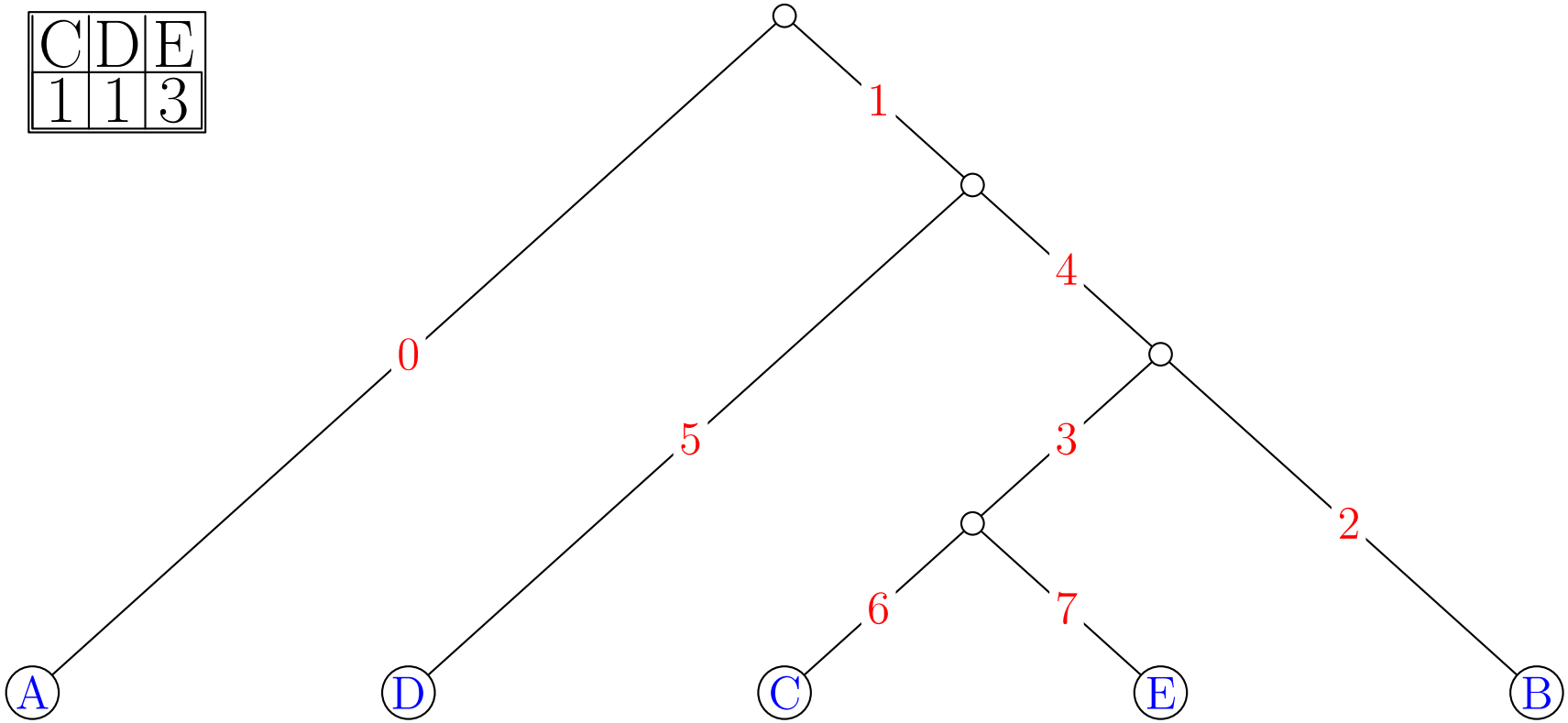
Enumerating Trees

C	D	E
1	1	2



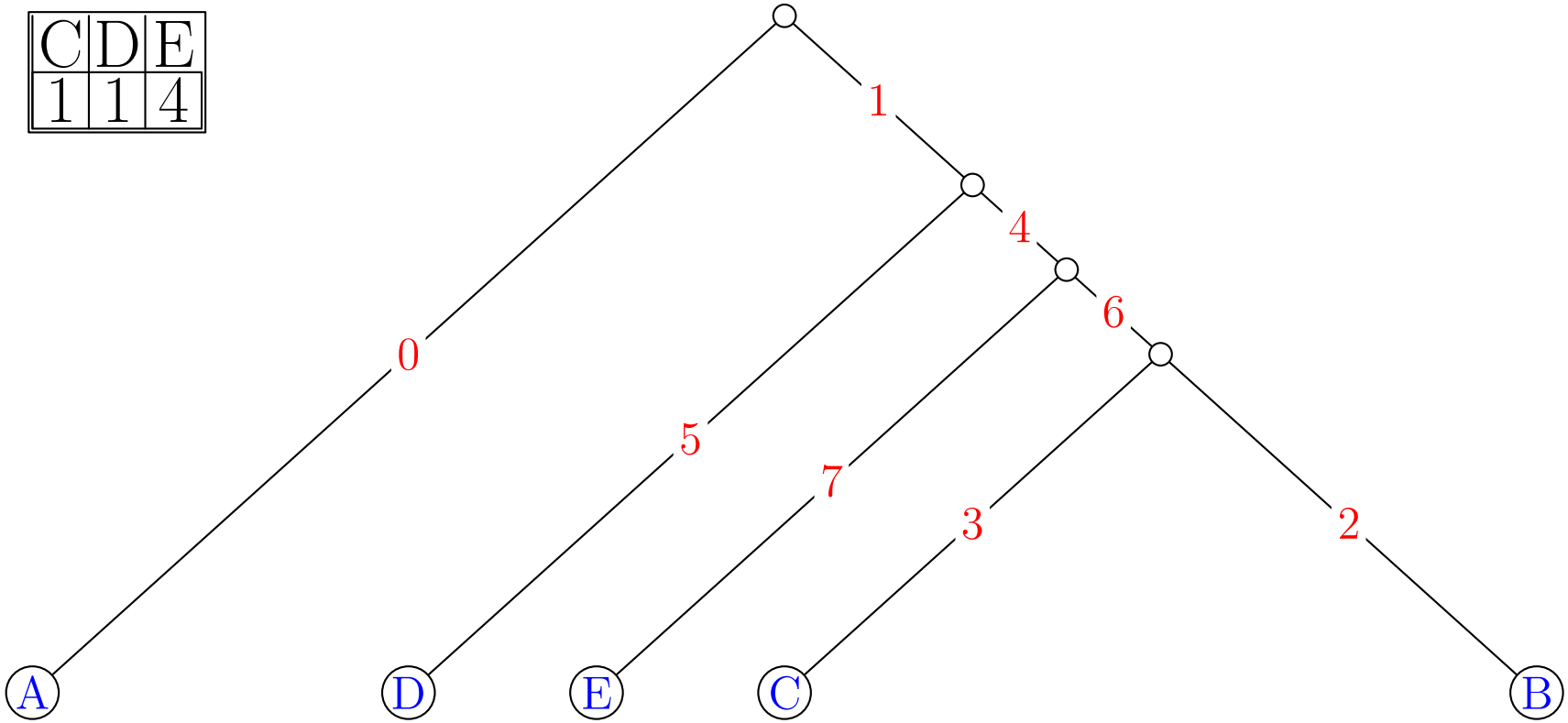
Enumerating Trees

C	D	E
1	1	3



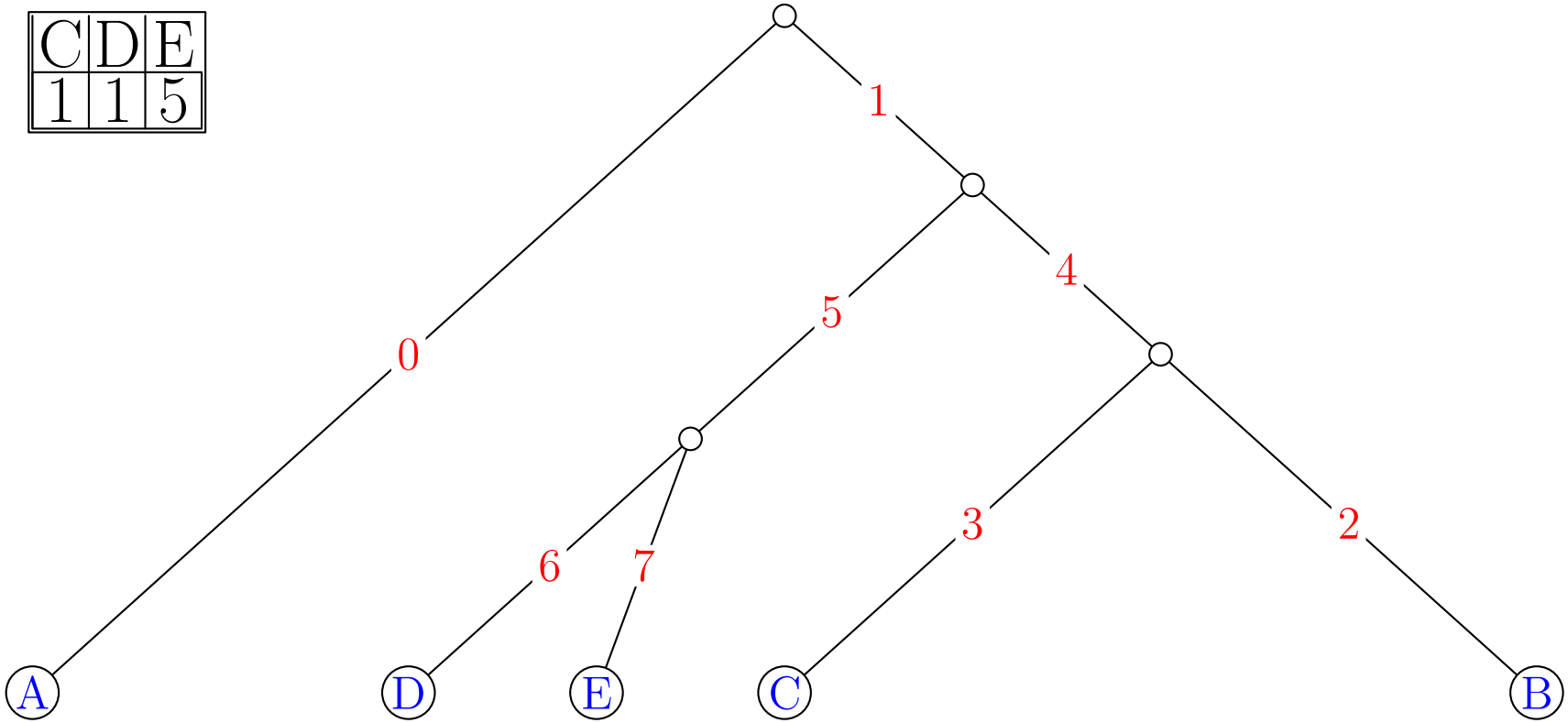
Enumerating Trees

C	D	E
1	1	4



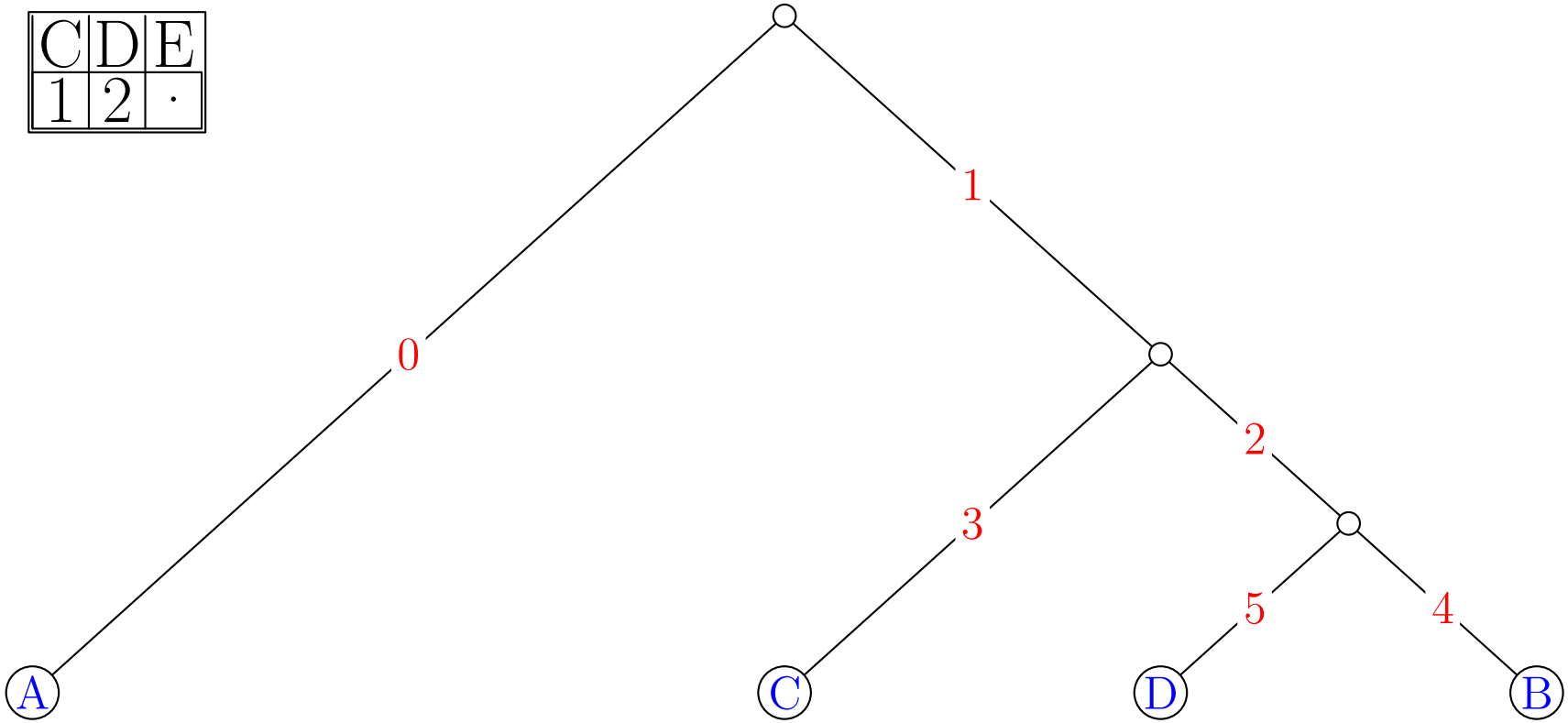
Enumerating Trees

C	D	E
1	1	5



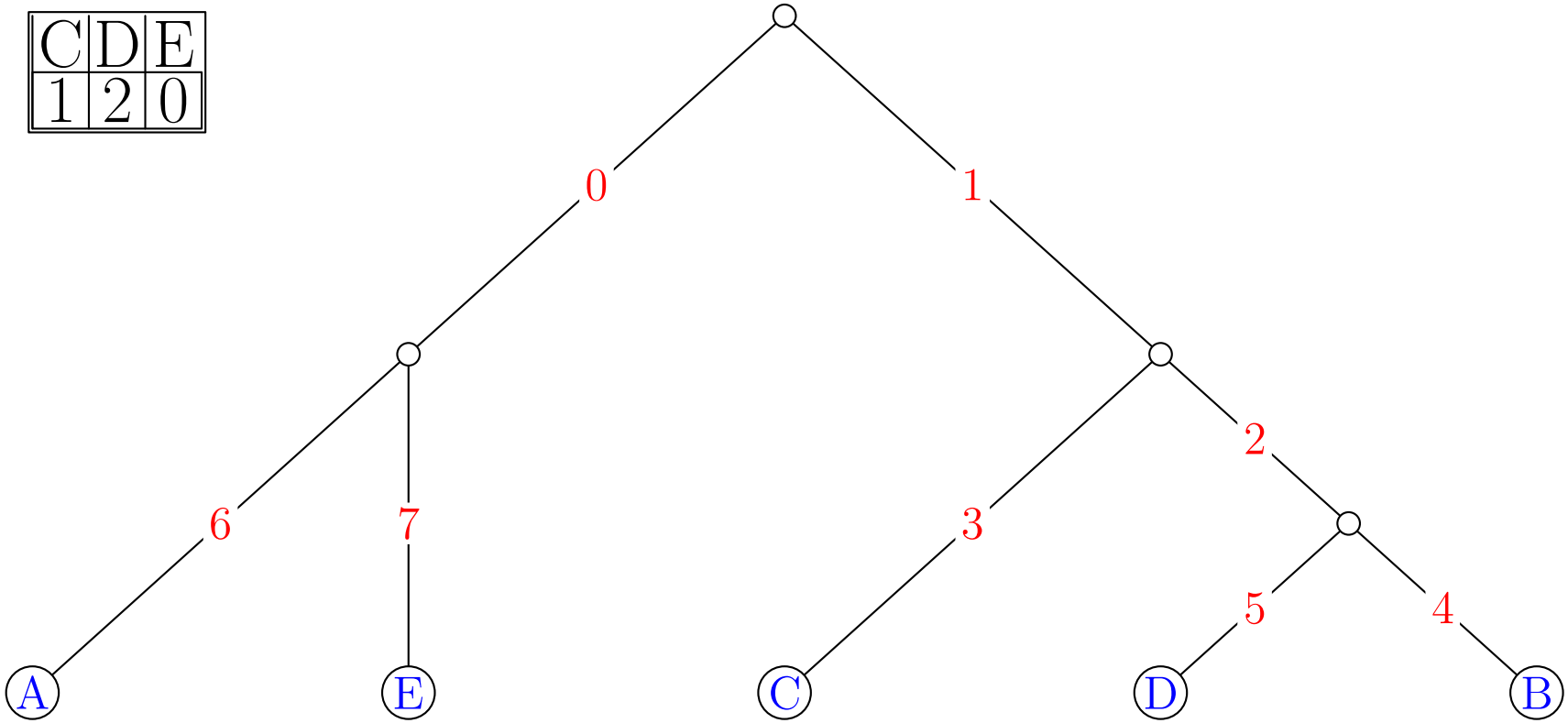
Enumerating Trees

C	D	E
1	2	·



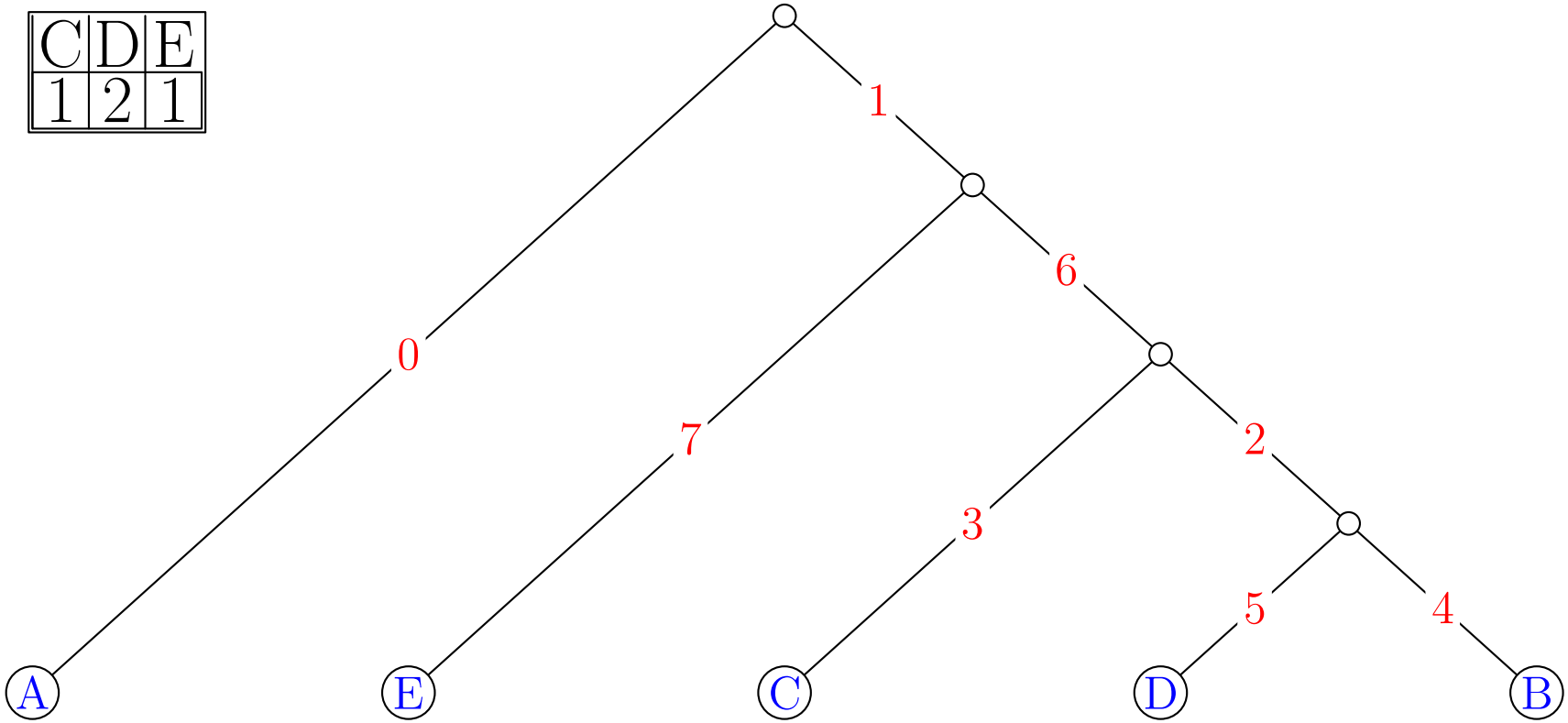
Enumerating Trees

C	D	E
1	2	0



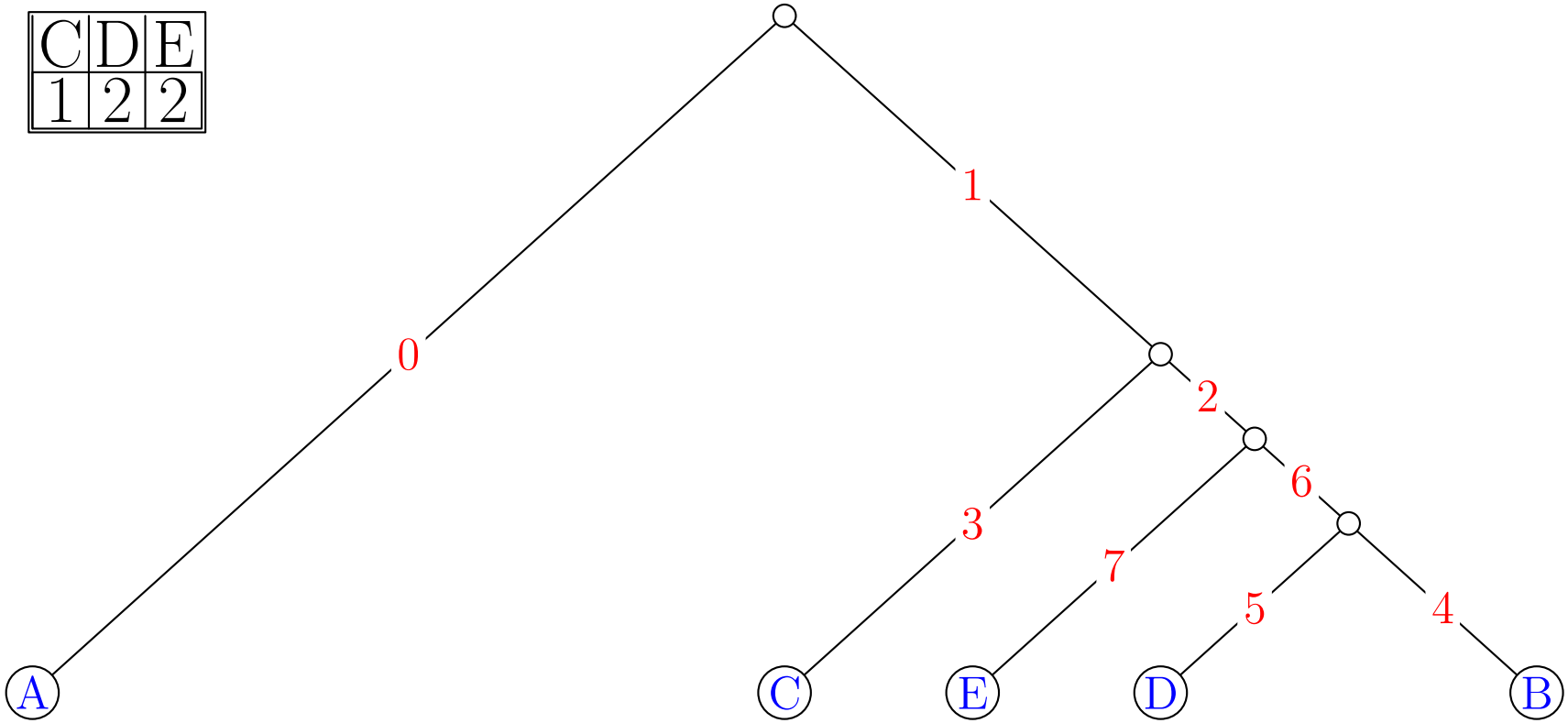
Enumerating Trees

C	D	E
1	2	1

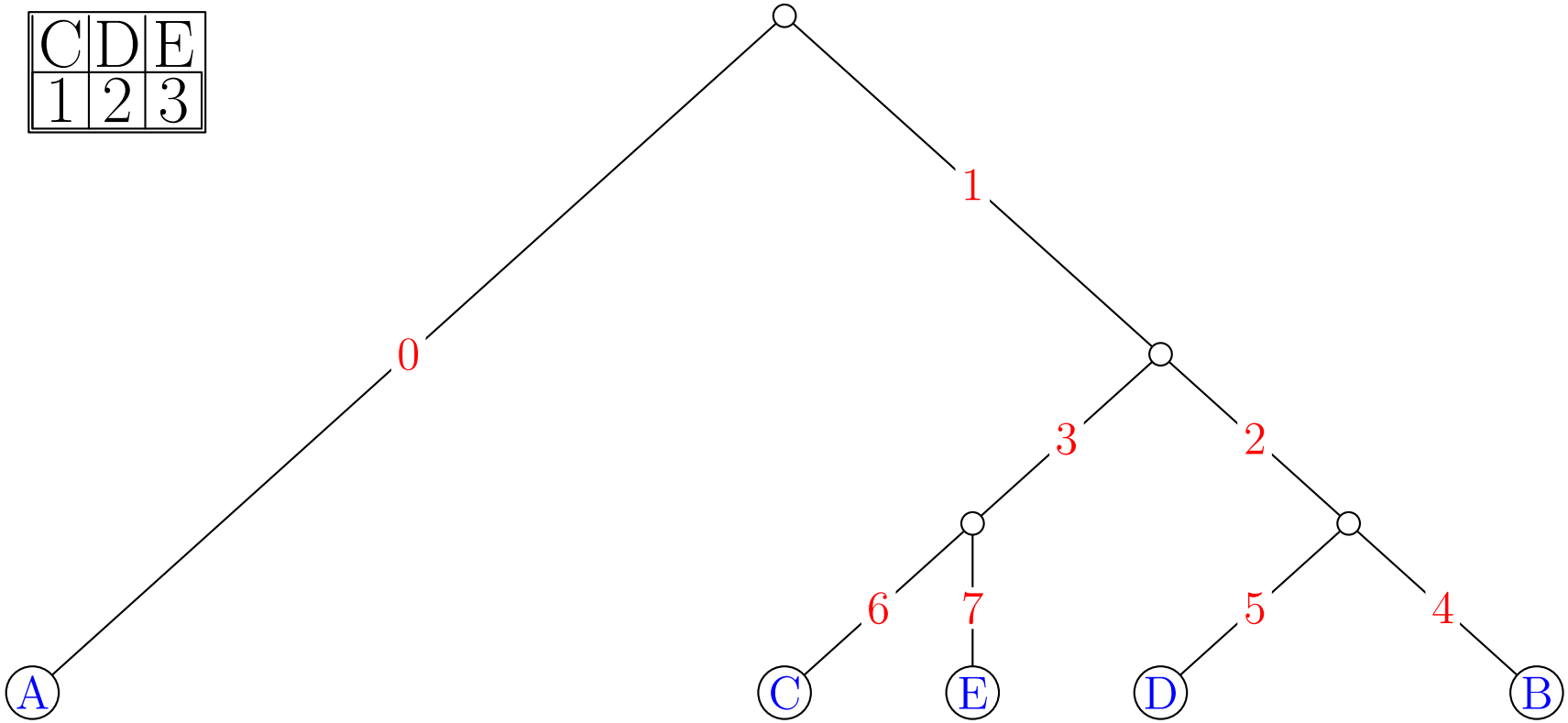


Enumerating Trees

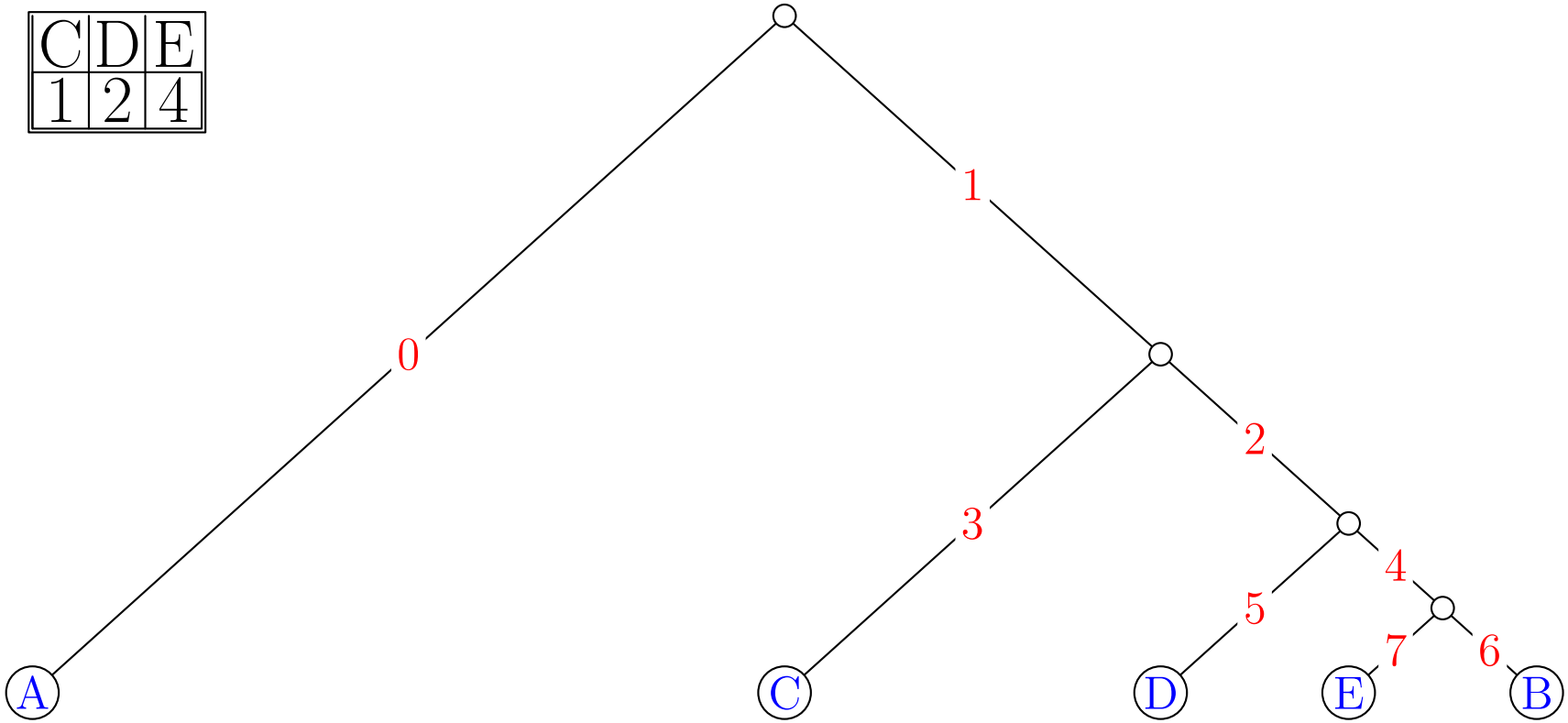
C	D	E
1	2	2



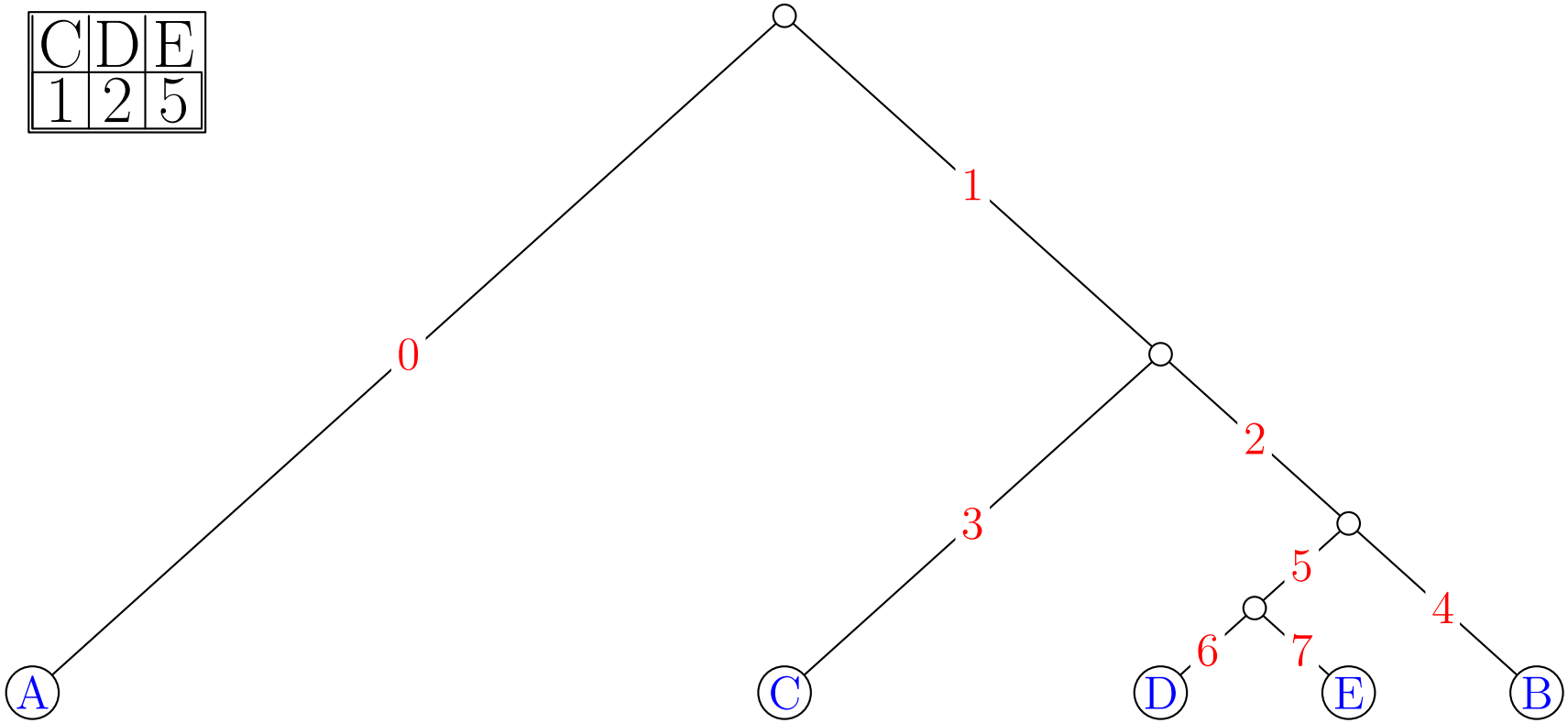
Enumerating Trees



Enumerating Trees

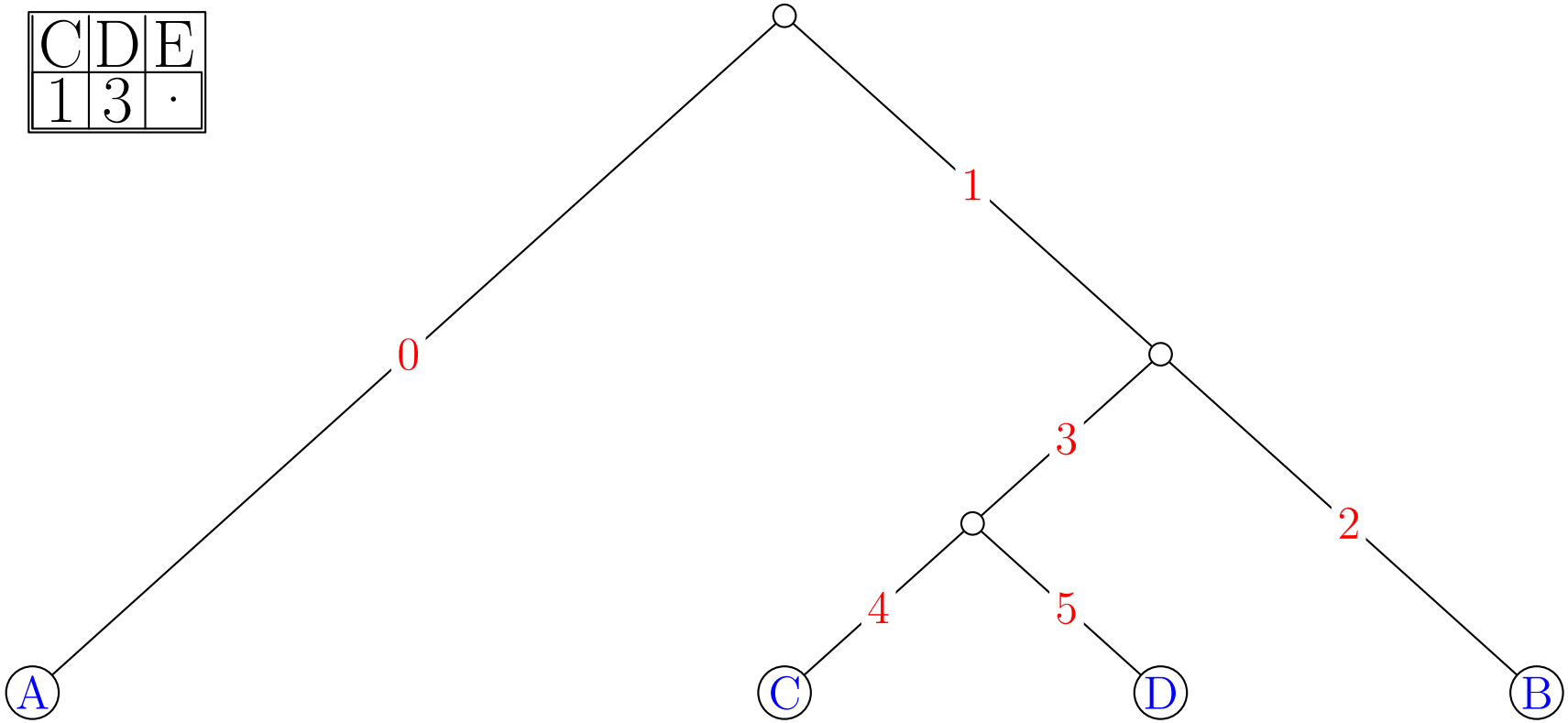


Enumerating Trees



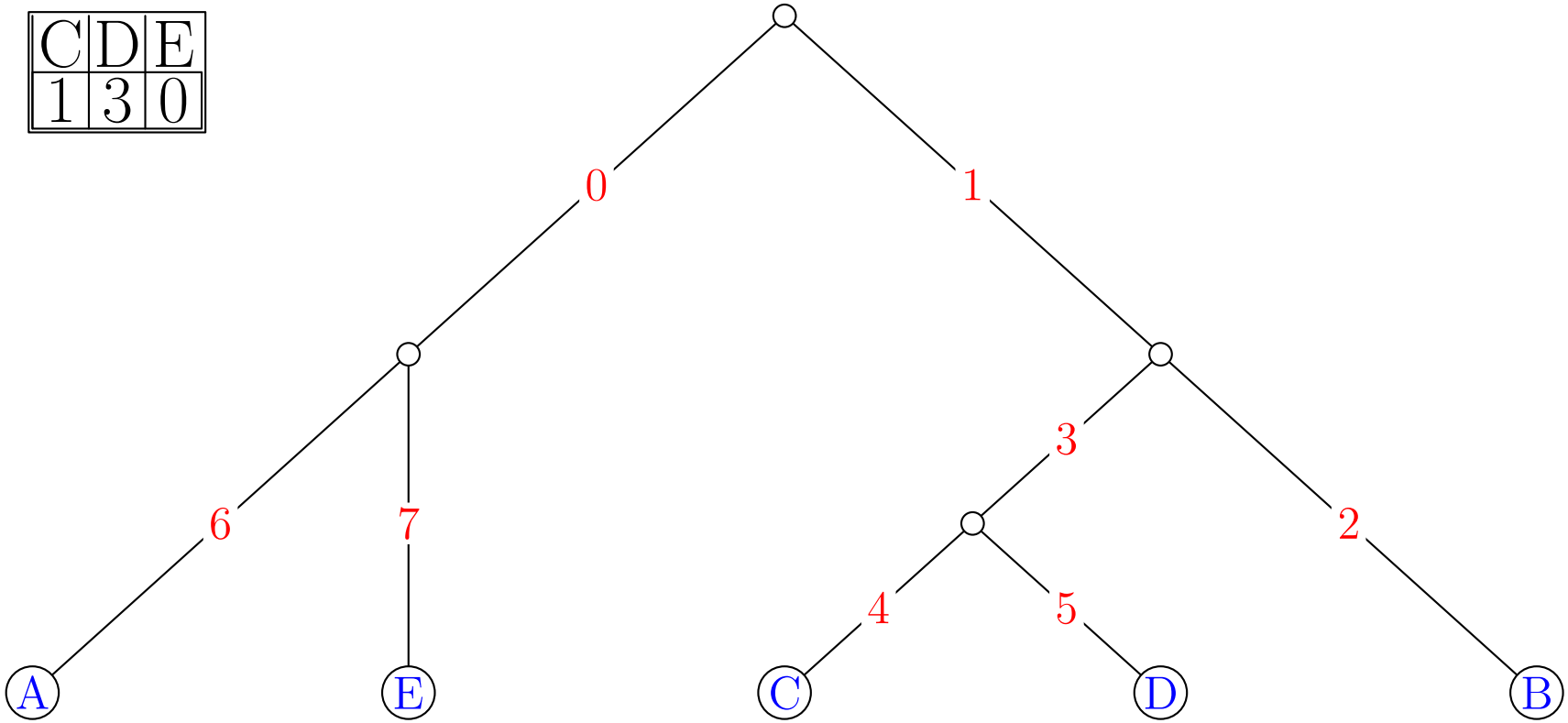
Enumerating Trees

C	D	E
1	3	·



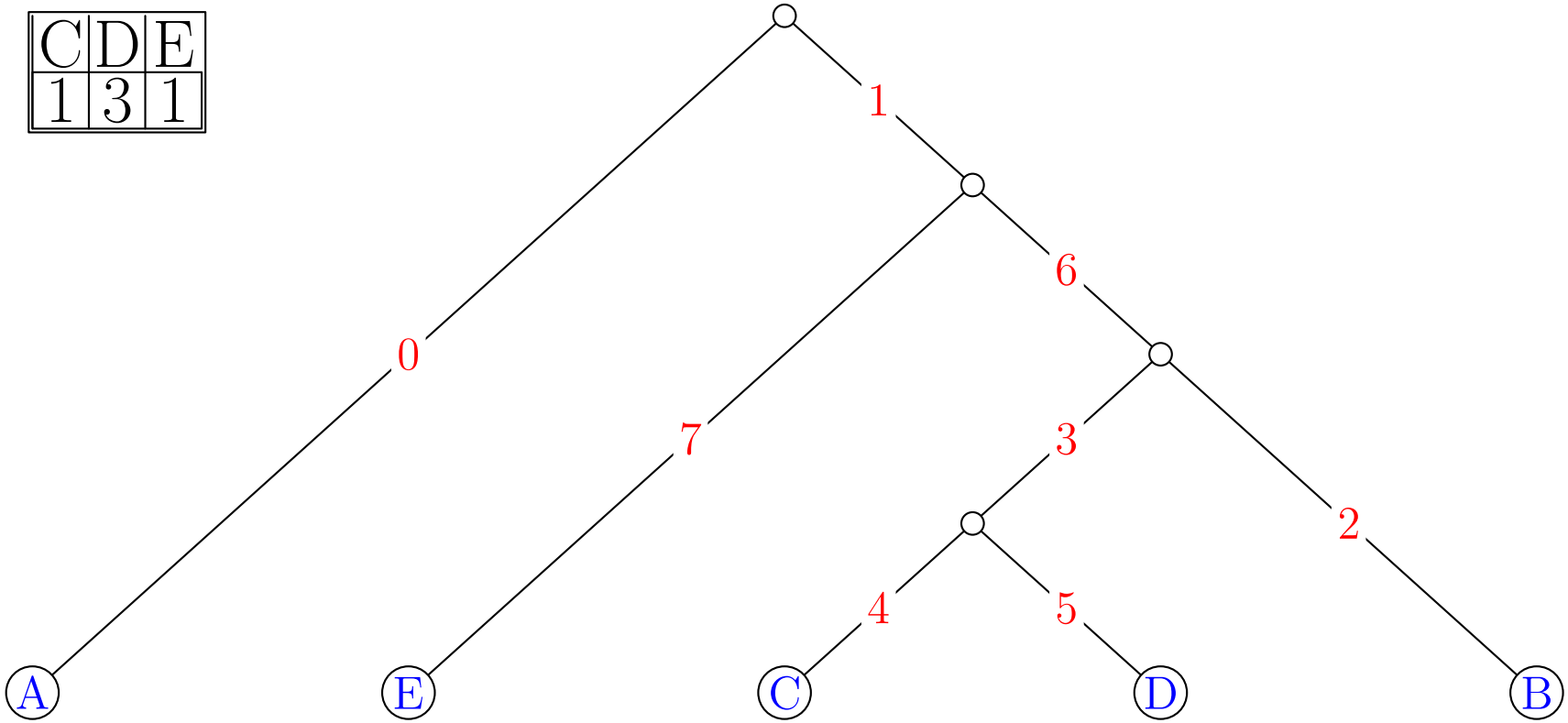
Enumerating Trees

C	D	E
1	3	0



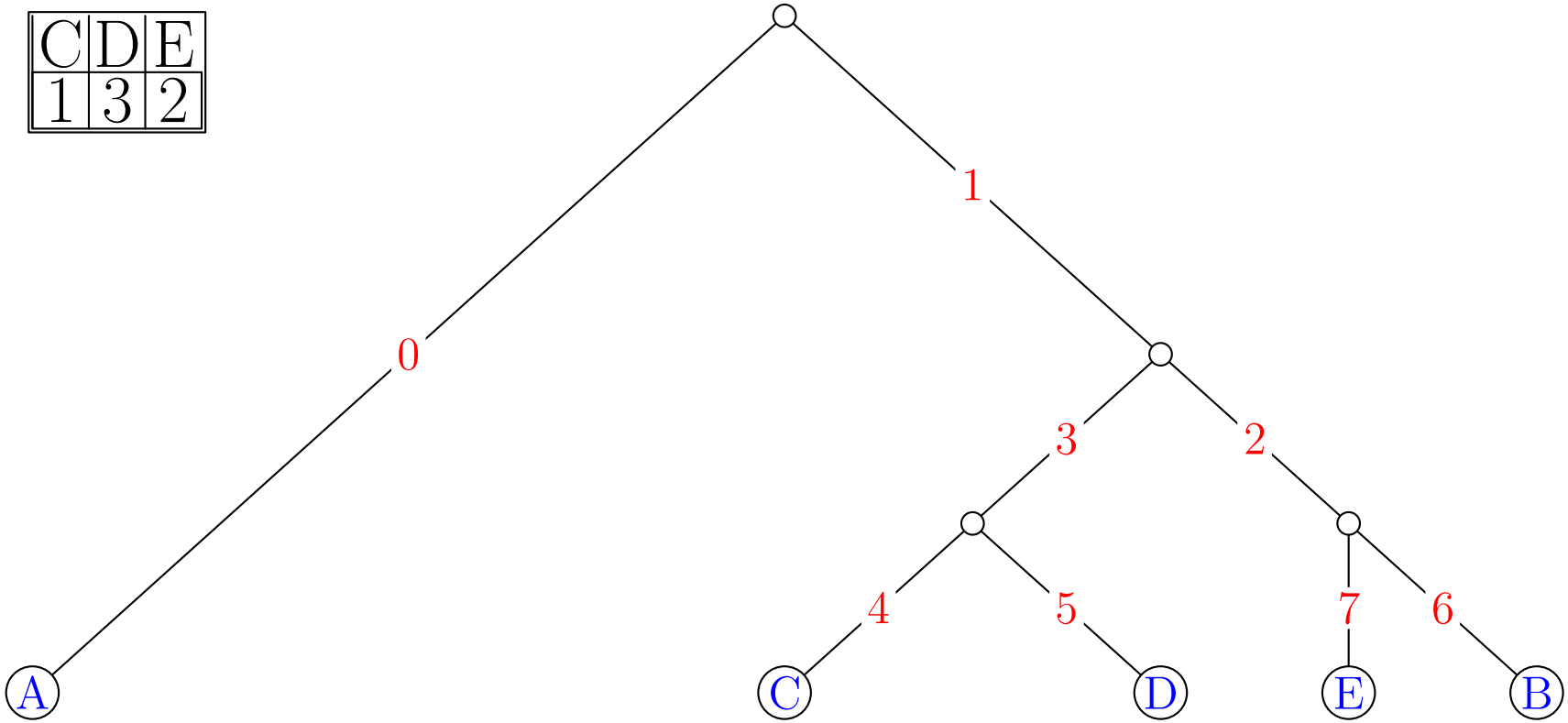
Enumerating Trees

C	D	E
1	3	1



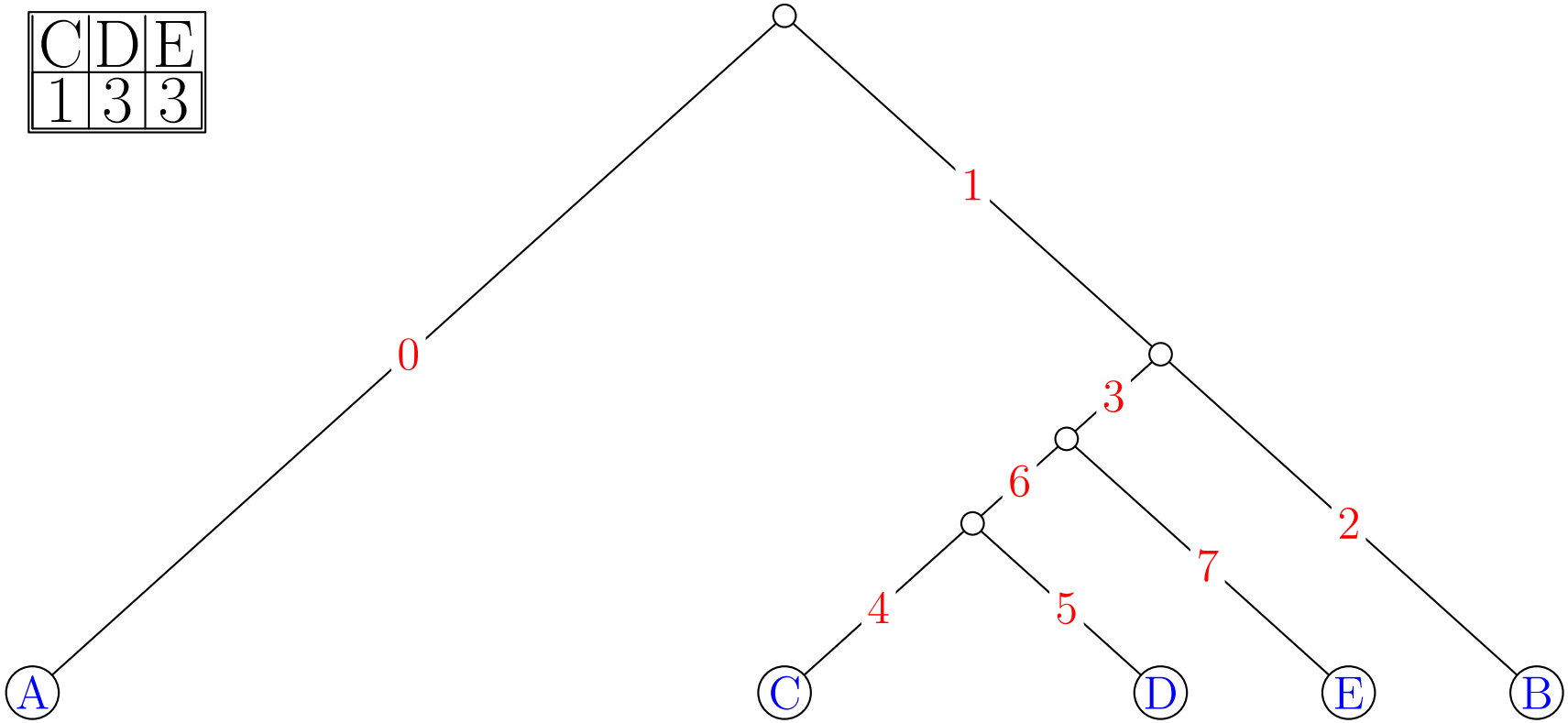
Enumerating Trees

C	D	E
1	3	2



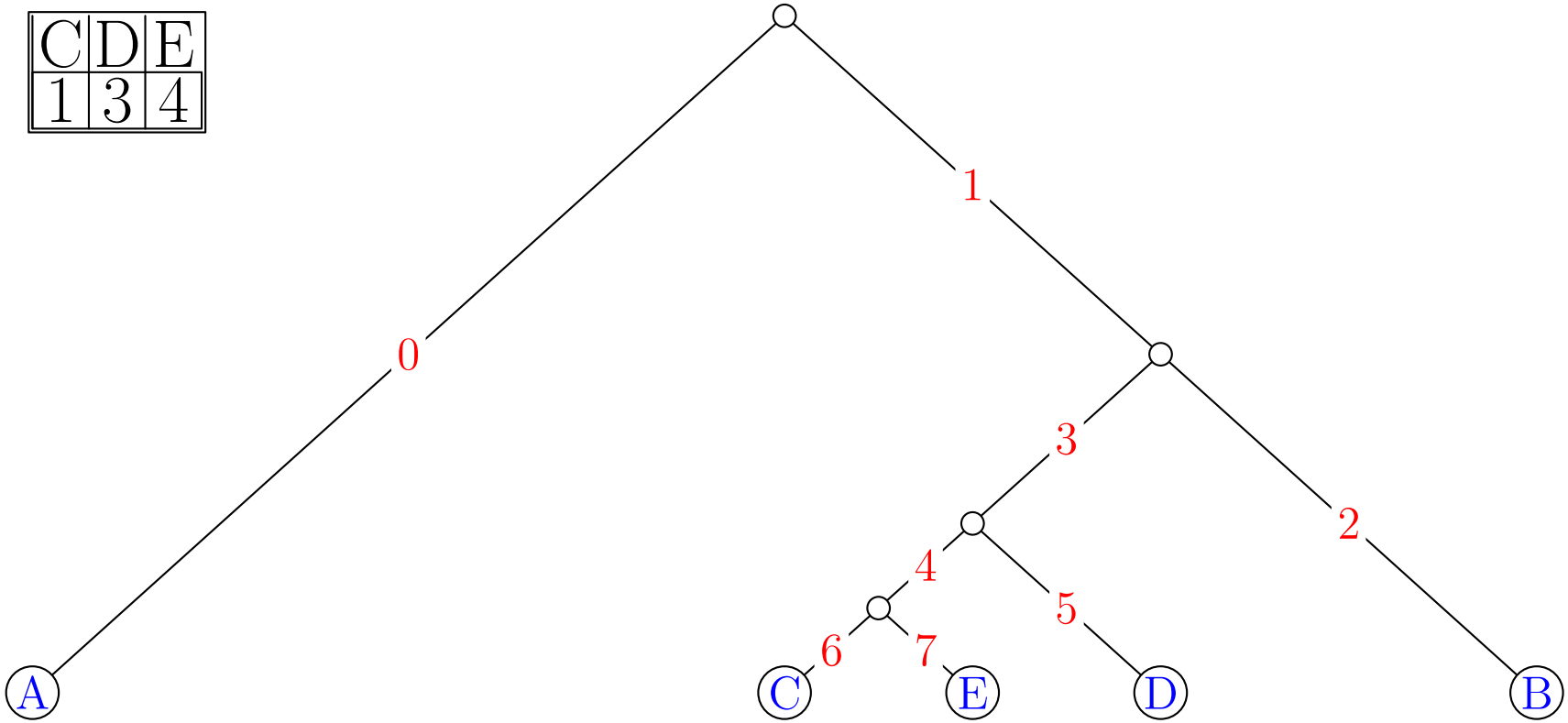
Enumerating Trees

C	D	E
1	3	3



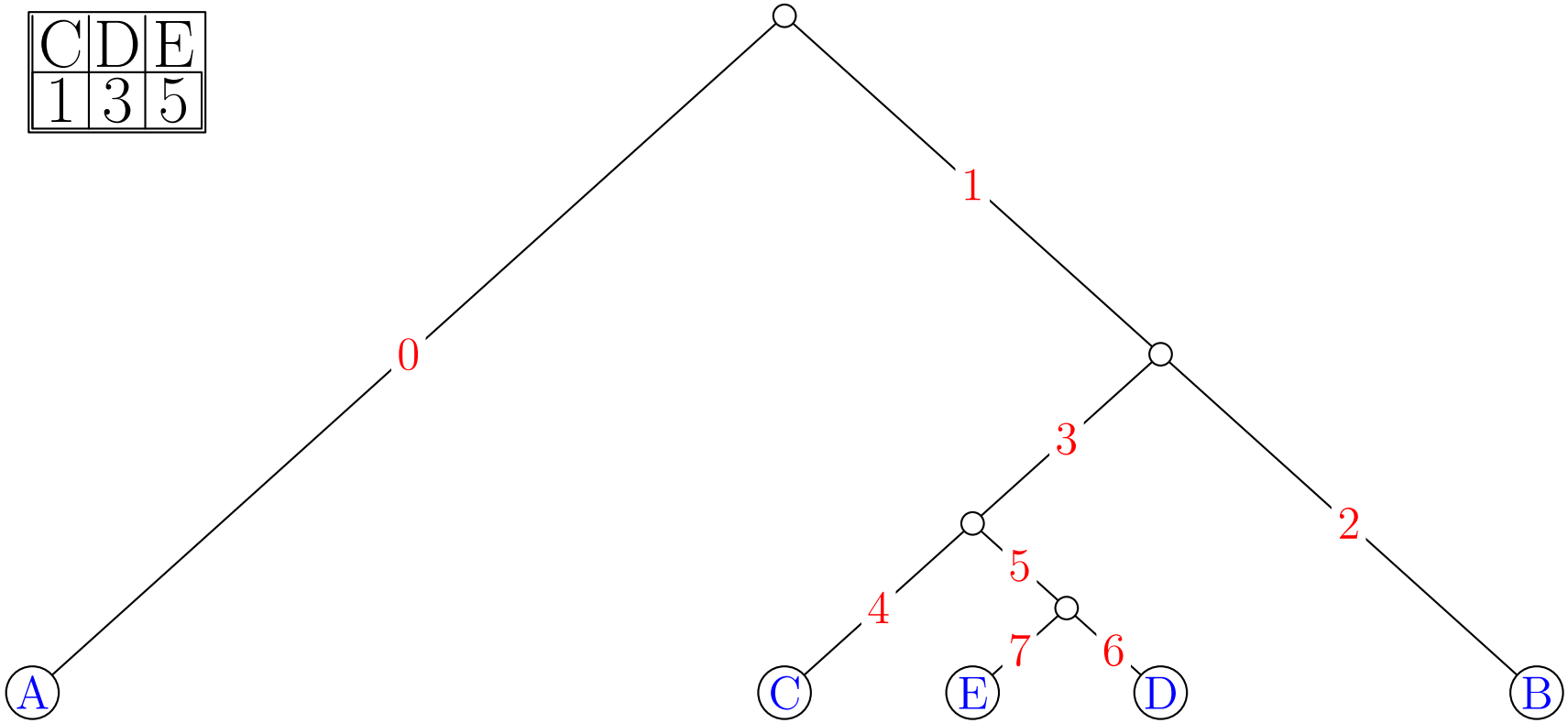
Enumerating Trees

C	D	E
1	3	4



Enumerating Trees

C	D	E
1	3	5



Branch and Bound

- Note that in the above procedure we generated partial trees
- Any tree generated from a partial tree will have at least as high a cost as the partial tree
- If the partial tree cost equals or exceeds the cost of the best found tree then we don't have to continue building the tree
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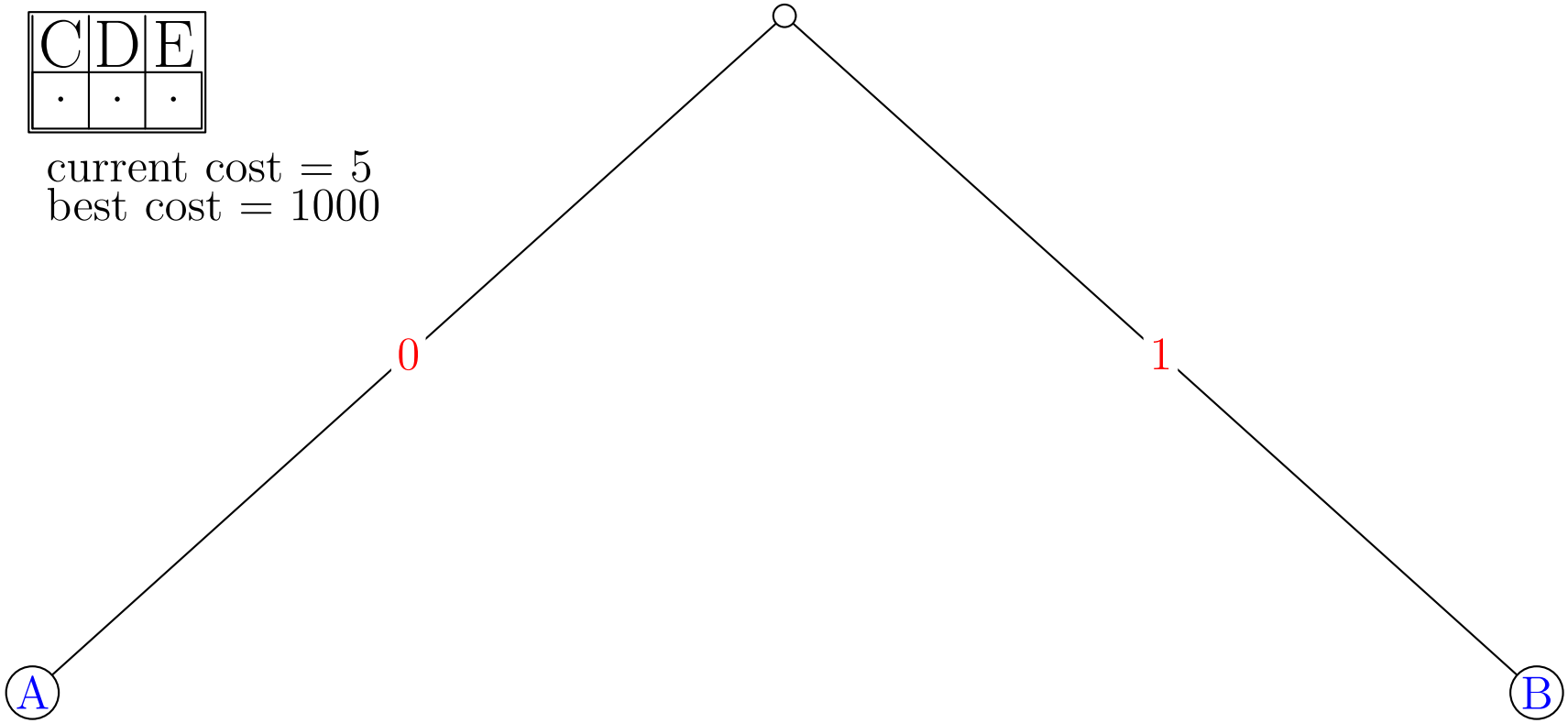
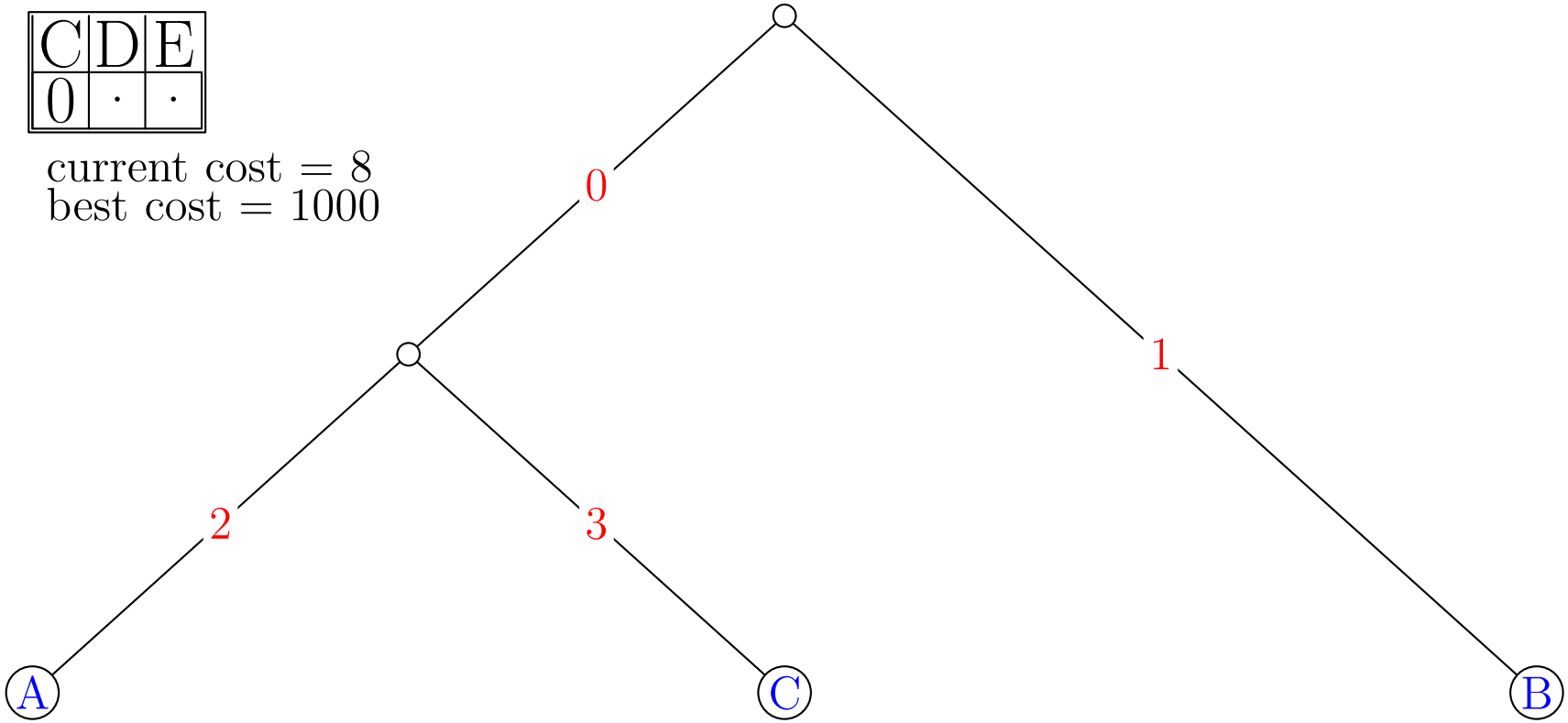


Branch and Bound

C	D	E
0	·	·

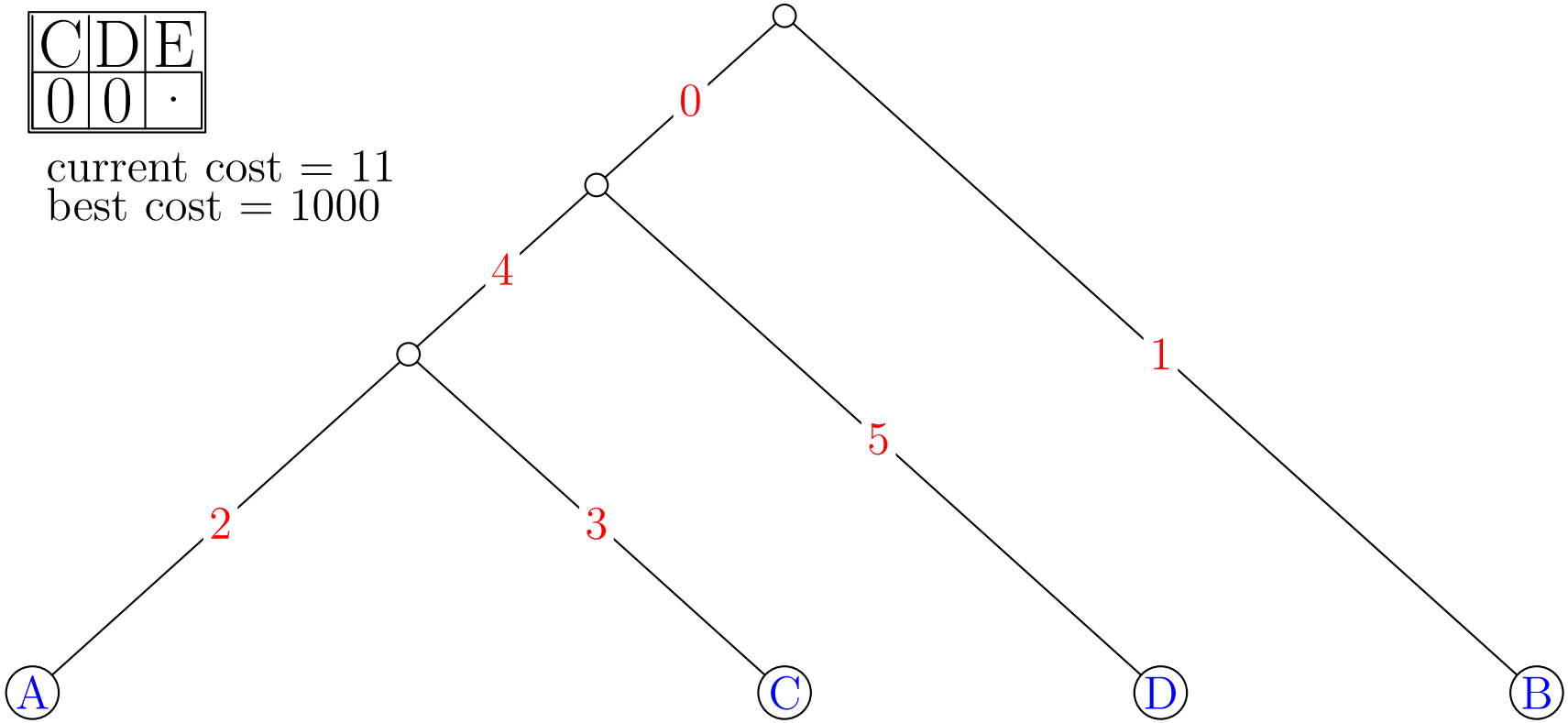
current cost = 8
best cost = 1000



Branch and Bound

C	D	E
0	0	·

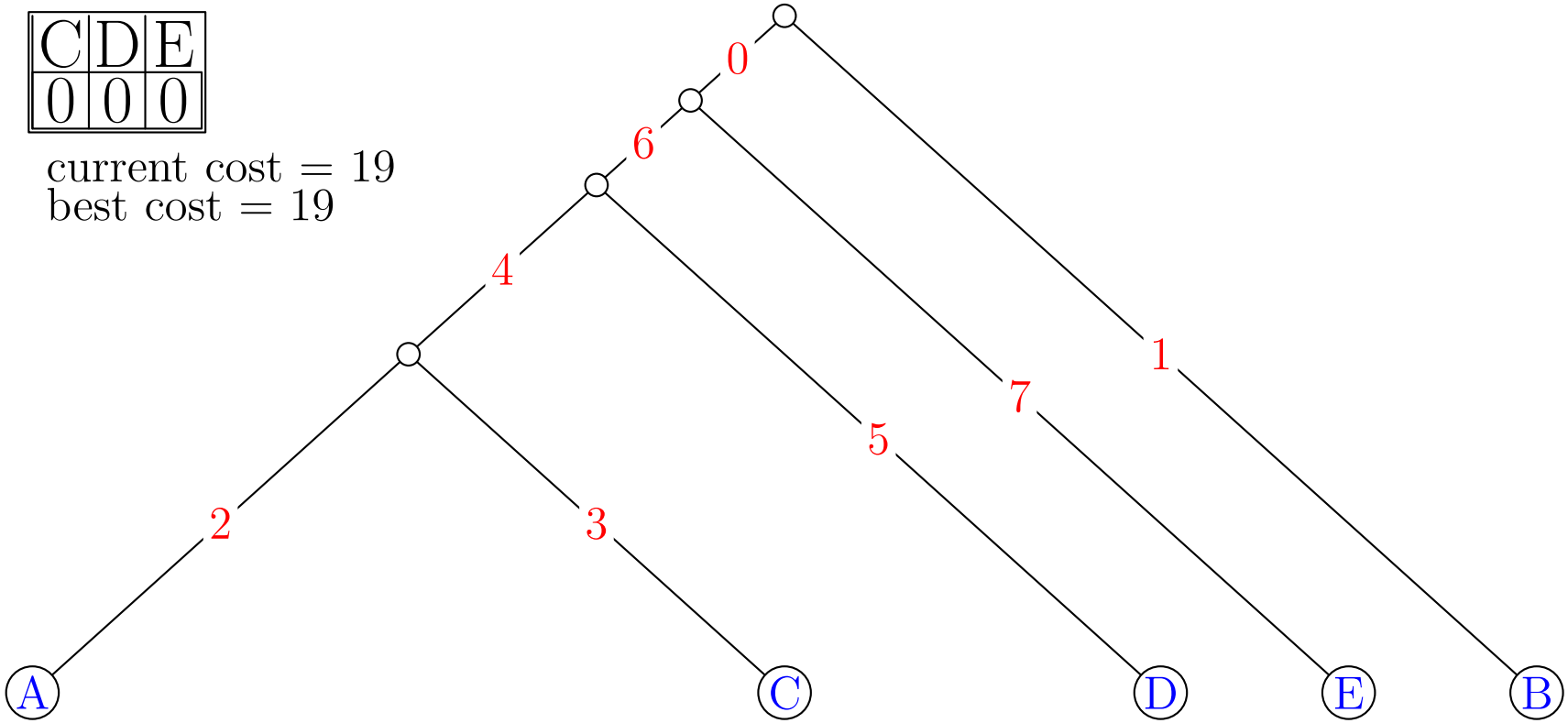
current cost = 11
best cost = 1000



Branch and Bound

C	D	E
0	0	0

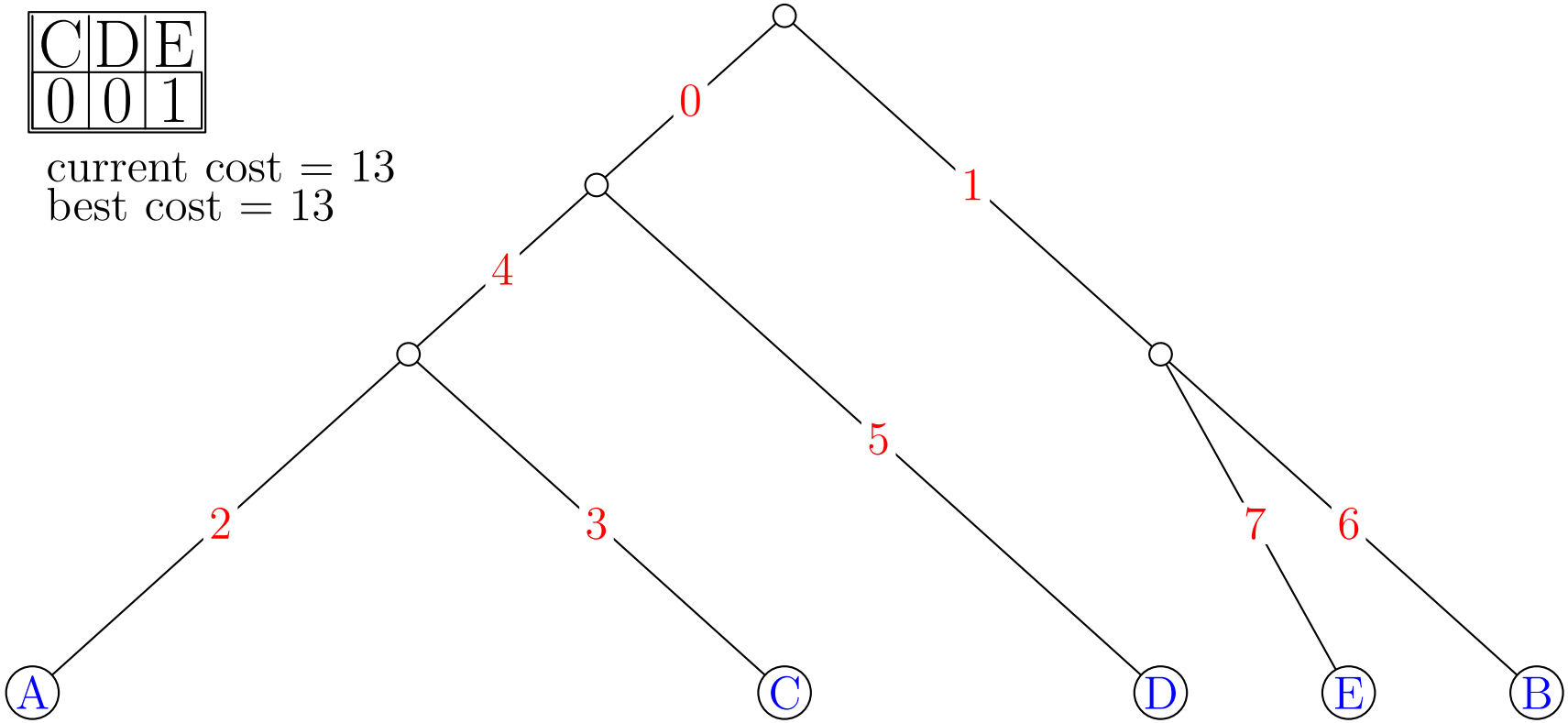
current cost = 19
best cost = 19



Branch and Bound

C	D	E
0	0	1

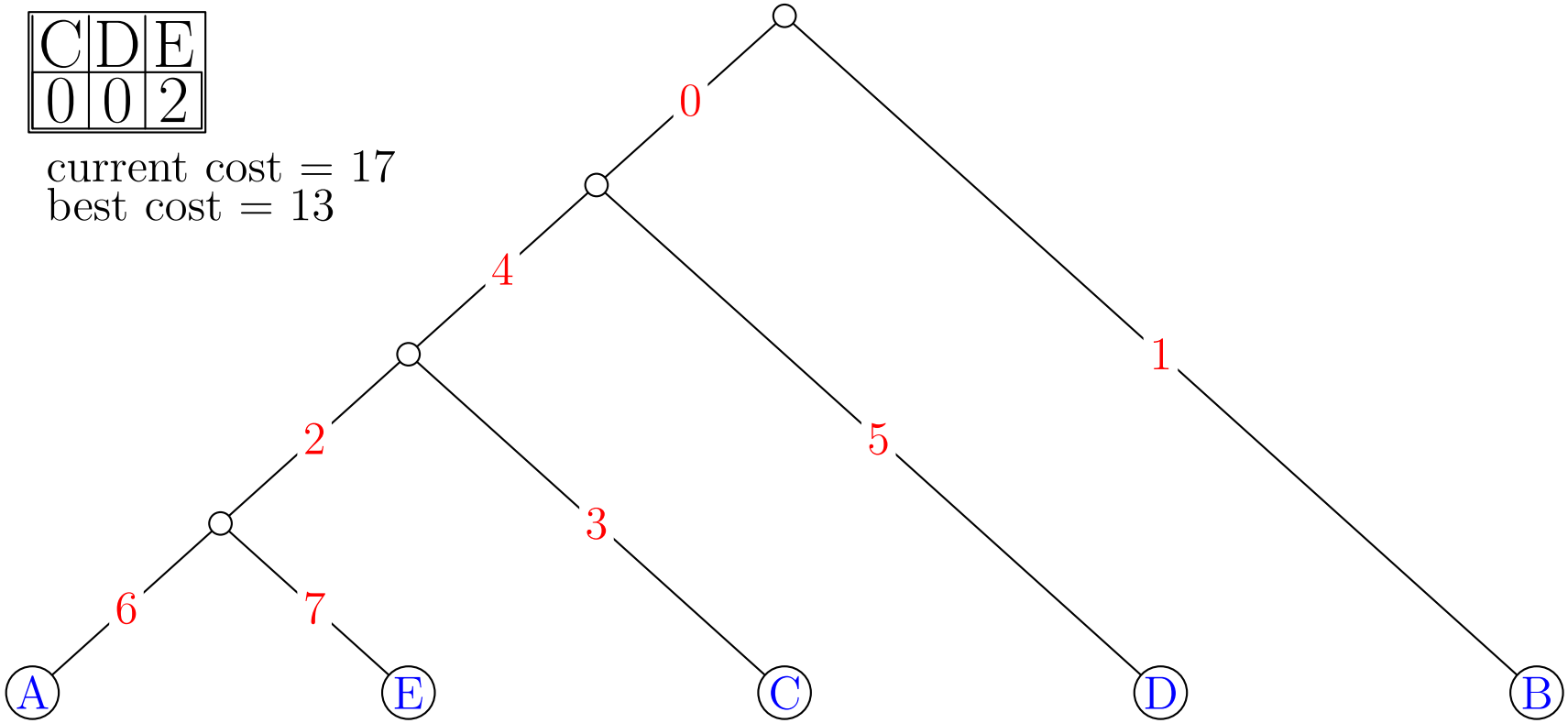
current cost = 13
best cost = 13



Branch and Bound

C	D	E
0	0	2

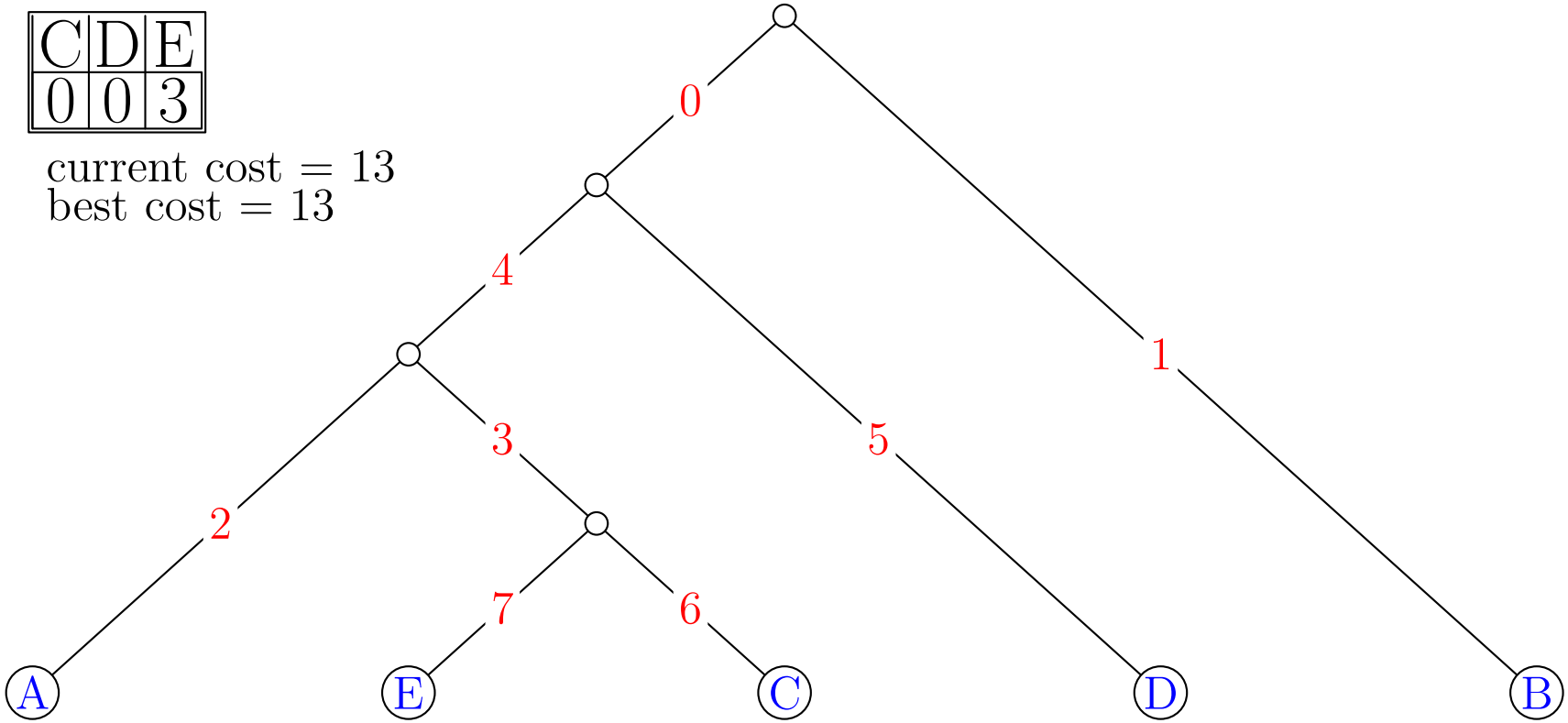
current cost = 17
best cost = 13



Branch and Bound

C	D	E
0	0	3

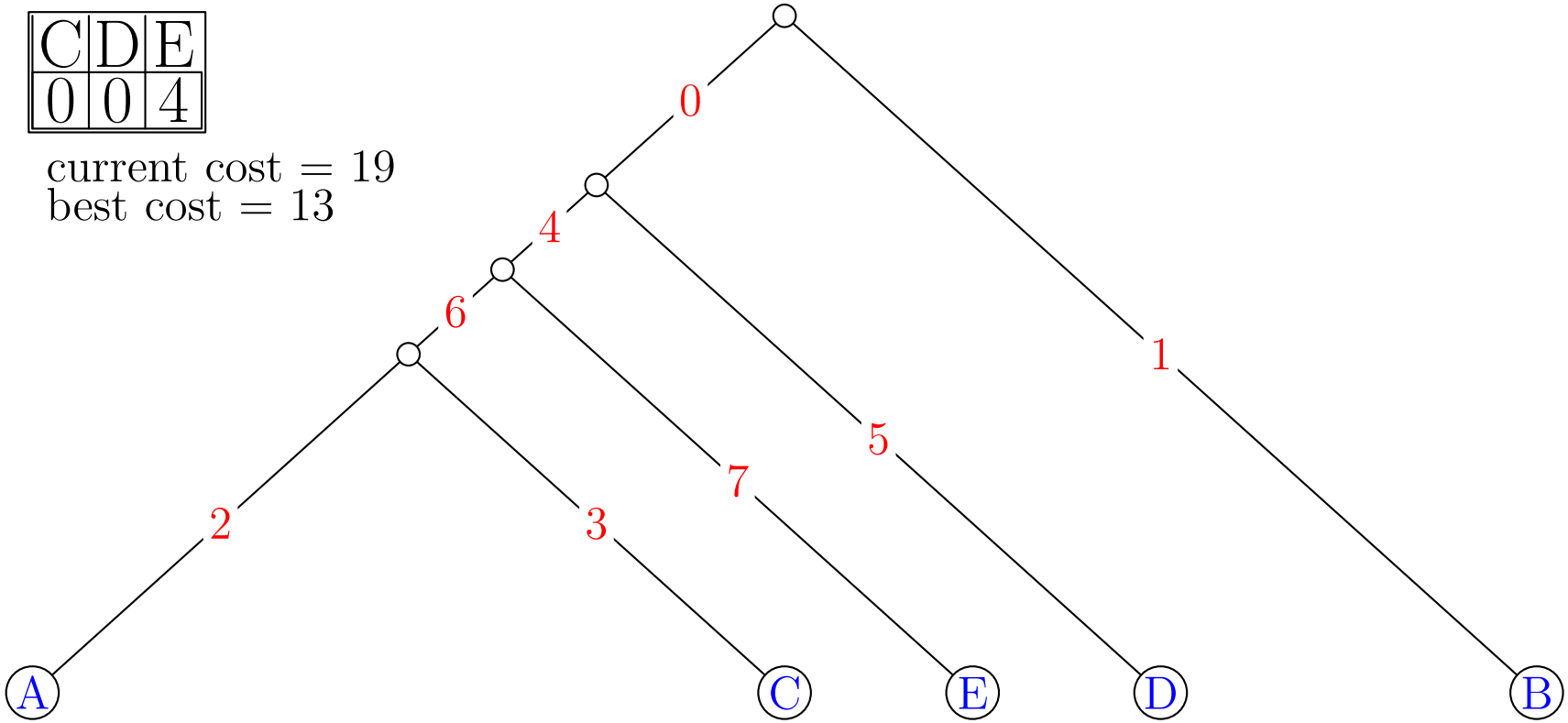
current cost = 13
best cost = 13



Branch and Bound

C	D	E
0	0	4

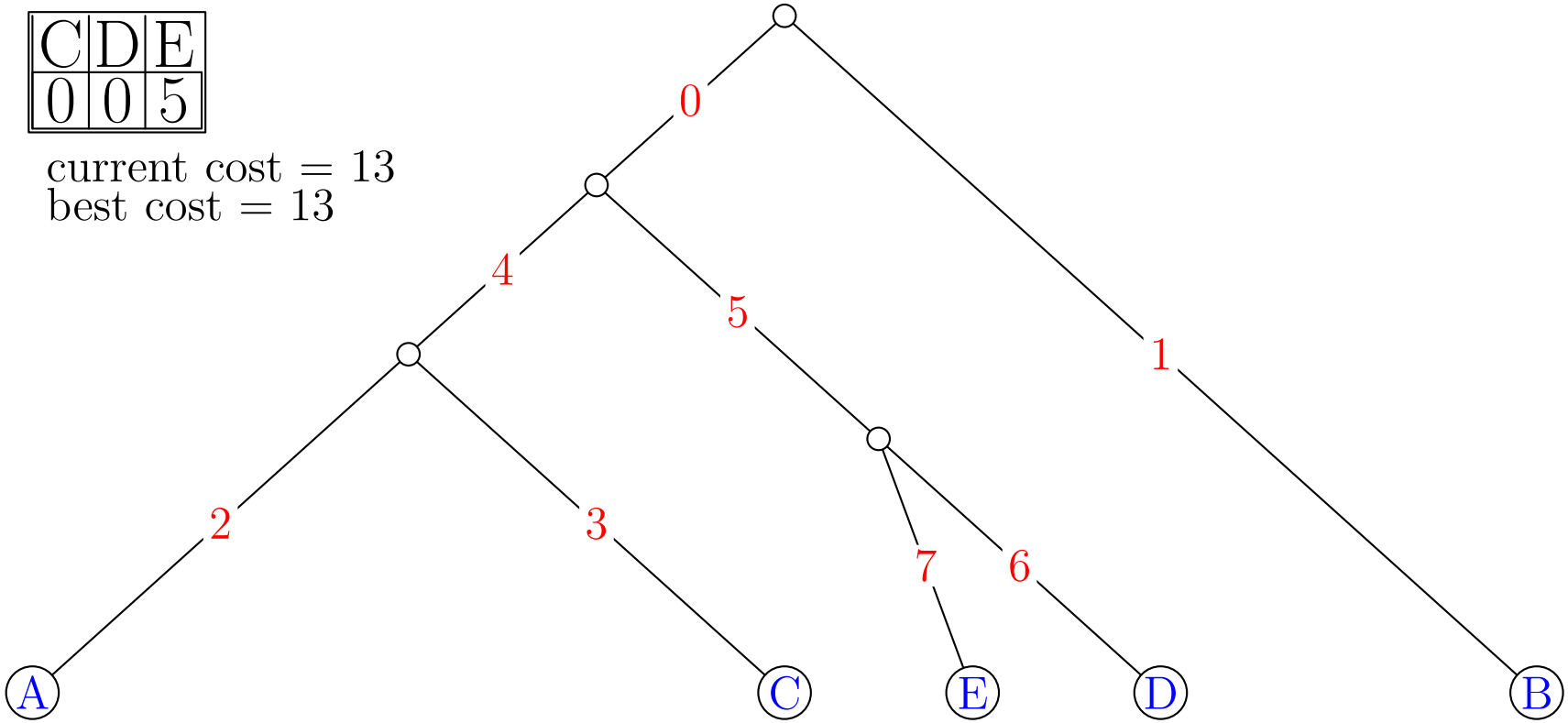
current cost = 19
best cost = 13



Branch and Bound

C	D	E
0	0	5

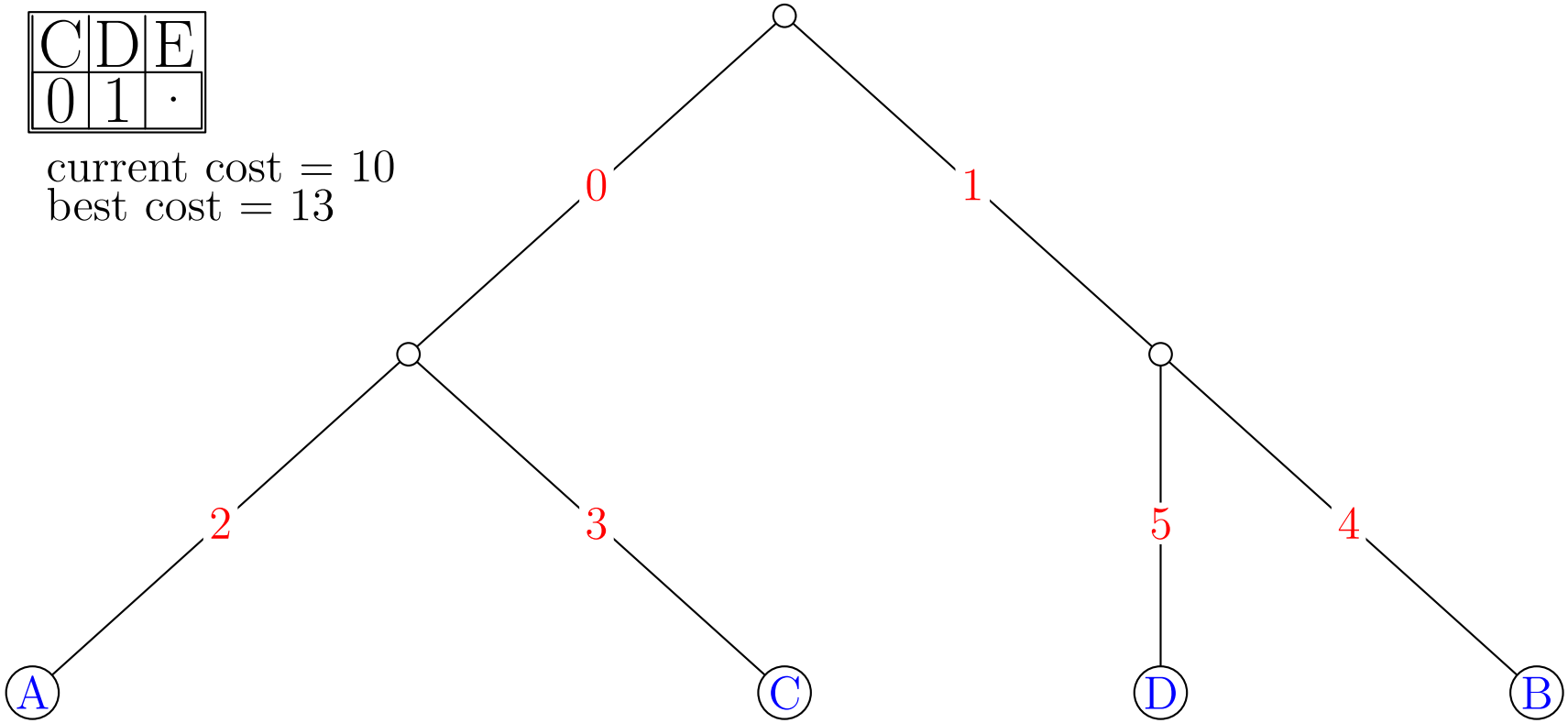
current cost = 13
best cost = 13



Branch and Bound

C	D	E
0	1	·

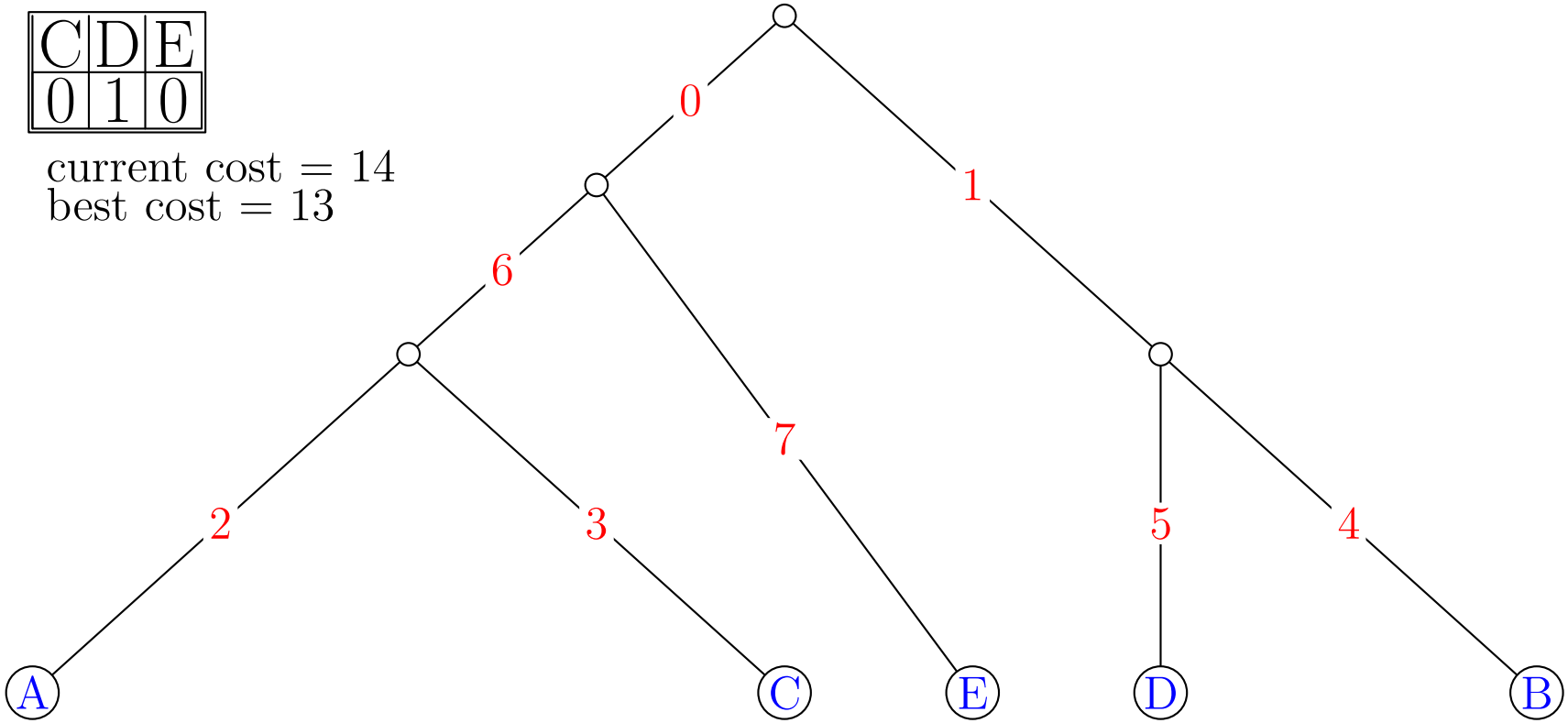
current cost = 10
best cost = 13



Branch and Bound

C	D	E
0	1	0

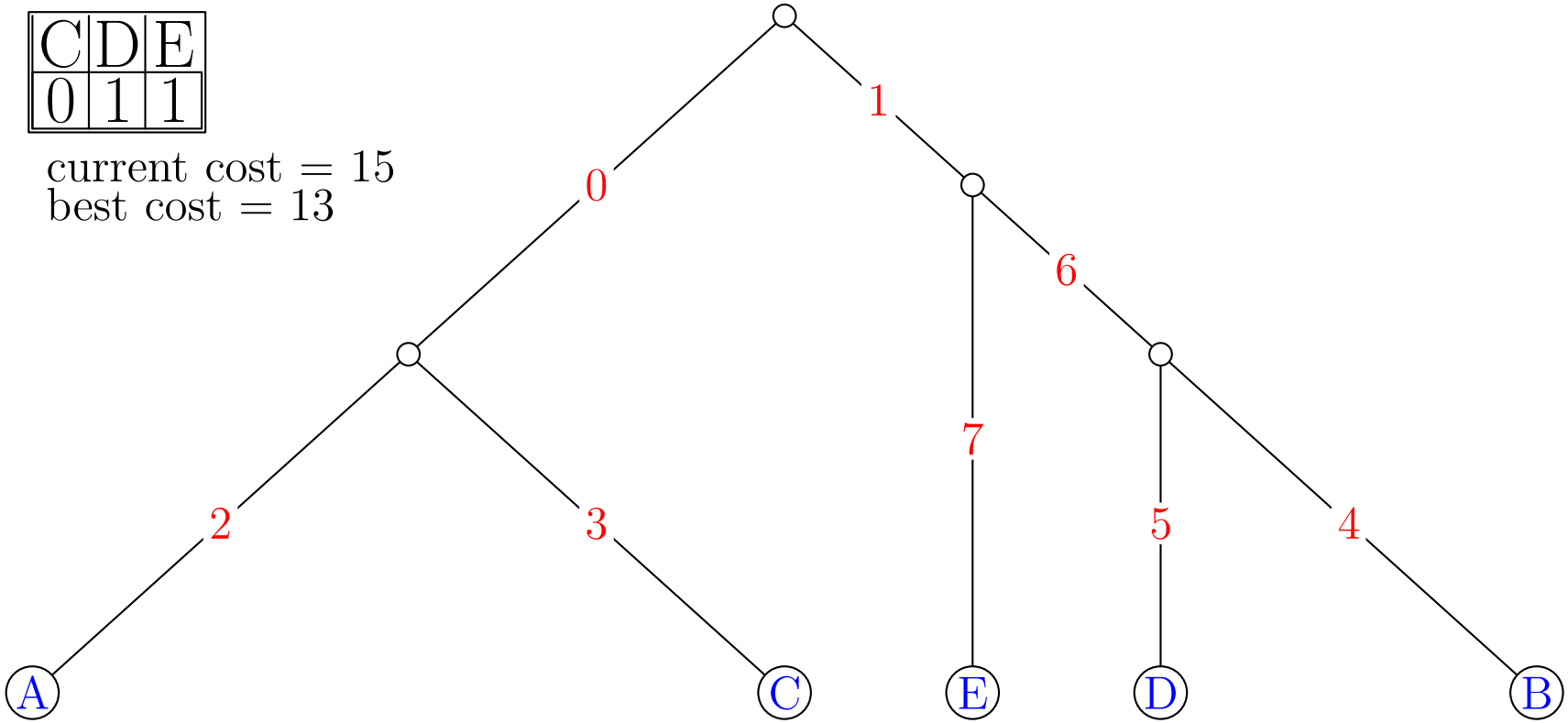
current cost = 14
best cost = 13



Branch and Bound

C	D	E
0	1	1

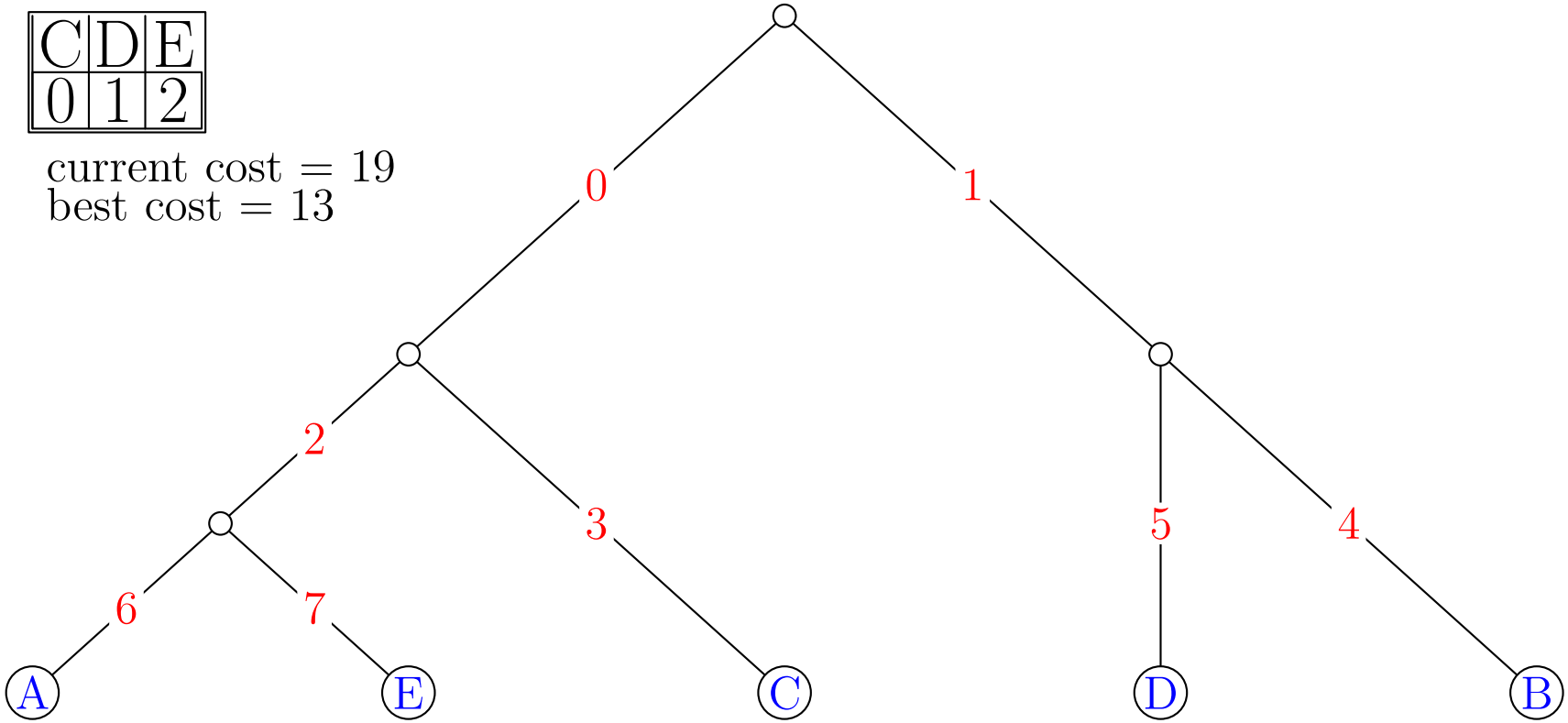
current cost = 15
best cost = 13



Branch and Bound

C	D	E
0	1	2

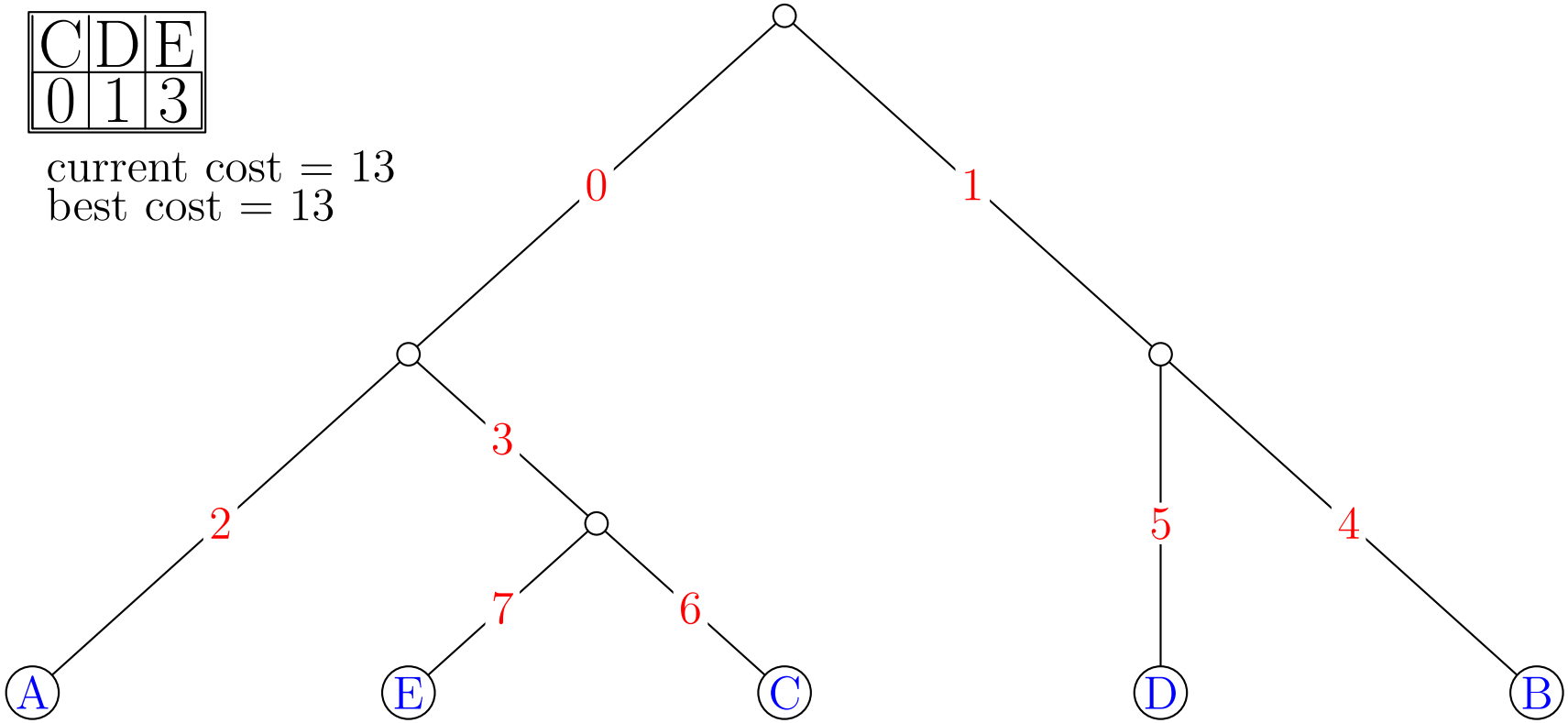
current cost = 19
best cost = 13



Branch and Bound

C	D	E
0	1	3

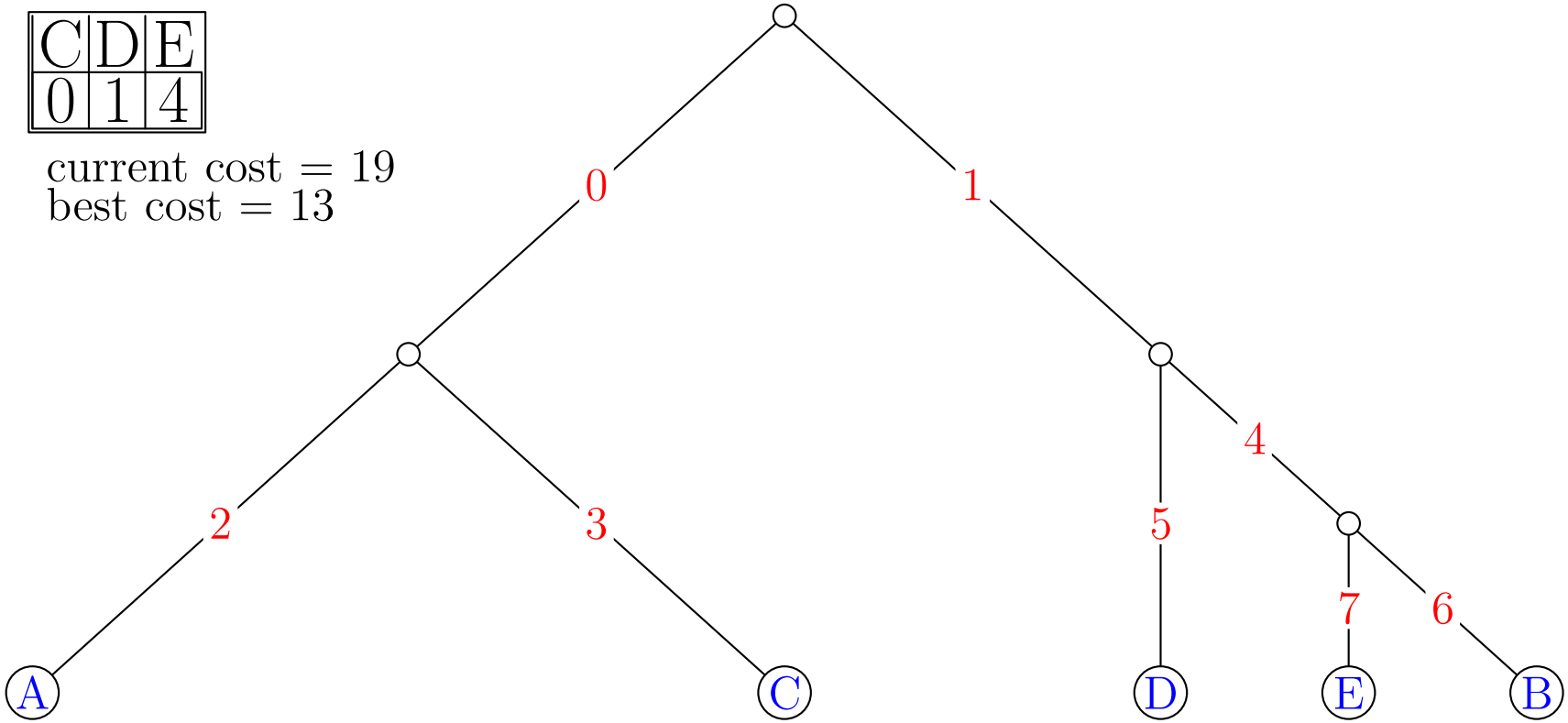
current cost = 13
best cost = 13



Branch and Bound

C	D	E
0	1	4

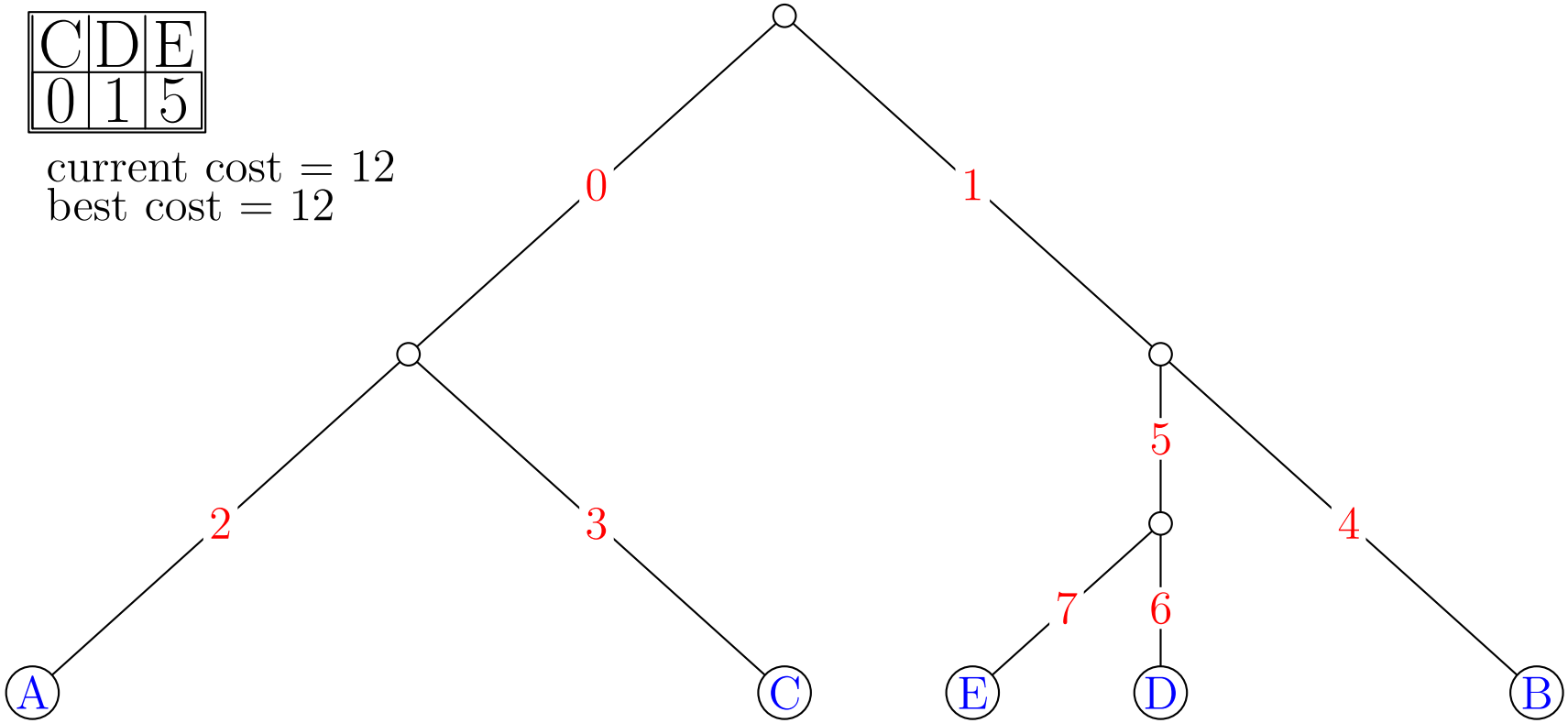
current cost = 19
best cost = 13



Branch and Bound

C	D	E
0	1	5

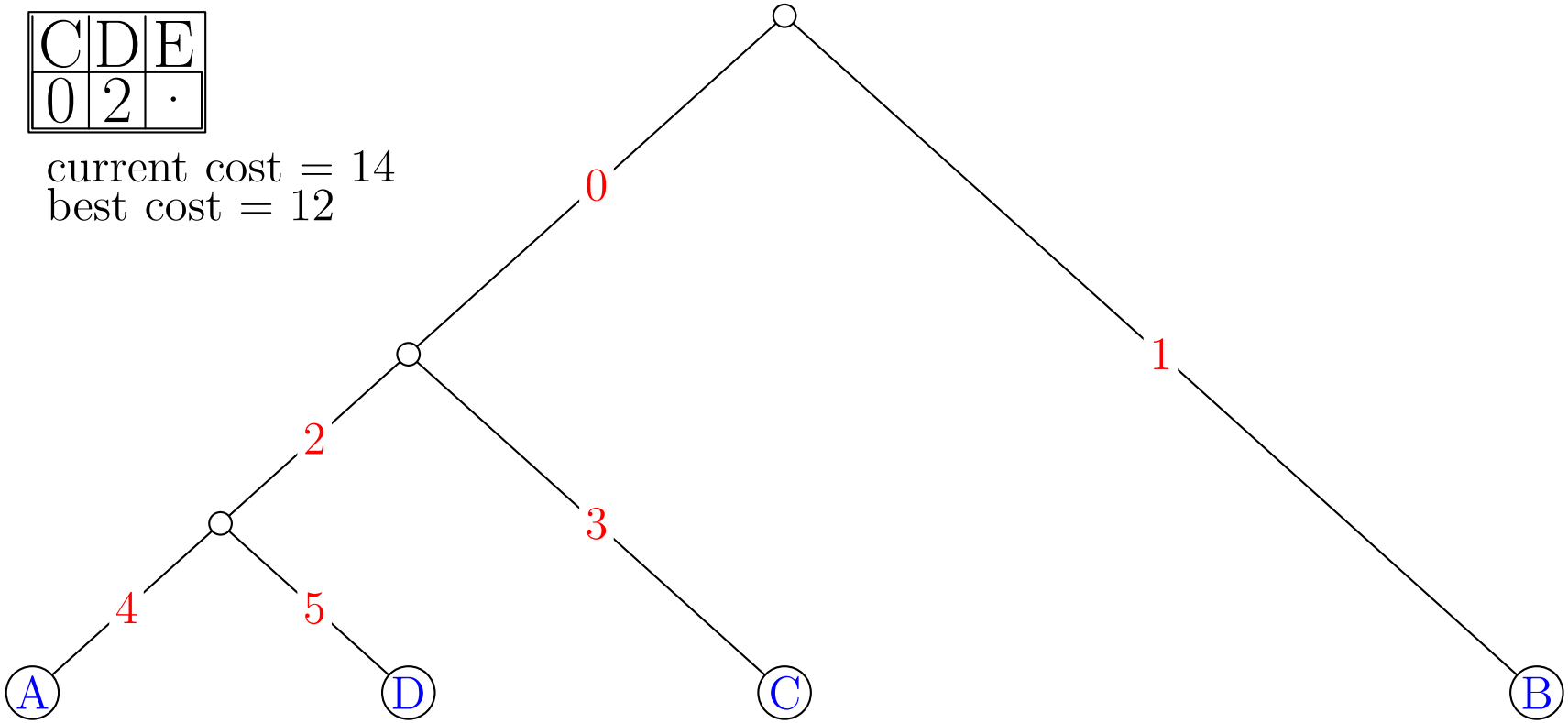
current cost = 12
best cost = 12



Branch and Bound

C	D	E
0	2	·

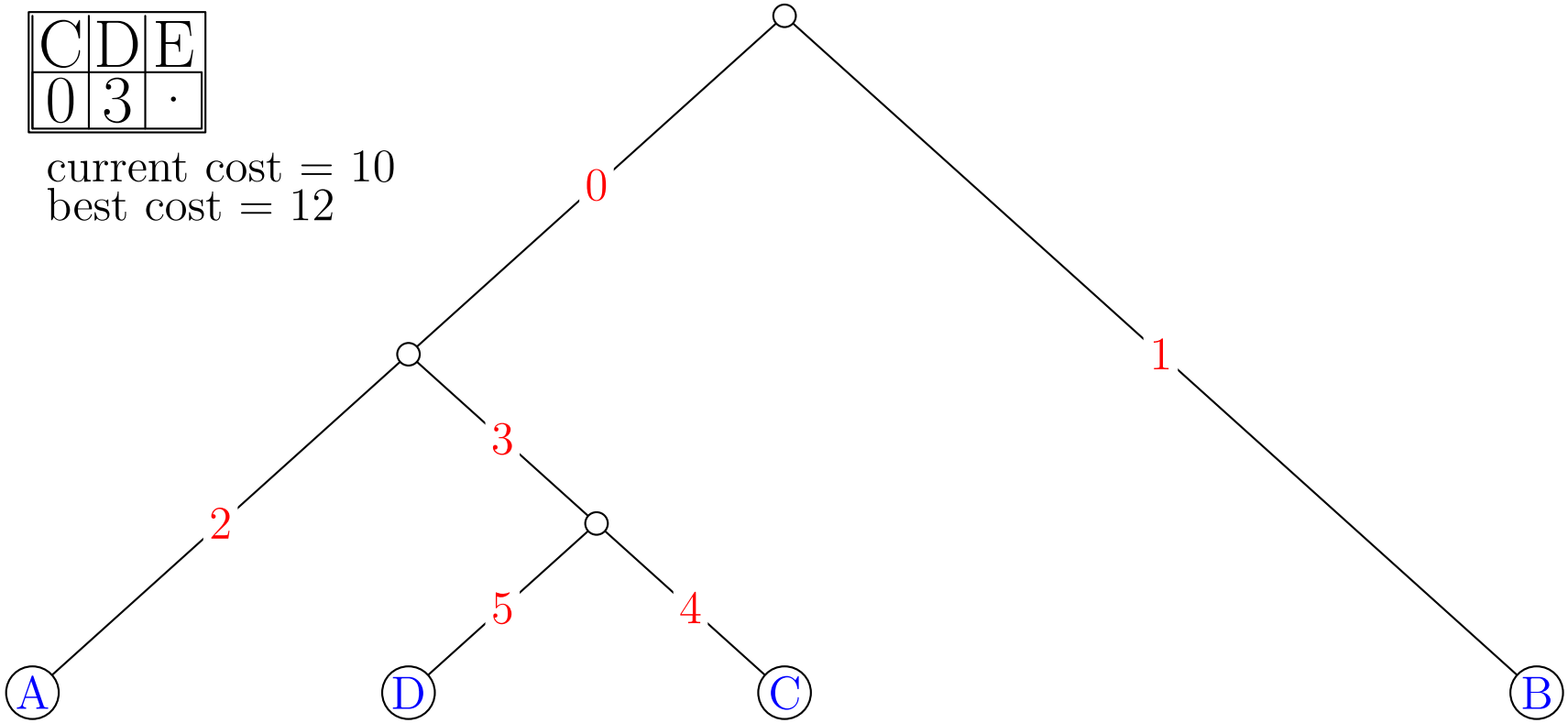
current cost = 14
best cost = 12



Branch and Bound

C	D	E
0	3	·

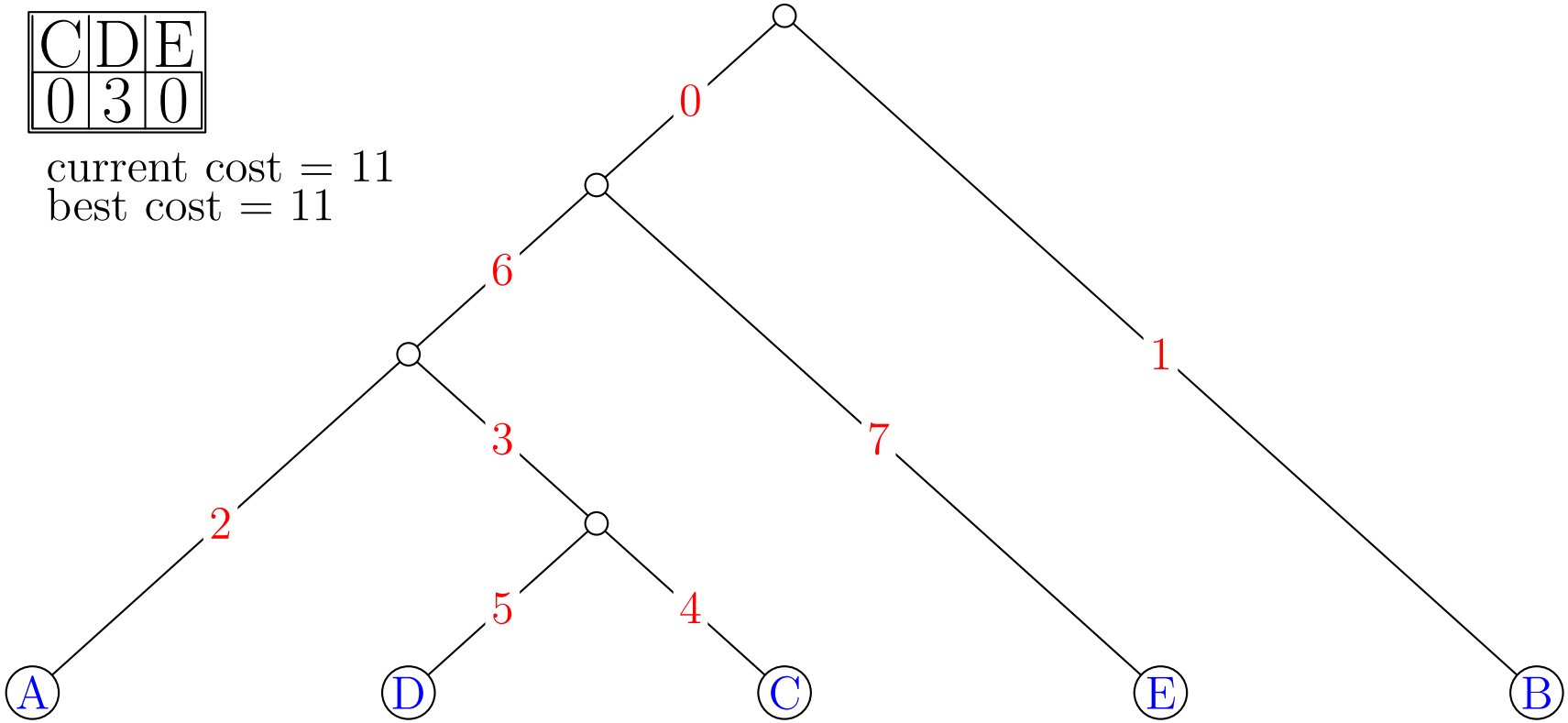
current cost = 10
best cost = 12



Branch and Bound

C	D	E
0	3	0

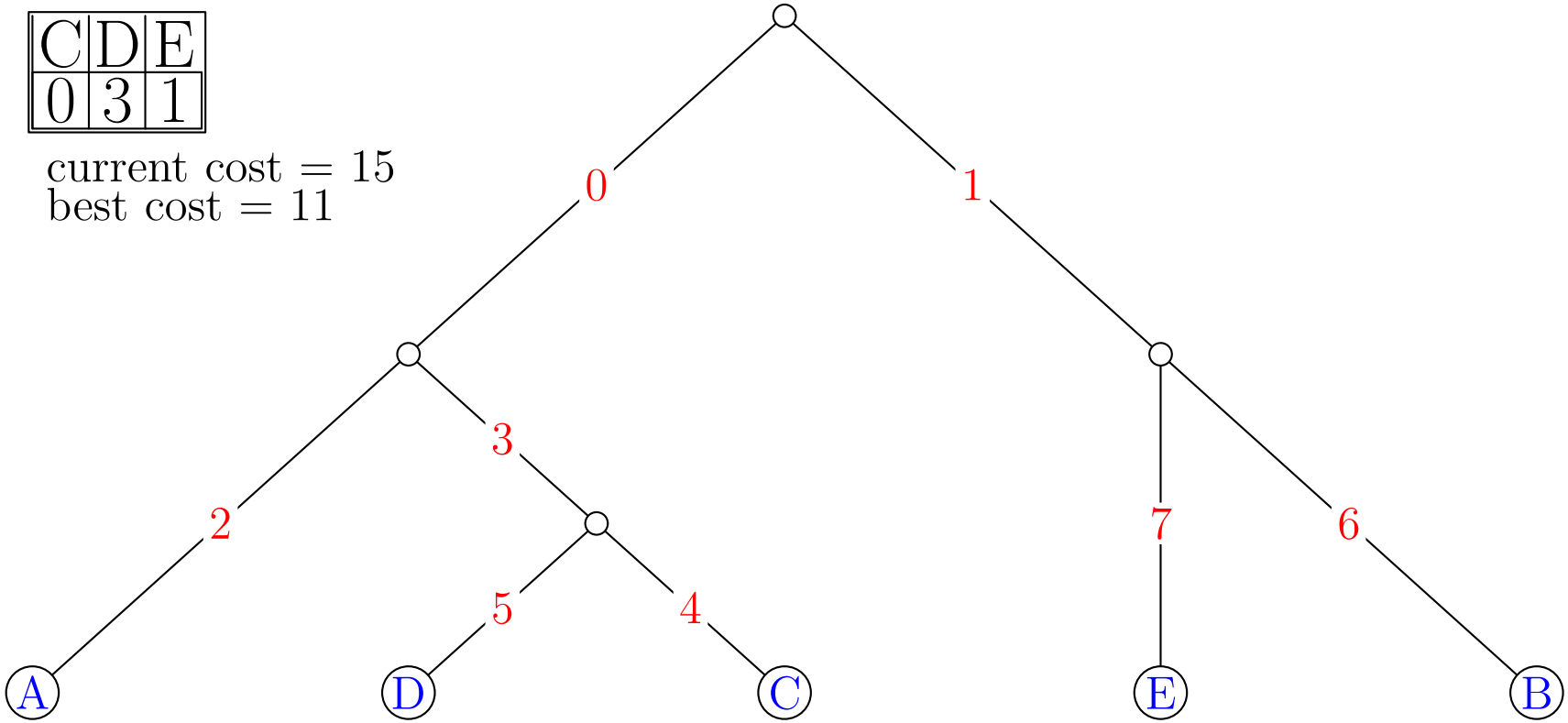
current cost = 11
best cost = 11



Branch and Bound

C	D	E
0	3	1

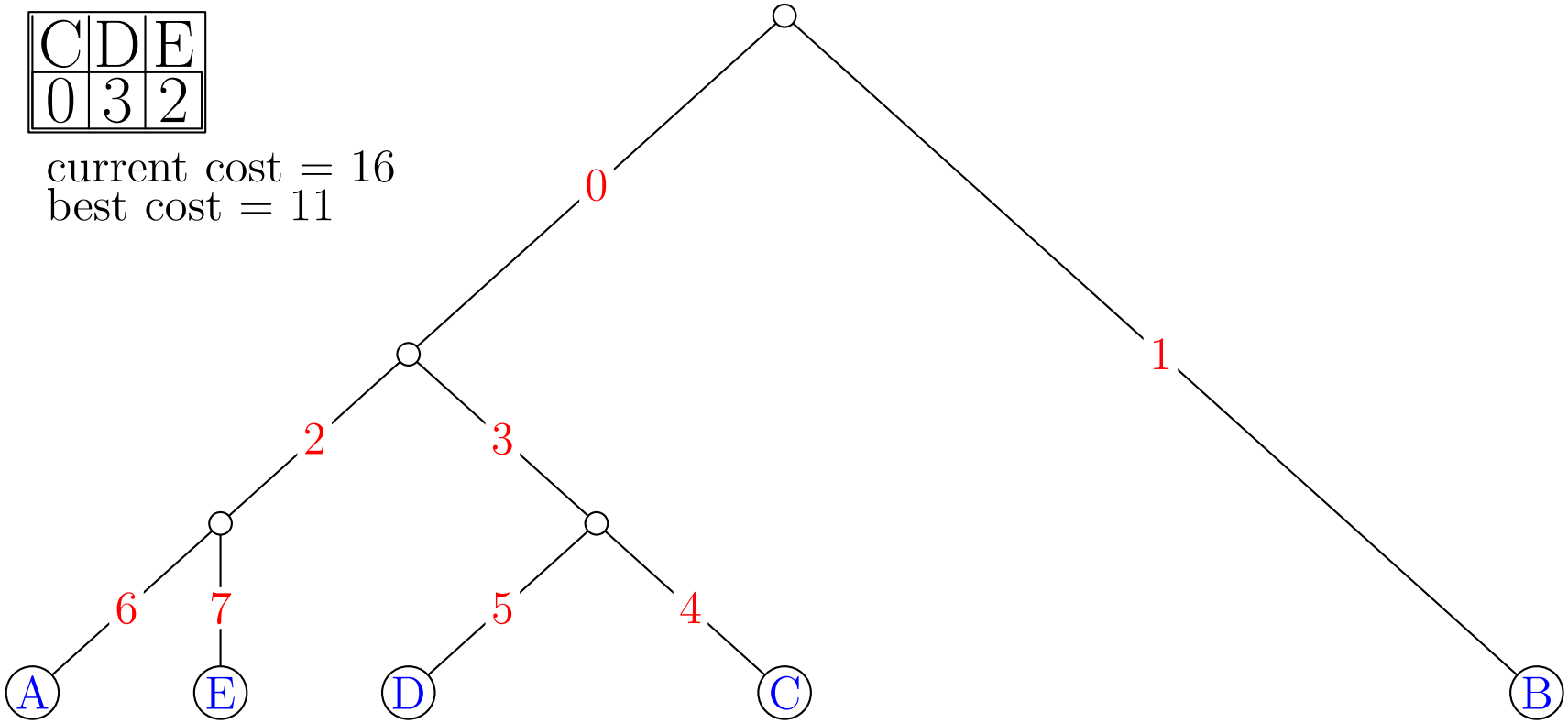
current cost = 15
best cost = 11



Branch and Bound

C	D	E
0	3	2

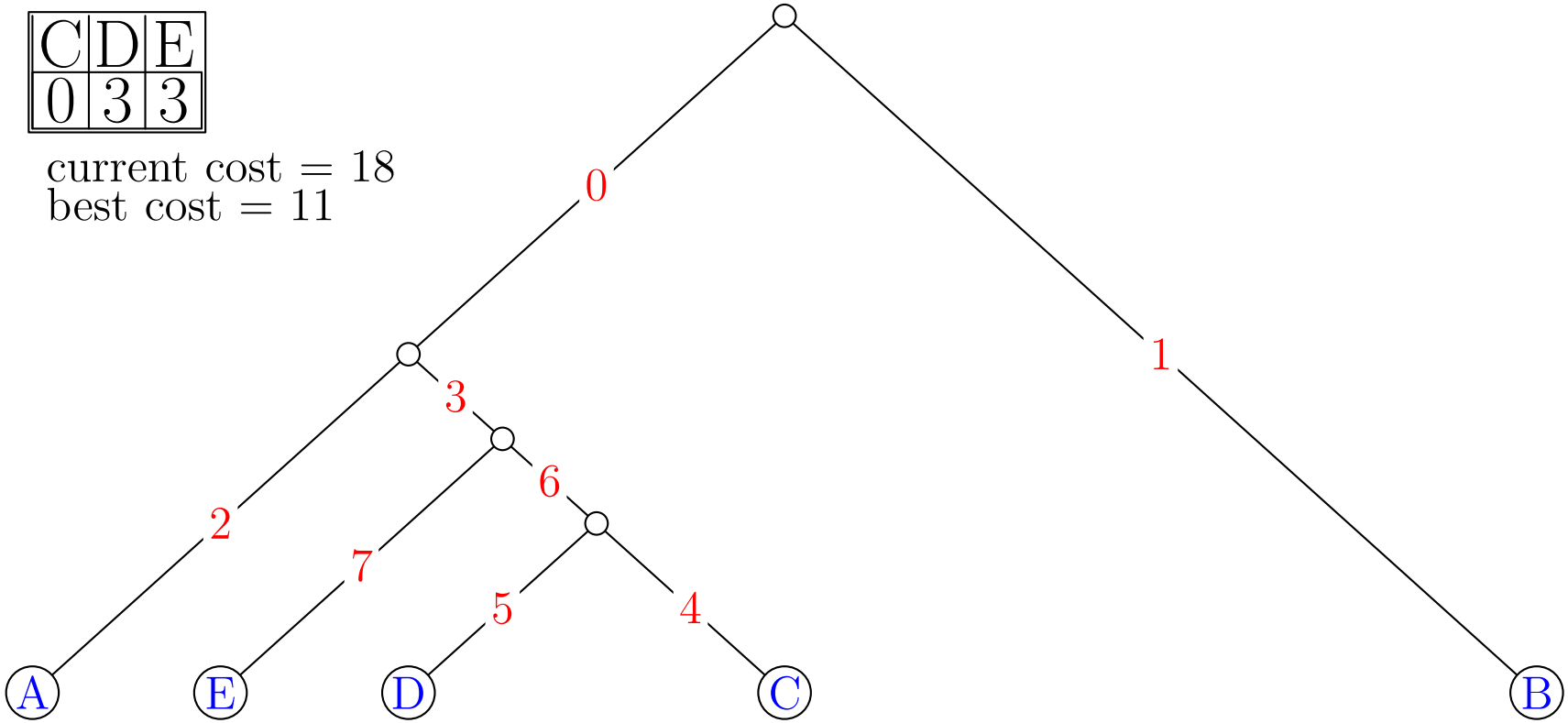
current cost = 16
best cost = 11



Branch and Bound

C	D	E
0	3	3

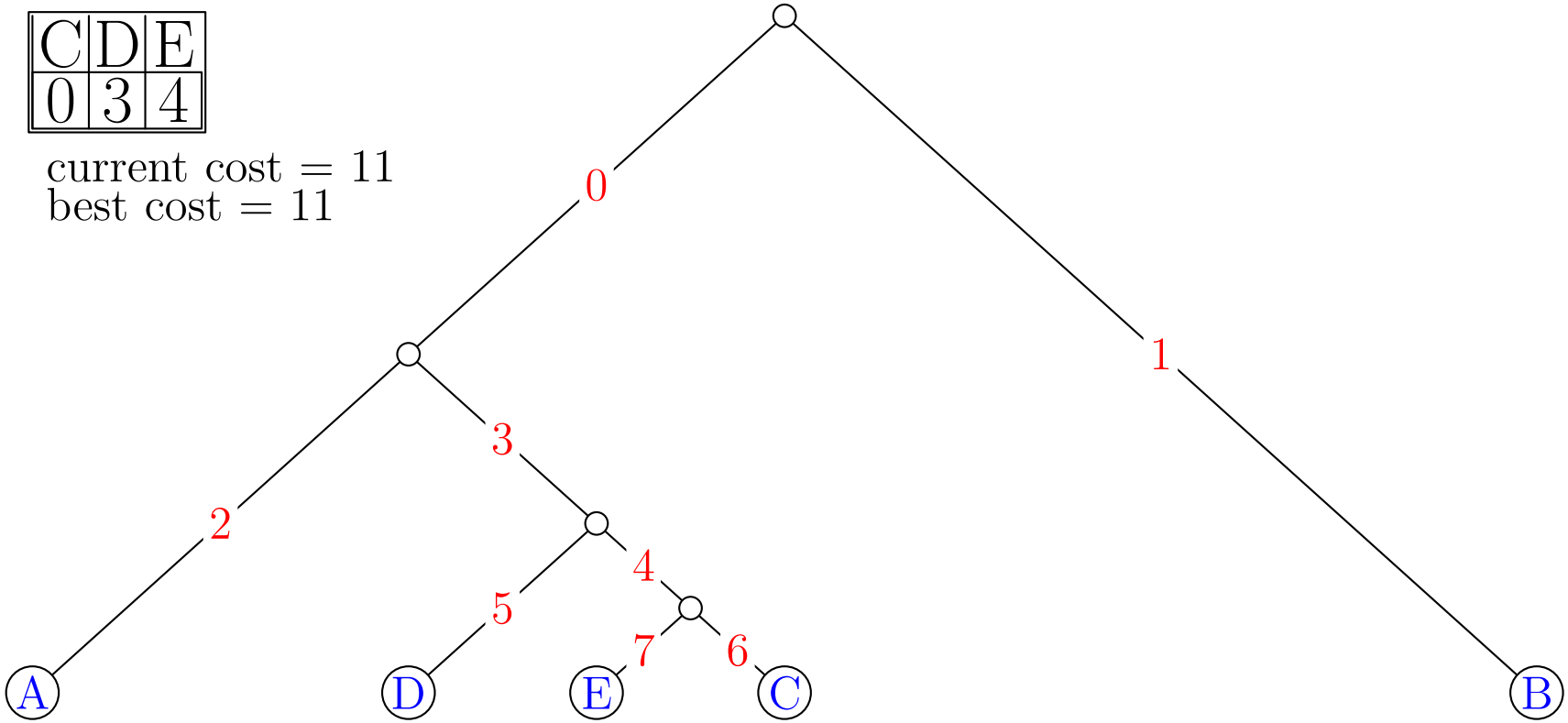
current cost = 18
best cost = 11



Branch and Bound

C	D	E
0	3	4

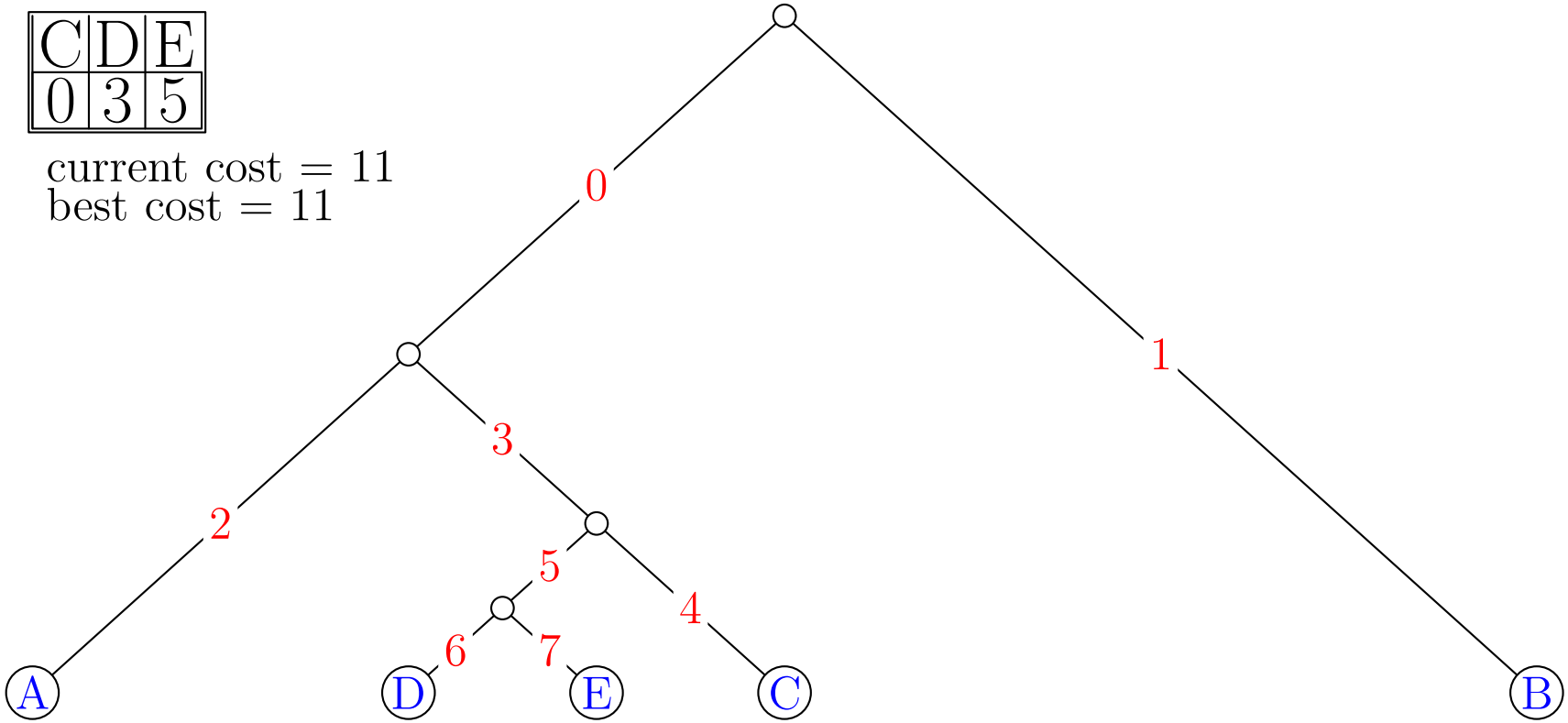
current cost = 11
best cost = 11



Branch and Bound

C	D	E
0	3	5

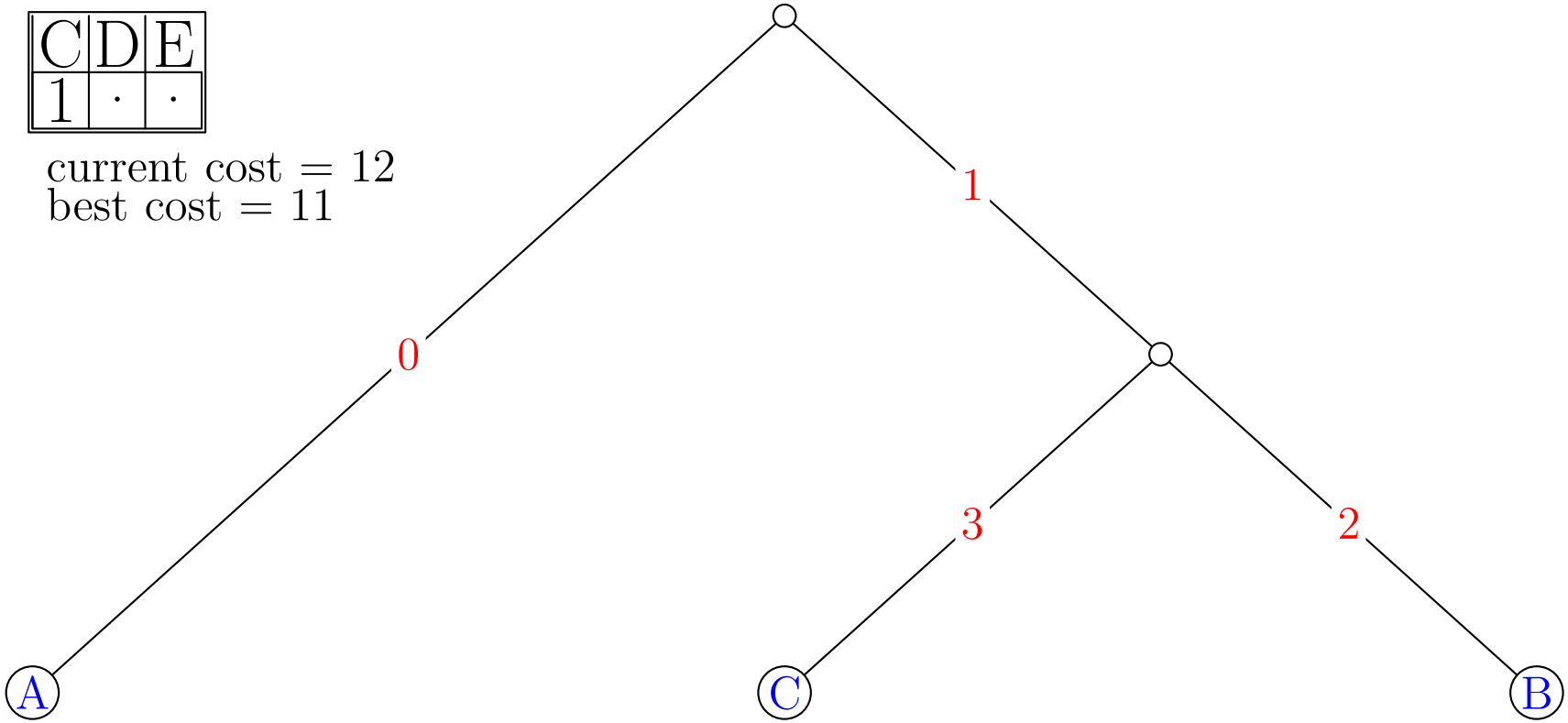
current cost = 11
best cost = 11



Branch and Bound

C	D	E
1	·	·

current cost = 12
best cost = 11



Assessing Trees

- How reliable are the maximum parsimony trees we find?
- Difficult to assess directly
- Can build different trees by **bootstrapping**

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ATCG AACT GATC CCGA CGCG CATC

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CGCG

ATCG AACT GATC CCGA CGCG CATC

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CGCG CCGA

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CGCG CCGA CGCG

ATCG AACT GATC CCGA CGCG CATC

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ATCG	CGCG	GATC	AACT	CGCG	AACT

ATCG	AACT	GATC	CCGA	CGCG	CATC

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ATCG	AACT	GATC	CCGA	CGCG	CATC

CGCG	CCGA	CGCG	CATC	AACT	ATCG
ATCG	CGCG	GATC	AACT	CGCG	AACT
CATC	ATCG	GATC	CATC	GATC	CCGA

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ATCG	AACT	GATC	CCGA	CGCG	CATC

CGCG	CCGA	CGCG	CATC	AACT	ATCG
ATCG	CGCG	GATC	AACT	CGCG	AACT
CATC	ATCG	GATC	CATC	GATC	CCGA
CCGA	GATC	ATCG	CATC	CCGA	CATC

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CGCG	CCGA	CGCG	CATC	AACT	ATCG
ATCG	CGCG	GATC	AACT	CGCG	AACT
CATC	ATCG	GATC	CATC	GATC	CCGA
CCGA	GATC	ATCG	CATC	CCGA	CATC
GATC	AACT	GATC	CCGA	CATC	AACT

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ATCG	AACT	GATC	CCGA	CGCG	CATC

CGCG	CCGA	CGCG	CATC	AACT	ATCG
ATCG	CGCG	GATC	AACT	CGCG	AACT
CATC	ATCG	GATC	CATC	GATC	CCGA
CCGA	GATC	ATCG	CATC	CCGA	CATC
GATC	AACT	GATC	CCGA	CATC	AACT

- See if trees look the same

Aligning and Tree Building

- To score the trees we want to know the alignment
- But to do an alignment we want to find related sequences
- It therefore makes sense to do the tree building and alignment together
- There is an algorithm due to Hein that does this, but this is getting too involved

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Conclusion

- When we have a set of sequences with a good measure of evolutionary distance we can use either an ultrametric model (UPGMA) or and additive model
- When given a set of (aligned) sequence and some substitution costs then we can use maximum parsimony to find good trees
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